Economics of Exchange Rates

Filippo Nardoni

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Chapter 1

Introduction to Exchange Rates and FX

1.1 Introduction

International macroeconomics and Finance are fields that focus on the analysis and understanding of real and financial inter-dependencies among open economies. Such subjects have already been reported and studied by scholars such as **David Hume**. When studying international macroeconomics, two important tools are basically used and considered – **Balance of Payment (BOP)** and **Exchange Rates**. Both tools will enable and allow for extensive analysis of the entire setting of International macroeconomics.

Two main approaches are generally used to view International Finance:

- **Macroeconomic Approach**: focusing on the macroeconomic determinants of exchange rates – GDP, Inflation, Interest Rates.
- **Microstructure approach**: starting from the behaviour and interactions of market participants, leading to the full picture of the microstructure of the foreign exchange market (forward-looking agents and expectation formation).

However, other dichotomies can arise. In fact, we have data-driven *vs* theory-driven and small open economy *vs* large open economy [1]. Especially for the former dichotomy, on one side we find the DSGE approach, which is microfounded and provides small-scale models, and on the other side we have linear correlational models, which are the VAR approach.

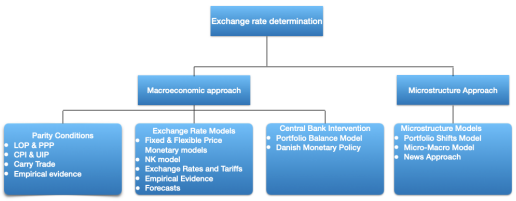


Figure 1.1: International Macroeconomics

1.1.1 Empirical Facts on FX Market

The FX is a financial market characterised by different market participants and different institutions. FX is by far the most liquid financial market in the world, even larger and more

[1] Small Open economy view assumes a country to be too small to influence RoW volumes and prices in world markets

liquid than the US government security market. From **BIS (2025)** (reporting comprehensive information on the size and structure of OTC FX and interest rate derivatives), we have that trading turnover [2] per day is on average $9.6 trillion in April 2025 (net-net so avoiding cross-border double counting), accounting for spot forward and swaps and excluding FX derivatives. We should also consider that, in order to get a relative idea of our numbers, the global total value of trade per year is expected to reach $35 trillion in 2026. The reasons that justify such an important role of the FX market are given by: **cross-border trade**, **Investment**, **Financial Transactions**.

Table 1.1: Currency distribution of reported foreign exchange market turnover. Percentage shares of average daily turnover in April.

	2007	2010	2013	2016	2019	2022	2025
US dollar	86.3	84.9	87.0	87.6	88.3	88.4	89.2
Euro	37.0	39.1	33.4	31.4	32.3	30.5	28.9
Japanese yen	16.5	19.0	23.0	21.6	16.8	16.7	16.8
Pound sterling	15.0	12.9	11.8	12.8	12.8	12.9	10.2
Swiss franc	6.8	6.3	5.2	4.8	4.9	5.2	6.4
Canadian dollar	4.2	5.3	4.6	5.1	5.0	6.2	5.8
Australian dollar	6.7	7.6	8.6	6.9	6.8	6.4	6.1
Swedish krona	2.8	2.2	1.8	2.2	2.0	2.2	1.6
Hong Kong dollar	2.8	2.4	1.4	1.7	3.5	2.6	3.8
Singapore dollar	1.2	1.4	1.4	1.8	1.8	2.4	2.4
Danish krone	0.9	0.6	0.8	0.8	0.6	0.7	0.7
All currencies	200.0	200.0	200.0	200.0	200.0	200.0	200.0

Source: BIS Triennial Central Bank Survey (2025).

From Table 1.1, we can see the distribution of foreign exchange market turnover[3], we see the USD dollar has gained some share in the market, whereas the EURO and sterling have dropped. *So one natural question would be why does the dollar have such strength?*

- The USD is used as a vehicle currency for cross-trading between other currencies (e.g., if you have DKK and want to buy the British Pound, we move from $DKK \rightarrow USD\ XrightarrowGBP$).

- The U.S. dollar is used in international trade and global payments, and important commodities such as oil, gold, and coffee are priced in dollars.

- About half of the debt and loans on the off-shore market are denominated in USD. Issuer/borrower and lender are non-US residents

From Table 1.2 we observe how the exchange of Dollars for Euros covers most of the FX market. As reported by BIS (2025), the most important FX are **London**, **New York**, **Tokyo**, **Singapore**, **Hong Kong**, and **Frankfurt**.

[2] Turnover = $\sum$ price $\times$ quantity traded.

[3] We have 200% since we evaluate two currencies, so we evaluate sell x and buy y, and sell y and buy x

Table 1.2: OTC foreign exchange turnover by currency pair (net-net basis). Daily averages in April in billions of US dollars and percentage share.

	2013		2016		2019*		2022*		2025	
Currency pair	Amount	%	Amount	%	Amount	%	Amount	%	Amount	%
USD/EUR	1,292	24.1	1,172	23.1	1,581	24.0	1,697	22.7	2,033	21.2
USD/JPY	980	18.3	901	17.8	871	13.2	1,009	13.5	1,372	14.3
USD/CNY	113	2.1	192	3.8	270	4.1	492	6.6	781	8.1
USD/GBP	473	8.8	470	9.3	630	9.6	710	9.5	731	7.6
USD/CAD	200	3.7	218	4.3	287	4.4	408	5.5	505	5.3
EUR/SEK	28	0.5	36	0.7	36	0.6	38	0.5	39	0.4
EUR/AUD	21	0.4	16	0.3	18	0.3	24	0.3	31	0.3
EUR/NOK	20	0.4	28	0.6	33	0.5	36	0.5	31	0.3
EUR/PLN	14	0.3	13	0.3	13	0.2	17	0.2	27	0.3
EUR/DKK	13	0.2	13	0.2	11	0.2	13	0.2	26	0.3

Note: Figures are reported on a net-net basis. *Source:* BIS Triennial Central Bank Survey.

Why the FX market is open 24 hours a day but not 24/7.

The foreign exchange (FX) market operates continuously for 24 hours a day during the working week, from Sunday 5:00 pm EST to Friday 5:00 pm EST, but it is not open seven days a week.
This is due to the global distribution of trading across time zones. Trading begins in the Asia–Pacific region (e.g. Hong Kong, Tokyo, Singapore), then moves to Europe (London, Frankfurt), and finally to North America (New York). When U.S. markets close, markets in New Zealand and Australia are already opening, and Asian markets reopen shortly thereafter. This "follow-the-sun" structure ensures continuous trading during the week.
However, FX trading relies on banks, dealers, and settlement systems that close on weekends for operational, regulatory, and risk-management reasons. Consequently, trading halts over the weekend and resumes when Asian markets open on Sunday evening (EST).

1.1.2 On the Foreign Exchange

Currencies are traded over the counter, so they are not regulated. There is no organisation supervision and no institution that sets the rules; however, we have a *Code of Conduct* whose Central Banks and market participants have agreed upon (around 1200 market participants). The FX market is **Decentralized** and **Fragmented**. An important role is played by the **Market Maker**, a bank that quotes prices and guarantees to buy and sell a minimum volume across one or more currency pairs. In so doing, market makers are exposed to different risks:

- **Liquidity risk**: bank is forced to match orders even in times of low liquidity.
- **Market Risk**: the value of a position in a currency varies over time.
- **Credit Risk**, which is subdivided into **Counterparty risk** (counterparty default – especially when dealing with forward contracts) and **Settlement risk** (risk that once the currency is bought, it is not received by the buyer).

FIGURE: Organization of the foreign exchange market

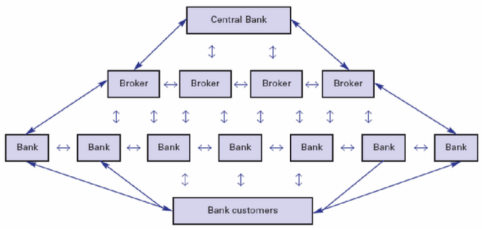


Figure 1.2: FX organization

Market Participants

In the FX Market, we might find different Market Participants who take part in the exchange of currencies, in particular, we have:

- **Retail clients**: businesses, international investors, multinational corporations and other participants that do not directly buy or sell foreign currencies but only place orders with commercial banks.
- **Commercial Banks**: Are the ones that carry out buy and sell orders on their own account and on behalf of customers. They trade *directly* – *Interbank Market* – with other banks or *FX Brokers*.
- **FX Brokers**: Those are specialised intermediaries whose role is to collect orders from different banks and efficiently match those. They operate in major financial centres and operate in a way to reduce transaction costs.
- **Central Banks**: Central banks intervene on the foreign exchange market to buy and sell its own currency in order to influence the value of the domestic currency.

In the overall structure of the foreign exchange market, two main sub-markets can be distinguished:

- **Retail Market**: Corporate customers do not trade foreign currencies directly among themselves. Instead, they access the foreign exchange market through commercial and investment banks, which act as intermediaries.
- **Interbank Market**: Banks trade foreign currencies with each other, either directly or through brokers. Hence, transactions can be dealer–to–dealer or broker–to–dealer. The interbank market can be further divided into:
  - the *direct market*, where trades occur directly between banks (bank–to–bank);
  - the *indirect* or *brokered market*, where banks trade via brokers (bank–to–broker).

1.1.3 The FX Market

We analyse in this section both *Bilateral* and *Multilateral* exchange rates.

Bilateral Exchange Rate

Whenever we introduce bilateral exchange rates, we need to account for nominal and real terms.

Nominal Exchange Rate The bilateral nominal exchange rate is the relative price between

two national currencies, we can express as follow:
The price quotation answers the question of how many units of home currency I need to buy 1 unit of foreign currency.

$$S_t^{H/F} = \underbrace{\frac{\text{n unit of domestic currency}}{\text{1 unit of foreign currency}}}_{\textbf{price quotation system or direct quotation}}$$

The volume quotation instead answers a different question: how many units of foreign currency do I need for 1 unit of home currency?

$$S_t^{F/H} = \underbrace{\frac{\text{n unit of foreign currency}}{\text{1 unit of home currency}}}_{\textbf{volume quotation system or indirect quotation}}$$

In the end, we can express both quotations as follows:

$$S_t^{H/F} = \frac{1}{S_t^{F/H}}$$

We generally refer to the direct quotation, we use the term **depreciation** of domestic currency to an increase in the nominal exchange rate in direct quotation, meaning that we need more units of domestic currency to buy 1 unit of foreign currency. On the other side, we talk about **Appreciation** when we have that $S_t^{H/F} \downarrow$. We can now introduce analysis regarding arbitrage.

Arbitrage on currencies implies simultaneous or quasi-simultaneous purchase and sales of currencies with the objective to realise profits from deviations in the exchange rate between the same pair of currencies occurring in different locations under the following assumptions:

- **Negligible or no transaction costs** (easy telecommunications)
- **No operation risk performed by the foreign exchange traders**

In case we get $S_t^{H/F} S_t^{F/H} \neq 1$, it is possible for arbitrageurs to start making profits as long as the law of demand and supply of the two currencies in each of the two financial centres does not equalise the temporarily diverging exchange rates.

Three-point Arbitrage and cross Rates

: Given three currencies i, j, k, we can find the exchange rate of two currencies just relying on the other two:

$$\underbrace{S_t^{i/j}}_{\text{direct route}} = \underbrace{S_t^{i/k} S_t^{k/j}}_{\text{indirect route}}$$

Generally, we have that under arbitrage conditions:

$$S_t^{i/j} S_t^{j/k} S_t^{k/j} = 1$$

This cross-currency exchange can be explained by a particular phenomenon, which is the **vehicle currency**, especially important during the Bretton Woods period (1946-1971).

Vehicle Currency is a major global currency traded across many financial centres and in large volumes, serving as a facilitator of currency exchange.

Real Exchange Rate

Unlike the nominal exchange rate the real exchange rate is a synthetic concept. The RER is a measure of relative price or relative cost of two separate economies when prices are expressed in the same currency. The RER can be thought as a measure of international competitiveness of one country against the other.

$$q_t^{PPP} = \frac{S_t P_t^*}{P_t}$$

P_t^* represents aggregate prices in foreign country [4]. $S_t P_t^*$ converts price index on basket of foreign goods using home currency and we divide it to the price level in home country, so if $q_t^{PPP} > 1$ foreign goods are more expensive so domestic currency is undervalued.

Multilateral or effective exchange rate

The multilateral or effective exchange rate attempts to capture the evolution of the value of the domestic currency with respect to the trading partners. SO the multilateral exchange rate at time t of a currency i is approximated by the weighted geometric average.

$$NEER_{it} = \prod_{\forall j \neq i}^{n} (S_t^{i/j})^{\omega_j}, \qquad \sum \omega_j = 1$$

The NEER is expressed with a base 100 and the weights are given by the respective share of foreign trade (export+imports). If NEER ↑ we have nomianl appreciation. The REER also takes into account the inflation rate through the price levels so we correct for relative prices so if REER ↑ we have a loss of competitiveness.

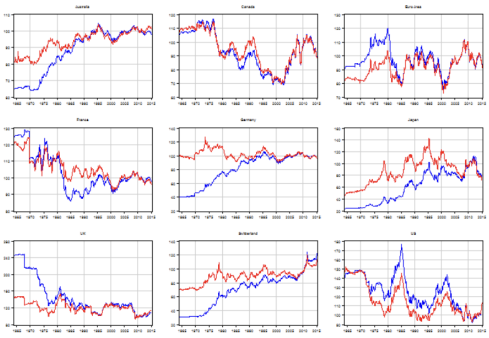


Figure 2: NEERs (blue/dashed) and REERs (red/solid) for Major World Currencies, Monthly Indexes, 2010=100: Increase (↑) Denotes Appreciation. Source: BIS (website).

Figure 1.3

From the figure above, both nominal and real effective exchange rates exhibit highly persistent dynamics that are largely consistent with a random-walk process. While real effective exchange rates appear less volatile and less trending than their nominal counterparts, they

[4] Edwards (1988), among others, argues that such a RER definition is conceptually problematic. It compares price levels, hence measures of relative standards of living in two countries, rather than relative production costs or the relative price of tradables to nontradables in each. It is the latter two RER concepts that relate more directly to international competitiveness.

nonetheless display little evidence of mean reversion. This visual impression is supported by augmented Dickey–Fuller unit-root tests, which in most cases fail to reject the null hypothesis of non-stationarity across standard specifications (with trend, with intercept only, and without deterministic components). Rejections, when present, are sensitive to the chosen significance level and are generally limited to the 10% level. Overall, these results suggest weak empirical support for mean-reverting behaviour in real exchange rates and, consequently, provide limited evidence in favour of Purchasing Power Parity as a long-run equilibrium condition, particularly at monthly frequencies.

Bid-Ask spread

The difference between the bid (willing to buy) and ask (willing to sell) is called the bid ask spread.

$$\frac{ask - bid}{\frac{ask+bid}{2}}$$

is what banks make on FX transactions. In the interbank market, the spread is on average between 0.05% and 0.07% (lower risk). While in retail market is about 3% (inventory costs). the bid ask spread has three main determinants:

- **Cost of dealer services** (know how)
- **Adverse selection costs**, they cannot identify liquidity-motivated trades and speculation.
- **Inventory costs**.

Empirically, we observe that there is a positive relation between exchange rate volatility and bid-ask spread, and a negative relation between volume and bid-ask spread. We also have the weekend effect.

1.1.4 Expectation Formation

As shown by [**?**], the authors provide a comprehensive overview of expectation formation in foreign exchange markets and of the empirical evidence based on survey data. A central finding is that macroeconomic fundamentals exhibit a weak and inconsistent explanatory power for floating exchange rate movements. In particular, fluctuations in exchange rates are typically much larger than those implied by their underlying macroeconomic fundamentals.

This discrepancy motivates alternative explanations based on expectation heterogeneity. The literature highlights several mechanisms through which heterogeneous expectations may arise, including differences in information flows, asymmetries between public and private information, and heterogeneous processing or interpretation of the same publicly available information. Consequently, models that allow for heterogeneity in agents' expectations appear more suitable for capturing exchange rate dynamics than representative-agent frameworks. The main conclusions of the survey can be summarized as follows:

- **Macroeconomic fundamentals have limited explanatory power for short-run exchange rate movements**. Exchange rates exhibit excess volatility relative to fundamentals.
- **Exchange rate expectations are heterogeneous and persistent**. Survey data reveal systematic and long-lasting differences in agents' beliefs across currencies and forecast horizons.

- **Expectation heterogeneity mainly reflects differences in interpretation rather than private information**. Agents observe largely the same public information but process and interpret it differently.
- **Central bank interventions increase expectation dispersion**. Interventions tend to raise disagreement among market participants, regardless of whether they are anticipated.
- **There is limited evidence of herding behavior**. While some coordination of beliefs exists, markets do not converge to a single representative expectation, and forecast leaders are not systematically more accurate.

Chapter 2

Foreign Market efficiency

2.1 Foreign Exchange and Market Efficiency

The first to give a representative definition of Market efficiency has been **Harry Roberts** in 1967 even though many tend to formally attribute it to Eugene Fama. In FAMA (1970) we can derive different forms of *market efficiency*:

- **Weak Form**: the current price incorporates all available information contained in past prices $\mathcal{F}^{\surd}_{\sqcup -\infty} = \{\sqrt{}^{\sqcup -\infty}, p_{t-2}, ...\}$ or $E[p_t|\mathcal{F}^p_t] = p_{t-1}$

- **Semi-strong**: current prices also include public information to the past prices.

- **Strong Form**: add to the public information also private information.

Since all information contained in past prices is fully incorporated into the current price, the effect of any past innovation cannot decay over time. If shocks were transitory, prices would exhibit mean-reverting dynamics, implying predictable price changes based on past information and thus violating weak-form market efficiency. Therefore, innovations to prices must have permanent effects, leading naturally to a random walk representation:

$$x_t = x_{t-1} + \varepsilon_t,$$

where ε_t is an unpredictable innovation with $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$. So basically $E[x_t|\mathcal{F}_{\sqcup -\infty}] = x_t$, therefore taking the difference between the actual price and the forecast, we have that:

$$x_t - E[x_t|\mathcal{F}_{t-1}] = x_{t-1} + \varepsilon_t - x_{t-1} = \varepsilon_t$$

Now we can conclude that if the asset price follows a random walk, then this market is efficient, so $RW \implies ME$, not vice versa.

We show why market efficiency does not imply that asset prices follow a random walk. We start by considering the following ARX(2) model:

$$x_t = \alpha x_{t-1} + \beta x_{t-2} + \gamma Z_t + \lambda Z_{t-1} + \varepsilon \tag{2.1}$$

Assuming that $|\alpha + \beta| < 1$ and rational expectations[a],

$$Z_t = E_{t-1}[Z_t] + \eta_t, \qquad E_{t-1}[\eta_t] = 0,$$

we can compute the expected value of the asset at time t:

$$E_{t-1}[x_t] = \alpha x_{t-1} + \beta x_{t-2} + \gamma E_{t-1}[Z_t] + \lambda Z_{t-1}. \tag{2.2}$$

Considering $\mathcal{F}_{t-1}$ and subtracting the expected price from x_t, we obtain:

$$x_t - E_{t-1}[x_t] = \gamma\big[Z_t - E_{t-1}[Z_t]\big] + \lambda\big[Z_{t-1} - Z_{t-1}\big] + \varepsilon_t - E_{t-1}[\varepsilon_t] \tag{2.3}$$

$$= \gamma\eta_t + \varepsilon_t. \tag{2.4}$$

Taking conditional expectations,

$$E_{t-1}\big[x_t - E_{t-1}[x_t]\big] = \gamma E_{t-1}[\eta_t] + E_{t-1}[\varepsilon_t] = 0.$$

[a] That is, forecasts are made with an error whose expected value is zero.

A large body of empirical evidence suggests that **nominal exchange rates** behave approximately as a random walk. As a consequence, classical unit root tests—such as the Augmented Dickey–Fuller (ADF), Phillips–Perron (PP), and KPSS tests—have traditionally been employed to assess whether exchange rates are characterized by a unit root process.

However, more recent contributions shift the focus away from testing the random walk hypothesis per se and instead rely on parity conditions, notably **Covered Interest Parity (CIP)** and **Uncovered Interest Parity (UIP)**. The rationale is that even if nominal exchange rates display near–random walk behavior, economic theory imposes equilibrium restrictions linking exchange rates to interest rate differentials.

In particular, parity conditions provide a structural benchmark grounded in no–arbitrage and expectation formation. While unit root tests are informative about the statistical properties of exchange rates, CIP and UIP allow researchers to assess whether exchange rate dynamics are consistent with financial market equilibrium, risk premia, and expectations, thereby offering a more economically meaningful framework than purely time-series-based tests.

2.1.1 Covered Interest Rate Parity

Covered Interest Rate Parity is an arbitrage relationship between interest rate differentials and forward and expected exchange rates, assuming *no transaction costs* or *barriers to arbitrage* among countries.

$$(1 + i_t) = (1 + i_t^*)\frac{F_t^{t+1}}{S_t}$$

The above equation states that, under no–arbitrage, the return on a domestic investment must be equal to the return on a foreign investment once exchange rate risk is eliminated. In particular, an investor converts domestic currency into foreign currency at the spot rate, invests in a foreign asset, and converts the proceeds back into domestic currency using the forward exchange rate. Considering the logarithmic transformation, we end up with:

$$i_t = i_t^* + f_t^{t+1} - s_t \tag{2.5}$$

From the literature, we know that, on average, CIP holds.

Testing CIP : We can empirically test CIP by providing the following regression:

$$f_t^{t+k} - s_t = \alpha + \beta(i_t^{t+k} - i_t^{*t+k}) + \varepsilon_t$$

where we construct a test as $H_0 : \beta = 1 \quad H_1 : \beta \neq 1$ (if there are transaction costs then $\alpha \neq 0$. Another way of doing that is to test whether the mean deviation of $f_t^{t+k} - s_t - i_t^{t+k} + i_t^{*t+k}$ is equal to 0.

Empirically we have that:

Table 2.1: Estimated β from CIP regressions

Maturity	Constant	(t–ratio)	β	(t–ratio)	p–value ($\beta = 1$)
1–month	0.009	(7.81)	1.02	(171.2)	0.001
3–month	0.026	(7.98)	0.983	(215.9)	0.000
6–month	0.035	(2.68)	0.953	(128.1)	0.000

Therefore, we have that the standard errors are very small and precise since we reject the null hypothesis even with coefficients very close to the one tested. These deviations are economically negligible and largely attributable to transaction costs, implying that arbitrage opportunities are not exploitable in practice. To assess whether CIP holds it is important to introduce and discuss the **bid-ask spread**, therefore we consider R_t^a and R_t^b respectively as ask and bid for interest rate in British pound while R_t^{*a} and R_t^{*b} denote the ask and bid interest rate in foreign currency:

$$\pi_{CIP} = S^b(1 + R_t^{*b})\frac{1}{F_t^a} - (1 + R_t^a) \leq 0$$

we refer as to "**borrowing pounds covered strategy**". So borrow one pound and exchange it into a foreign currency to buy a foreign bond at the spot rate and convert the payoff at a forward rate. So to avoid arbitrage, we imply that π_{CIP} must be negative.

$$\pi^*_{CIP} = \frac{1}{S_t^a}(1 + R_t^b)F_t^b - (1 + R_t^{*a}) \leq 0.$$

Equation (30) characterizes the payoff of the *borrowing foreign currency covered strategy*. The strategy consists of borrowing one unit of foreign currency at the foreign interest rate R_t^{*a}, converting it into domestic currency at the spot exchange rate S_t^a, investing the proceeds at the domestic interest rate R_t^b, and finally converting the payoff back into foreign currency at the forward rate F_t^b.

The resulting payoff is

$$\pi^*_{CIP} = \frac{1}{S_t^a}(1 + R_t^b)F_t^b - (1 + R_t^{*a}).$$

The first term represents the foreign-currency value of the domestic investment after hedging exchange rate risk through the forward contract, while the second term is the repayment of the foreign-currency loan.

In the absence of arbitrage opportunities, this payoff must be non-positive, implying

$$\pi^*_{CIP} \leq 0.$$

The strategy consists of borrowing one unit of foreign currency at the foreign interest rate R_t^{*a}, converting it into domestic currency at the spot exchange rate S_t^a, investing the proceeds

at the domestic interest rate R_t^b, and finally converting the payoff back into foreign currency at the forward rate F_t^b.

This condition ensures that borrowing in foreign currency and covering the exchange rate risk cannot generate a risk-free profit once bid–ask spreads in interest rates and exchange rates are taken into account.

Table 2.2: Covered Interest Arbitrage at the 1-Month Horizon

Currency	Median return to borrowing covered in		Fraction of periods with positive returns		Median of positive returns to borrowing	
	Pounds percent	FX percent	Pounds percent	FX percent	Pounds percent	FX percent
Full Sample						
Belgium	-0.21	-0.22	1.92	2.19	0.12	0.14
Canada	-0.11	-0.08	0.37	1.38	0.06	0.02
France	-0.14	-0.12	1.00	1.00	0.26	0.07
Germany	-0.23	-0.22	0.15	0.04	0.09	0.37
Italy	-0.16	-0.13	0.81	0.66	0.10	0.04
Japan	-0.26	-0.27	0.43	0.11	0.09	0.31
Netherlands	-0.30	-0.29	0.06	0.15	0.11	0.10
Switzerland	-0.32	-0.32	0.30	0.18	0.20	0.46
USA	-0.07	-0.07	0.72	0.67	0.01	0.11
Average	-0.20	-0.19	0.64	0.71	0.11	0.18
1994:1–2005:1						
Belgium	-0.18	-0.19	2.07	2.76	0.05	0.05
Canada	-0.11	-0.09	0.48	1.00	0.12	0.01
France	-0.10	-0.10	0.92	0.61	0.22	0.05
Germany	-0.11	-0.11	0.31	0.08	0.14	0.28
Italy	-0.16	-0.13	0.31	0.23	0.07	0.21
Japan	-0.10	-0.12	0.83	0.24	0.19	0.31
Netherlands	-0.11	-0.11	0.23	0.08	0.11	0.20
Switzerland	-0.12	-0.12	0.42	0.31	0.17	0.17
USA	-0.05	-0.05	1.25	0.62	0.01	0.13
Average	-0.12	-0.12	0.76	0.66	0.12	0.16

We can see from the table above that we do have negative results for the median return to borrowing covered (otherwise, there would be an arbitrage opportunity). Also, the fraction of periods with positive returns is relatively small. This empirically confirms on average CIP hold.

2.1.2 Uncovered Interest Rate Parity

So far, we have introduced CIP using predetermined variables at time t; now, instead, we introduce the risk-neutrality assumption, which implies that domestic and foreign bonds are perfect substitutes (a very strong assumption). So $F_t^{t+1} - S_{t+1} = 0$, they arbitrarily choose between the current forward rate or exposure to the variation of the exchange rate by S_{t+1}, so we have **forward market efficiency condition** $F_t^{t+1} = E[S_{t+1}]$. By inserting this into the CIP we would end up with:

$$1 + i_t = (1 + i_t^*)\frac{E_t[S_{t+1}]}{S_t}$$

so we can rewrite terms:

$$i_t - i_t^* = E[s_{t+k}] - s_t$$

Using CIP we get:

$$E[s_{t+k}] - s_t = f_t^{t+k} - s_t$$

Empirical testing UIP : to empirically test UIP, we can run the following regression. Let s_t denote the (log) spot exchange rate at time t (domestic price of one unit of foreign currency), and let f_t^{t+k} be the (log) k-period forward exchange rate quoted at time t for delivery at $t+k$. A common way to test *Uncovered Interest Parity (UIP)* is to run the regression

$$\mathbb{E}_t[s_{t+k}] - s_t = \alpha + \beta\big(f_t^{t+k} - s_t\big) + \varepsilon_t, \tag{2.6}$$

and test the joint null hypothesis

$$H_0: \ \alpha = 0 \quad \text{and} \quad \beta = 1.$$

The left-hand side, $\mathbb{E}_t[s_{t+k}] - s_t$, is the *expected depreciation* of the domestic currency over the next k periods. The regressor $f_t^{t+k} - s_t$ is the *forward premium*. Under UIP (in its simplest form), the forward premium should be a one-for-one predictor of the expected depreciation, hence $\alpha = 0$ and $\beta = 1$.

Why this is not "just" a UIP test. To take (2.6) to the data, we must replace the conditional expectation $\mathbb{E}_t[s_{t+k}]$ with something observable. Typically, we assume *rational expectations*, i.e.

$$\mathbb{E}_t[s_{t+k}] = s_{t+k} + \eta_{t+k}, \qquad \text{with} \quad \mathbb{E}_t[\eta_{t+k}] = 0, \tag{2.7}$$

where η_{t+k} is the forecast error.

Moreover, the clean UIP link also requires (at least) *no transaction costs* and *risk neutrality* (or, equivalently, a zero currency risk premium). Therefore, the null $H_0 : \alpha = 0, \beta = 1$ is best interpreted as a *joint hypothesis*:

UIP holds + rational expectations + risk neutrality (no risk premium) (+ negligible frictions).

Econometric meaning of ε_t. The regression error ε_t absorbs deviations from the ideal UIP relation: expectation errors (if (2.7) fails), time-varying risk premia, and market frictions. Hence, rejecting H_0 does not automatically tell you *which* component failed (UIP itself, expectations, or risk neutrality/frictions).

See Appendix A for UIP through Utility maximization

Now from the UIP and CIP we can obtain the following relation so:

$$F_t = E_t(S_{t+1})$$

This basically comes from the risk neutrality property assumed in UIP. We can empirically test the UIP as suggested by [**?**].

We consider the following regression:

$$\frac{S_{t+1} - S_t}{S_t} = \alpha + \beta\left(\frac{F_t - S_t}{S_t}\right) + \varepsilon_{t+1}$$

Here the error terms can be interpreted as the forecast error assuming rational expectations. we test the null hypothesis of $\alpha = 0, \beta = 1$. Hower we do systematic departure from $\beta = 1$ with empirical negative values.

What we have analysed so far brings us to consider a **forward premium puzzle** which

is largely present in high income economies and less evident in industrial economies. So **WE REJECT MARKET EFFICIENCY** when we base our tests on UIP.

We now consider the regression equation used for testing **market efficiency**, hence we have:

$$\Delta_k s_{t+k} = \alpha + \beta(f_t^{(k)} - s_t) + \eta_{t+k}$$

So we are basically testing whether deviations of the forward rate from the current spot rate affect the prediction of spot rate changes. If we assume rational expectations, then $s_{t+k}^e = s_{t+k} + \varepsilon_t$, so the null hypothesis of market efficiency would imply $\beta = 1$. If we suppose that the spot rate is a random walk instead, then $E_t[\Delta_k s_{t+k}] = 0$ so $\beta = 0$. The regression used to test market efficiency is

$$\Delta_k s_{t+k} = \alpha + \beta \left(f_t^{(k)} - s_t \right) + \eta_{t+k}.$$

Under CIP, UIP, and rational expectations we have

$$E_t[s_{t+k}] = f_t^{(k)}.$$

Hence,

$$s_{t+k} = f_t^{(k)} + \eta_{t+k},$$

which implies

$$\Delta_k s_{t+k} = (f_t^{(k)} - s_t) + \eta_{t+k}.$$

Therefore, under market efficiency,

$$\alpha = 0, \qquad \beta = 1.$$

If instead the spot rate follows a random walk,

$$s_{t+1} = s_t + \varepsilon_{t+1},$$

then future exchange rate changes are unpredictable:

$$E_t[\Delta s_{t+1}] = 0,$$

so OLS typically yields

$$\hat{\beta} \approx 0.$$

Thus, market efficiency theoretically implies $\beta = 1$, even though empirically one often finds $\hat{\beta} \approx 0$.

Risk Premium

So far, we have assumed that individuals are risk-neutral; however, this is not always the case, as they may require a higher return when investing abroad. So we have $r_t^{t+1} > 0$, therefore:

$$i_t = i_t^* + E[s_{t+1}] - s_t + r_t^{t+1}$$

If domestic and foreign assets are perfect substitutes, then we would have $r_t^{t+1} = 0$. If we reject the CIP and UIP, we are also rejecting the underlying assumptions of risk neutrality and rational expectations; we should further investigate the question. But what do we know about the risk premium:

- **Does Risk premium exist?**
- **Is it constant over time?**

So as written before

$$i_t = i_t^* + E[s_{t+1}] - s_t + r_t^{t+1}$$

Or we can write, assuming the CIP holds:

$$f_t^k - s_t = \Delta_k s_{t+k}^e + r_t^{t+1}$$

We now consider two equations:

$$\begin{cases} f_t^k - s_t = \gamma + \delta(f_t^k - s_t) + v_{t+k} & \text{previous equation but considering a forward position open} \\ s_{t+k}^e - s_t = \alpha + \beta(f_t^k - s_t) + \eta_{t+k} & \text{UIP standard regression} \end{cases}$$

Now, combining the two, we would end up with:

$$f_t^k - s_{t+k}^e + s_{t+k}^e - s_t = \gamma + \alpha + (\delta + \beta)\left(f_t^k - s_t\right) + \nu_{t+k} + \eta_{t+k}$$

Therefore, we would end up with an identity where $\gamma + \alpha = 0$ and $\beta + \delta = 1$ to let this identity be true. So now we can find that:

$$\beta = \frac{\operatorname{var}\left(\Delta_k s_{t+k}^e\right) + \operatorname{cov}\left(r_t^{t+1}, \Delta_k s_{t+k}^e\right)}{\operatorname{var}\left(r_t^{t+1}\right) + \operatorname{var}\left(\Delta_k s_{t+k}^e\right) + 2\ \operatorname{cov}\left(r_t^{t+1}, \Delta_k s_{t+k}^e\right)}$$

$$\delta = \frac{\operatorname{var}\left(r_t^{t+1}\right) + \operatorname{cov}\left(r_t^{t+1}, \Delta_k s_{t+k}^e\right)}{\operatorname{var}\left(r_t^{t+1}\right) + \operatorname{var}\left(\Delta_k s_{t+k}^e\right) + 2\ \operatorname{cov}\left(r_t^{t+1}, \Delta_k s_{t+k}^e\right)}$$

Empirically, the β is in general negative, therefore we must have a negative numerator, in particular $Coc(r_t, \Delta s^e) < 0$, therefore risk premium and expected depreciation move in opposite directions. Assuming $\beta < 1/2$ would imply that $Var(\rho_t) > Var(\Delta_k s_{t+k}^e)$, meaning that the risk premium should be different from zero and **Time Varying**!! What Fama did was to estimate the regression using both simple regression and SUR[1], which is more robust. We obtain the result of β being negative and statistically different from zero.

Conclusions We conclude that UIP fails empirically, so the forward premium does not only reflect expected depreciation as predicted by the UIP. We have strong evidence of a time-varying risk premium. Since UIP fails, the carry trade becomes possible; borrowing in low-interest currencies and investing in high-interest currencies can generate positive average returns. This does not reflect arbitrage but risk.

[1]By using separate regressions, we assume the errors are independent across the different regressions. Since we are basically computing the same regression, the errors are related, so SUR accounts for that

Exploiting the Failure of UIP

Burnside et al. (2006) provide both a theoretical and empirical analysis of trading strategies that exploit the failure of UIP. Two strategies are considered:

- **Carry Trade:** Borrow in the low-interest-rate currency and lend in the high-interest-rate currency, hedging exchange rate risk via forward contracts.
- **BGT (Backus–Gregory–Telmer) Strategy:** Use a predictive regression on forward premia to signal the direction of currency speculation.

Carry Trade Empirically, low-interest-rate currencies tend to *depreciate* on average rather than appreciate as UIP predicts. Under the assumption of zero bid–ask spread and indirect quotation, let y_t denote the number of pounds borrowed at the start of period t. The carry trade rule is:

$$y_t = \begin{cases} > 0 & \text{if } R_t < R_t^*, \\ < 0 & \text{if } R_t > R_t^*. \end{cases} \tag{2.8}$$

That is, a positive position is taken when the domestic interest rate is *below* the foreign rate. The period-$(t+1)$ payoff (in pounds) is:

$$y_t \left[\frac{S_t(1+R_t^*)}{S_{t+1}} - (1+R_t) \right]. \tag{2.9}$$

For each unit borrowed, one exchanges pounds for foreign currency, invests abroad at $(1+R_t^*)$, converts back at the future spot rate S_{t+1}, and repays the domestic borrowing.

An equivalent formulation exploits the forward premium directly: sell pounds forward when $F_t > S_t$, so that

$$x_t = \begin{cases} > 0 & \text{if } F_t > S_t, \\ < 0 & \text{if } F_t < S_t, \end{cases} \tag{2.10}$$

with pound-denominated payoff:

$$x_t \left(\frac{F_t}{S_{t+1}} - 1 \right). \tag{2.11}$$

Under CIP, the two formulations are identical.

Accounting for Bid–Ask Spreads. In practice, one distinguishes between ask prices S_t^a, F_t^a and bid prices S_t^b, F_t^b. The trading signal becomes:

$$x_t = \begin{cases} +1 & \text{if } F_t^b / S_t^a > 1, \\ -1 & \text{if } F_t^a / S_t^b < 1, \\ \ \ 0 & \text{otherwise.} \end{cases} \tag{2.12}$$

A position of $+1$ corresponds to selling one unit of the domestic currency forward when the bid forward rate exceeds the ask spot rate (i.e. the forward premium net of the spread is profitable). The resulting period-$(t+1)$ payoff is:

$$z_{t+1} = \begin{cases} x_t \left(\dfrac{F_t^b}{S_{t+1}^a} - 1 \right) & \text{if } x_t > 0, \\ x_t \left(\dfrac{F_t^a}{S_{t+1}^b} - 1 \right) & \text{if } x_t < 0, \\ 0 & \text{if } x_t = 0. \end{cases} \tag{2.13}$$

As seen in Table 1, transaction costs are systematically larger for forward rates than for spot rates, and declined substantially after 1999 across all currencies in the sample.

TABLE 1

Median Bid-Ask Spreads

	Spot	1 month Forward	3 month Forward	Spot	1 month Forward	3 month Forward	Units	
	100 x ln(Ask/Bid)			foreign currency units				
	Full Sample Period							Dates
Belgium	0.159	0.253	0.291	10.00	15.93	20.00	Centimes	76:01-98:12
Canada	0.053	0.096	0.111	0.10	0.20	0.23	Cents	76:01-05:12
France	0.100	0.151	0.176	1.00	1.50	1.88	Centimes	76:01-98:12
Germany	0.213	0.311	0.319	1.00	1.12	1.13	Pfennig	76:01-98:12
Italy	0.063	0.171	0.208	1.00	4.00	5.00	Lire	76:01-98:12
Japan	0.216	0.272	0.280	1.00	1.08	1.13	Yen	78:06-05:12
Netherlands	0.234	0.344	0.359	1.00	1.25	1.25	Cents	76:01-98:12
Switzerland	0.255	0.412	0.456	1.00	1.13	1.13	Centimes	76:01-05:12
USA	0.055	0.074	0.082	0.10	0.12	0.13	Cents	76:01-05:12
Euro*	0.043	0.060	0.070	0.04	0.06	0.07	Cents	99:01-05:12
	1999-2005							
Canada	0.066	0.071	0.076	0.15	0.16	0.17	Cents	
Japan	0.061	0.066	0.070	0.11	0.12	0.13	Yen	
Switzerland	0.087	0.094	0.103	0.21	0.22	0.24	Centimes	
USA	0.023	0.027	0.027	0.04	0.04	0.05	Cents	
Euro*	0.043	0.060	0.070	0.04	0.06	0.07	Cents	

*Euro quotes are Euro/USD, whereas other quotes are originally in FCU/British pound
Note: Results are based on daily data

Figure 2.1: Median bid–ask spreads for spot and forward rates (Burnside et al., 2006, Table 1).

BGT Strategy Backus, Gregory, and Telmer (BGT) propose using the following predictive regression to forecast future carry-trade payoffs:

$$\frac{F_t - S_{t+1}}{S_{t+1}} = a + b\,\frac{F_t - S_t}{S_t} + \varepsilon_{t+1}. \tag{2.14}$$

Empirically, b is not significantly different from 3. Intuitively, whenever $F_t - S_t > 0$ (the pound is at a forward premium), the regression predicts a positive future excess return, so the agent *sells* pounds forward. The estimated coefficient b is closely related to the Fama (1984) coefficient β from the UIP regression

$$\frac{F_t - S_{t+1}}{S_t} = \alpha + \beta\,\frac{F_t - S_t}{S_t} + \varepsilon_{t+1}. \tag{2.15}$$

Under a random-walk approximation for the spot rate, one can show that $a = \alpha$ and $b = 1 - \beta$. Since the literature typically finds $\beta \approx -2$, this implies $b \approx 3$, consistent with Table 3.

Although the **Sharpe ratio**[2] of both strategies is comparable to that of the S&P 500 (approximately 0.14 for 1976–2005), average payoffs are economically small: generating an annual payoff of \$1 million requires betting 28.6 million pounds per month. Bid–ask spreads meaningfully erode profits for speculators.

To exploit diversification gains, Burnside et al. construct both equally weighted and optimally weighted (maximum Sharpe ratio) portfolios across currencies.

Figure 2.3 reveals that carry trade payoffs are spread relatively uniformly across the sample, whereas BGT payoffs are more concentrated in the early part of the sample. Both return distributions exhibit excess kurtosis—a hallmark of financial data—providing clear evidence of fat tails. Normality is strongly rejected by the Jarque–Bera test (Table 7).

Conclusions The Covered Interest Parity condition very closely approximates the actual relationship between spot rates, forward rates, and interest rates for major currencies under normal market conditions.

[2]Excess return per unit of standard deviation.

TABLE 4

Payoffs to the Carry Trade Strategies 76:01-05:12

	No Transactions Costs			With Transactions Costs		
	Mean	Standard Deviation	Sharpe Ratio	Mean	Standard Deviation	Sharpe Ratio
Belgium*	0.0044	0.028	0.157	0.0029	0.021	0.140
	(0.0019)	(0.002)	(0.068)	(0.0015)	(0.002)	(0.072)
Canada	0.0053	0.032	0.169	0.0042	0.026	0.161
	(0.0018)	(0.002)	(0.059)	(0.0014)	(0.002)	(0.055)
France*	0.0054	0.027	0.201	0.0031	0.023	0.134
	(0.0016)	(0.002)	(0.060)	(0.0015)	(0.002)	(0.066)
Germany*	0.0011	0.028	0.038	0.0014	0.024	0.060
	(0.0018)	(0.002)	(0.066)	(0.0016)	(0.002)	(0.068)
Italy*	0.0029	0.028	0.105	0.0024	0.024	0.102
	(0.0017)	(0.002)	(0.058)	(0.0014)	(0.002)	(0.056)
Japan†	0.0022	0.036	0.061	0.0017	0.034	0.049
	(0.0022)	(0.003)	(0.063)	(0.0020)	(0.003)	(0.060)
Netherlands*	0.0024	0.028	0.087	0.0014	0.023	0.062
	(0.0018)	(0.002)	(0.068)	(0.0015)	(0.002)	(0.067)
Switzerland	0.0019	0.030	0.063	0.0008	0.027	0.028
	(0.0017)	(0.002)	(0.060)	(0.0015)	(0.002)	(0.057)
USA	0.0039	0.031	0.124	0.0030	0.029	0.103
	(0.0017)	(0.002)	(0.058)	(0.0016)	(0.002)	(0.059)
Euro‡	0.0014	0.021	0.066	0.0024	0.016	0.153
	(0.0017)	(0.002)	(0.083)	(0.0013)	(0.002)	(0.090)
Average	0.0031	0.029	0.107	0.0023	0.025	0.099
Equally-weighted portfolio	0.0031	0.017	0.183	0.0029	0.020	0.145
	(0.0009)	(0.001)	(0.061)	(0.0011)	(0.001)	(0.057)

* Euro legacy currencies available 76:1-98:12
† Japanese yen available 78:7-05:12
‡ Euro available 99:1-05:12
Notes: Other currencies and the equally-weighted portfolio are available for 76:1-05:12. Standard errors in parentheses.

Figure 2.2: Payoffs to the BGT strategy (Burnside et al., 2006, Table 5).

The empirical implications of Uncovered Interest Parity and rational expectations are consistently at odds with the data: high-interest-rate currencies tend, on average, to *appreciate* rather than depreciate, constituting the *forward premium puzzle*.

Carry trade and BGT strategies exploit this anomaly. Their average payoffs are low, but their Sharpe ratios are on par with the S&P 500. Transaction costs reduce—but do not eliminate—profitability, and no standard risk factor explains the returns, suggesting the gains are not simply compensation for bearing risk.

Transaction Costs Matters a lot!!!

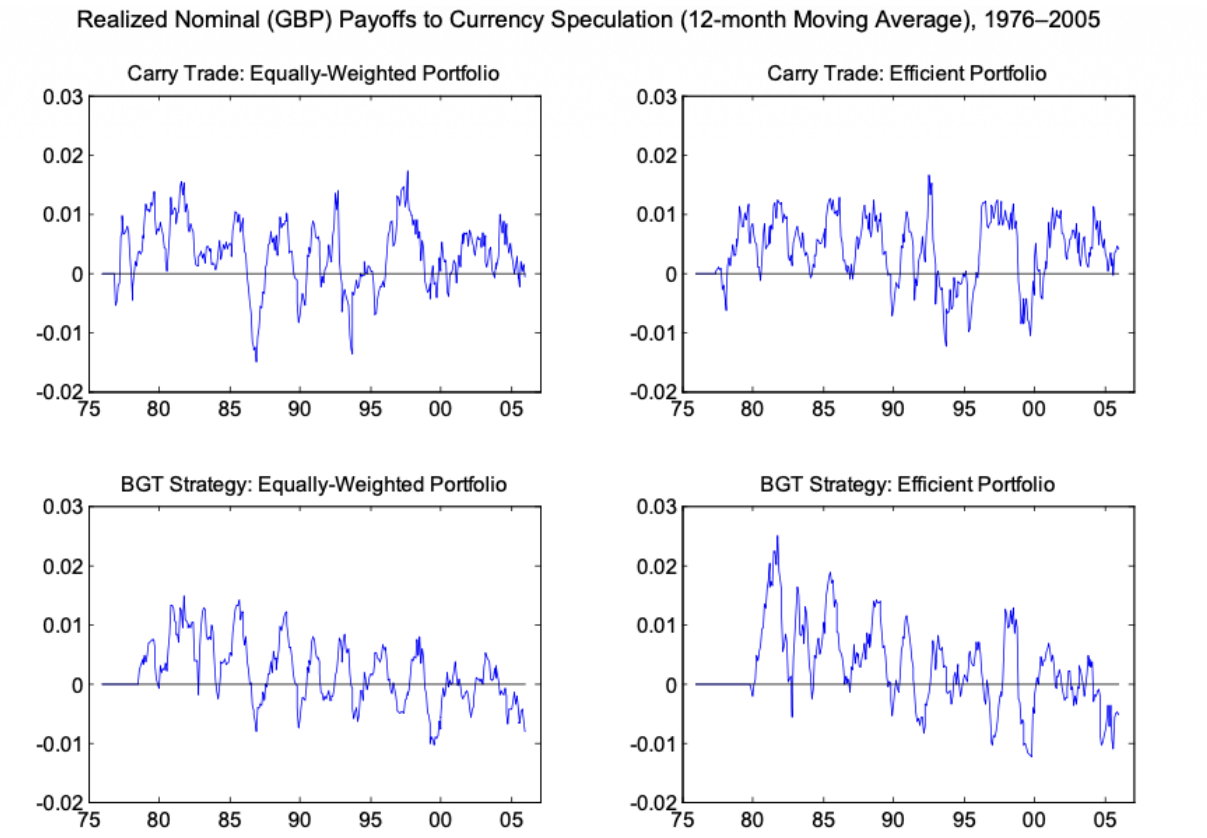


Figure 2.3: Realized payoffs and Sharpe ratios for carry trade and BGT portfolios (Burnside et al., 2006, Figures 1 and 4).

2.1.3 Real Interest Rate Parity

We have previosuly defined the real exchange parity as follows:

$$\ln q_t = s_t + p_t^* - p_t$$

now using that definition one period ahead we get:

$$E[s_{t+1}] - s_t + E[p_{t+1}^*] - p_t^* - (E[p_{t+1} - p_t]) = 0$$

using the definition of Fisher we end up with:

$$i_t - i_t^* = \pi_t^e - \pi_t^{e*}$$

Therefore, we define this as the real interest rate parity. Such relations have been studied in the literature and present mixed results, depending on the currency of the sample period and on the time horizon. Now, if we believe that both inflation expectations and the nominal interest rate are stationary, then we should also expect real interest rates to be stationary, supporting the idea of real interest rate parity. Even if i_t and π_t are not stationary, they could be cointegrated. So far, models rely on rational expectations and risk neutrality; therefore, a model accounting for risk premiums is suggested.

2.1.4 Forward Premium Puzzle

The forward premium puzzle refers to the empirical failure of forward exchange rates as predictors of future spot rates. The logic chain of tests proceeds as follows.

Step 1: Is the Forward Rate Unbiased?

The starting point is the regression:

$$E[s_{t+k}] - s_t = \alpha_1 + \beta_1(f_{t,t+k} - s_t) + \varepsilon_{t+k}$$

The unbiasedness hypothesis is $H_0 : \alpha_1 = 0,\ \beta_1 = 1$, which can be rewritten as:

$$E[s_{t+k}] = \alpha_1 + \beta_1 f_{t,t+k} + (1 - \beta_1)s_t + \varepsilon_{t+k}$$

so the expected future spot rate is a weighted average of the forward rate and the current spot rate. Empirically, $\beta_1 = 1$ is almost always rejected. Since this test jointly assumes rational expectations and risk neutrality, rejection tells us one or both assumptions fail, but not which.

Step 2: Isolating the Risk Premium

To separate the two explanations, we replace unobservable rational expectations with survey-based expectations s^e_{t+k} and run:

$$s^e_{t+k} - s_t = \alpha_2 + \beta_2(f_{t,t+k} - s_t) + \varepsilon_{t+k}$$

If no risk premium exists: $\beta_2 = 1$. Empirically, $\beta_2 \in [0, 1)$, so the null is rejected. Since we no longer impose rational expectations, this rejection is attributable to a genuine risk premium.

Step 3: Are Expectations Biased?

We can also test directly whether survey expectations are biased:

$$s_{t+k} - s_t = \alpha_3 + \beta_3(s^e_{t+k} - s_t) + \varepsilon_{t+k}$$

Under unbiasedness: $H_0 : \alpha_3 = 0,\ \beta_3 = 1$. The literature robustly rejects this across currencies, horizons, and samples. However, biased expectations do not necessarily imply irrationality — agents may form expectations rationally while realized outcomes differ due to low-probability events (the peso problem) or regime changes.

Step 4: Is the Forecast Error Orthogonal to Available Information?

A second condition for rational expectations is that forecast errors must be orthogonal to all variables in the information set at time t. Two standard tests are:

$$s_{t+k} - s^e_{t,t+k} = \alpha_4 + \beta_4(s_t - s^e_{t-k,t}) + \varepsilon_{t+k}$$

$$s_{t+k} - s^e_{t,t+k} = \alpha_5 + \beta_5(f_{t,t+k} - s_t) + \varepsilon_{t+k}$$

In both cases, rational expectations require $\alpha = 0$ and $\beta = 0$. Empirically, rejection is rare at very short horizons (one to two weeks), becomes more frequent at one month, and is strongest beyond three months. Estimates of β_5 are typically negative, implying agents could have reduced forecast errors by betting against the forward rate.

Step 5: How Are Expectations Actually Formed?

The findings motivate an extrapolative expectations model:

$$s^e_{t,t+k} - s_t = \alpha_6 + \beta_6(s_t - s_{t-l}) + \varepsilon_t$$

where $\beta_6 > 0$ indicates bandwagon behavior (expected depreciation continues) and $\beta_6 < 0$ indicates stabilizing expectations (depreciation expected to reverse). Empirically, bandwagon effects dominate at short horizons while stabilizing behavior emerges at longer horizons.

Overall Conclusion

The evidence points to both a genuine time-varying risk premium and systematic expectational biases that depend on the forecast horizon. No single model fully explains all findings, motivating the use of structural models such as ARCH/GARCH specifications for the risk premium.

Chapter 3

Law of One Price and Real Exchange Rates

3.1 Purchasing Power

Gustav Cassel (1916)

"The rate of exchange is primarily an expression for the value in the money of one country put upon the money of another country."

"According to the quantitative theory of money the general level of prices varies, other things being equal, in direct proportion to the quantity of the circulating medium in a country."

Gustav Cassel (1918)

"At every moment the real parity between two countries is represented by this quotient between the purchasing power of the money in the one country and the other. I propose to call this parity the 'purchasing power parity'."

3.1.1 Law of One Price

The **Law of One Price** in its absolute terms can be determined by the following:

$$P_{i,t} = S_t P^*_{i,t}, \quad \text{Absolute}$$

In the absence of transportation costs and trade barriers, arbitrage in the goods market ensures that the price of a homogeneous good is the same in any location measured in the same currency. We can state the relative version as follow:

$$\frac{P^*_{i,t} S_{t+1}}{P_{i,t+1}} = \frac{P^*_{i,t} S_t}{P_{i,t}}, \quad \text{Relative}$$

So the LOP should hold at different periods in time. We should say that **Absolute** $\implies$ **Relative** not viceversa. Empirically, we have evidence suggesting rejection of LOP because of:

- transportation costs
- tariffs and trade barriers
- price discrimination
- national borders

Big mac Index: The big mac index is an indicator of LOP which is used to assess whether a currency is overvalued or undervalued. This indicator relies on some assumptions:

- Big Mac Homogeneous and non tradable (assuming that all other factors are the same).

Here we observe the price in local currency in the first column, the second is the price converted in dollars using actual exchange rate. the third column is the exchange rate implied by the LOP and finally we register the over-undervaluation as $S^{PPP} - S^{actual}/S^{actual}$. We should consider that differently from what we have assumed the non tradable component of the BIG Mac is only 50% of the total.

Table 1: *The Economist's* Big Mac Index, December 2010.

	Big Mac Price		Implied PPP	Actual $	% over- under-
Country	Local currency	In Dollars	of the Dollar	Exchange Rate	valued
United States	3.73	$3.73	.	.	.
Argentina	14	$3.56	3.75	3.93	-5
Australia	4.35	$3.84	1.17	1.13	+3
Brazil	8.71	$4.91	2.33	1.77	+31
Britain	2.29	$3.48	1.63	1.52	-7
Canada	4.17	$4.00	1.12	1.02	+7
Chile	1,750	$3.34	469	524	-10
China	13.2	$1.95	3.54	6.78	-48
Colombia	8,200	$4.39	2196	1,868	+18
Czech Republic	67.6	$3.43	18.1	19.7	-8
Denmark	28.5	$4.90	7.63	5.81	+31
Egypt	13	$2.28	3.48	5.7	-39
Estonia	32	$2.62	8.57	12.2	-30
Euro Area	3.38	$4.33	1.1	1.28	+16
Hong Kong	14.8	$1.90	3.96	7.77	-49
Hungary	740	$3.33	198	222	-11
Indonesia	22,780	$2.51	6,102	9,063	-33
Israel	14.9	$3.86	3.99	3.86	+3
Japan	320	$3.67	85.7	87.2	-2
Lithuania	7.3	$2.71	1.96	2.69	-27
Malaysia	7.05	$2.19	1.89	3.21	-41
Mexico	32	$2.50	8.57	12.8	-33
New Zealand	5	$3.59	1.34	1.39	-4
Norway	45	$7.20	12.1	6.25	+93
Pakistan	210	$2.46	56.3	85.5	-34
Peru	10	$3.54	2.68	2.83	-5
Philippines	102	$2.19	27.3	46.5	-41
Poland	8.3	$2.60	2.22	3.2	-30
Russia	71	$2.33	19	30.4	-38
Saudi Arabia	10	$2.67	2.68	3.75	-29
Singapore	4.23	$3.08	1.13	1.37	-18
South Africa	18.5	$2.45	4.94	7.54	-34
South Korea	3,400	$2.82	911	1,204	-24
Sri Lanka	210	$1.86	56.3	113	-50
Sweden	48.4	$6.56	13	7.37	+76
Switzerland	6.5	$6.19	1.74	1.05	+66
Taiwan	75	$2.34	20.1	32.1	-37
Thailand	70	$2.17	18.8	32.3	-42
UAE	11	$2.99	2.95	3.67	-20
Ukraine	14.5	$1.84	3.88	7.9	-51
Uruguay	79	$3.74	21.2	21.1	nil.

Figure 3.1

The usual pattern we can register is that we have large negative deviations for poor countries and positive for rich countries. We also remark that the **Non-traded** component explains deviations from LOP.

3.1.2 Purchasing Power Parity

So far we have considered the LOP for only one good however we might be interesting to evaluate the same for a basket of goods therefore aggregating all the good $i \in N$. What we culd do is to have:

$$\sum_{i=1}^{N} \alpha_i P_{i,t} = S_t \sum_{i=1}^{N} \alpha_i P^*_{i,t}$$

or taking the logs:

$$\sum_{i=1}^{N} \gamma_i p_{i,t} = s_t + \sum_{i=1}^{N} \gamma_i p^*_{i,t}$$

so deviations from the PPP is the real exchange rate:

$$q_t = s_t - p_t + p^*_t$$

therefore we have found a way to relate PPP to Real Exchange Rates. In this setting we should be able to distinguish between prices related to tradable goods and prices related to non tradable goods. So we consider the overal domestic price as an arithmetic average:

$$p_t = \alpha p_t^N + (1-\alpha) p_t^T$$

same for foreign country, hence by aggregating we can obtain the following real exchange rate expression:

$$q_t = s_t + p_t^{T*} - p_t^T + \alpha \left[\left(p_t^{N*} - p_t^{T*} \right) - \left(p_t^N - p_t^T \right) \right]$$

Now we should consider the **terms of trade** in particular we refer to it as:

$$s_t + p_t^{T*} - p_t^T = \underbrace{s_t + p_{IM_t} - p_{EX_t}}_{q_t^{ToT}} + \left(p_{EX_t} - p_t^T \right) + \left(p_t^{T*} - p_{IM_t} \right)$$

The choice of weights is crucial when evaluating PPP. It may happen that the LOP holds while PPP does not, simply because we apply different aggregation systems across countries. We can observe the following:

$$\sum_{i=1}^{N} \gamma_i p_{i,t} = s_t + \sum_{i=1}^{N} \gamma_i^* p_{i,t}^* + \underbrace{\sum_{i=1}^{N} \left(\gamma_i - \gamma_i^* \right) p_{i,t}^*}_{\text{difference between weights}} .$$

How to test PPP

To test whether PPP does hold we can use different approaches:

- Classic regression based tests.
- Recent time series tests of a unit root in real exchange rates.
- Cointegration tests, spot exchange rates and relative price cointegrate.

We start considering the following equation:

$$s_t = \alpha + \beta p_t + \beta^* p_t^* + \omega_t$$

and verify the null hypothesis of PPP by $H_0 : \beta = 1, \beta^* = -1$, however here basic OLS assumptions are violated leading to spurious regressions.

Another way to test this would be to assume $\beta = 1$ and $\beta = -1$ such that:

$$q_t = s_t + p_t^* - p_t$$

if it PPP holds then $q_t = 0$ so constant therefore stationary. therefore we test $I(1)$ of q_t. In a difficult way we reject the null hypothesis of unit root, since results might also mix fixed exchange rates and structural breaks. Another possibility would be using multilateral tests so tests of several different currencies at time. No consensus yet on the results.

A final test could be using:

$$z_t = s_t + k\pi_t$$

where z_t is deviations from PPP, and test z_t to be stationary if s_t, π_t (relative price) cointegrate and PPP holds if $k = -1$. Empirical studies often find that exchange rates and price levels are cointegrated, suggesting the existence of a long-run relationship consistent with Purchasing Power Parity (PPP). However, the strict proportionality restriction implied by strong PPP (e.g., $\kappa = -1$) is frequently rejected.

Evidence tends to be more supportive of PPP under fixed exchange rate regimes, while PPP is commonly rejected for floating exchange rates, particularly in the post–Bretton Woods period. Furthermore, tests based on Consumer Price Index (CPI) data reject PPP more often than those using Wholesale Price Indices (WPI) or GDP deflators, likely because CPI measures include a larger share of non-tradable goods.

Why Do We Reject PPP So Often? Empirical rejections of Purchasing Power Parity (PPP) are pervasive in the literature and stem from several distinct sources. First, standard unit root and cointegration tests have **low statistical power** in short samples, making it difficult to distinguish a slowly mean-reverting process from a random walk. Second, **structural breaks**—whether permanent (e.g., productivity differentials as in the Harrod–Balassa–Samuelson effect) or temporary (e.g., policy regime changes)—distort inference if left unaccounted for. Third, **price index construction** introduces measurement error: broad indices such as the CPI include a large share of non-tradable goods, which are not subject to arbitrage and therefore need not conform to PPP.

Real Exchange Rate Volatility and Slow Mean Reversion. Real exchange rates display two seemingly contradictory empirical regularities that together constitute what Rogoff (1996) termed the *PPP puzzle*:

1. **High short-run volatility.** Real exchange rate fluctuations are far more volatile than movements in relative price levels, suggesting that nominal exchange rate dynamics—driven by asset market forces—dominate in the short run, while goods markets adjust sluggishly.

2. **Slow mean reversion.** Despite this volatility, real exchange rates do eventually revert toward their long-run equilibrium. However, the adjustment is extremely gradual: Rogoff (1996) surveys the literature and finds consensus half-life estimates of **three to five years**, meaning that a shock to the real exchange rate takes between three and five years to be half absorbed.

The puzzle lies in the *coexistence* of these two facts. High volatility points to nominal rigidities or asset-market frictions as the dominant short-run force, yet slow mean reversion implies that real factors and goods market frictions govern the long-run. Standard models struggle to reconcile both simultaneously.

Several explanations have been proposed. **Limited goods market integration**—arising from trade costs, tariffs, and non-tariff barriers—prevents the law of one price from equalising goods prices across borders quickly. **Market segmentation** allows firms to price-discriminate internationally, further delaying arbitrage. **Aggregation bias** may also be important: when the real exchange rate is measured using broad price indices (e.g., the CPI), the inclusion of non-tradable goods mechanically induces slow adjustment, since those prices are not disciplined by international arbitrage at all.

Evidence from **micro-level price data** (e.g., Bergman et al., 2018) supports this last channel. When individual goods prices are used rather than aggregate indices, estimated half-lives fall dramatically—to between **two and twelve months**—suggesting that much of the apparent persistence in aggregate real exchange rates is a statistical artefact of aggregation rather than a reflection of true economic frictions.

3.2 Price Convergence in Denmark (Bergman)

Bergman et al. (2018) begin from the observation that PPP is widely regarded as the canonical anchor for the long-run real exchange rate. Yet the empirical record is deeply uncomfortable: real exchange rates are highly persistent—some studies fail to reject a unit root altogether—and even when mean reversion is detected, deviations from PPP dissipate at a glacial pace, with consensus half-life estimates of three to five years (Rogoff, 1996).

This sluggishness is, however, difficult to reconcile with two well-established empirical regularities:

- **Microeconomic evidence on price adjustment.** A large body of recent empirical work documents that individual goods prices change frequently—often within weeks or a few months—across a wide range of sectors and countries. If prices at the micro level adjust rapidly, aggregate real exchange rates should also converge quickly.

- **Short-run real exchange rate volatility.** Real exchange rates exhibit far greater short-run volatility than relative price levels. This implies that nominal exchange rate movements dominate in the short run, which in turn requires prices to be *sticky* enough to prevent immediate real adjustment. Yet if prices are sticky, they should also be slow to correct deviations from PPP — implying *fast* volatility and *slow* convergence simultaneously, a combination that standard models struggle to produce with plausible parameter values.

The resulting tension can be stated sharply: *if individual prices adjust frequently, why do aggregate real exchange rates converge so slowly?* This is the PPP puzzle in its most pointed form.

The central conjecture of Bergman et al. (2018) is that conventional half-life estimates are **biased upward due to aggregation**. Standard aggregate estimates compute a simple (unweighted) average of half-lives across goods and regions, treating all estimates as equally informative. This procedure ignores the fact that individual goods-level estimates differ substantially in their precision—that is, they carry very different standard errors. A simple average conflates noisy, unreliable estimates with precise ones, potentially inflating the aggregate half-life relative to its true value.

To sidestep a further confound—namely, that bilateral cross-country real exchange rates embed nominal exchange rate movements, which are themselves highly volatile and persistent—the authors exploit **intra-Denmark price data**. By studying price convergence *within* a single country sharing a common currency and monetary policy, they isolate the pure goods-market adjustment process, free from any exchange rate noise. This identification strategy allows them to obtain clean estimates of how quickly price gaps across regions close, and to assess whether aggregation bias explains the discrepancy between micro and macro estimates of PPP convergence.

3.2.1 Data

Consider Danish data from January 1997 to May 2010, which are collected twice a month (7th and 15th) for specific items sold at specifis store at a given point in time. The store is tracked for two years, and they record codes both for products and for the store. For some products, prices are not recorded monthly as they are assumed to be constant (housing rent, financial services...). Also, the Danish Statistics Department edits the data for possible error correction (extreme price observations).

In our dataset, we consider 13 regions and select 124 different consumer goods. It was not possible to track specific homogeneous goods or brands, so they have aggregated them in broad categories.

The authors have also proposed a data cleaning process to eliminate noise from short-term sale offers. If the six-month average price of a good both before and after a given date is more than 10 percent higher (lower), then the price is equal to the average (removing short-term sale offers). If the permanent price change exceeds 75 percent at a given date, the price change is set to zero (removing price changes due to quality or quantity changes).

Methodology

So they start estimating AR models and the implied half-life using the impulse response functions:

$$P_{ij,t}^{k} = \alpha_{ij}^{k} + \sum_{l=1}^{n} \rho_{ij,l}^{k} P_{ij,t-l}^{k} + \varepsilon_{ij,t}^{k}$$

where ij (implying 78 unique relative prices) relates to the relative price between region i and j, k is the good and t is the time. As shown from previous studies, the bias is related to the effect of aggregation of the degree of persistence in PPP. So they consider a correlated effect estimator to take cross-sectional dependence into account.

For each relative price series, the half-life is estimated through the following procedure:

1. **AR(n) estimation and lag selection.** Estimate an autoregressive model AR(n) for each relative price series and determine the optimal lag length n using the Schwarz Information Criterion (SIC).

2. **Correction for cross-sectional dependence.** Re-estimate the AR(n) model accounting for cross-sectional dependence by applying the Common Correlated Effects (CCE) estimator proposed by Pesaran (2006).

3. **Half-life computation.** Derive the moving-average representation of the estimated AR(n) model and compute the implied half-life from the impulse response function.

4. **Bootstrap bias correction (Kilian, 1998).** Apply a bootstrap-after-bootstrap procedure with bias correction:

   (a) Estimate the AR(n) model and apply the Shaman and Stine (1988, 1989) first-order bias correction.

   (b) If the bias-corrected model is stationary (i.e., $\sum \hat{\rho}_i < 1$), retain the estimate; otherwise, exclude the non-stationary relative price.

   (c) Generate bootstrap replications using non-parametric bootstrap methods based on the bias-corrected model.

   (d) Recompute the half-life for each bootstrap replication.

   (e) Construct confidence intervals using the empirical percentiles of the bootstrap distribution.

Finally, the estimated half-lives are aggregated across relative prices using a **random-effects meta-analysis framework**, which weights each estimate by the inverse of its variance to account for estimation precision and avoid aggregation bias.

Empirical Results

The authors begin by computing, for each good within each region, the **frequency of price change**—defined as the fraction of monthly observations in which the price of a given good changes—and the **size of price change**, measured as the absolute value of the monthly percentage change. These statistics are then aggregated across regions using a simple unweighted

mean, yielding a summary measure per good category. From the frequency of adjustment one can directly recover the **implied duration**—the expected number of months a price remains unchanged—via the standard relationship $d = 1/f$, where f denotes the monthly frequency of price change.

Table 2. *Frequency and duration of price changes by groups of goods*

Type of goods	Mean freq.	Median freq.	Fraction price incr.	Mean size	Implied duration in months (frequency)	(duration)
Food	69.6	70.1	58.6	1.7	1.1	1.7
Fruit and vegetables	92.1	93.0	54.9	3.7	0.3	1.1
Confectionery and non-alcoholic bev.	47.8	46.2	62.9	1.0	2.6	3.3
Alcoholic beverages	47.7	46.7	65.0	0.6	1.9	2.4
Clothing and footwear	25.6	24.1	53.2	1.0	5.1	5.1
Furnishings	23.6	22.1	70.7	0.4	5.4	5.5
Household equipment and maintenance	22.1	19.6	65.3	0.4	6.7	5.9
Transport	14.9	14.4	76.9	0.3	20.8	15.2
Audio-visual equipment	15.2	15.5	70.2	0.4	12.4	9.1
Recreation and culture	17.5	16.8	60.2	0.5	10.9	9.9
Personal services and care	11.0	10.2	87.1	0.4	11.9	10.8
Jewelry, clocks, and watches	12.6	11.0	84.2	0.4	12.9	10.9
Other good and services	11.5	11.2	84.6	0.4	13.3	9.8
All goods	43.0	42.5	63.2	1.3	5.8	5.3

Notes: All frequencies are reported in percent per month across all goods within each type of goods. The size of price changes refers to the absolute value of the percentage change in the price on a monthly basis. The last two columns show the implied duration using the frequency approach and the duration approach. The duration is measured in months.

Figure 3.2: Frequency and size of price changes by good category. Frequencies are expressed in percent per month, aggregated across goods within each category. Size refers to the absolute value of the monthly percentage price change.

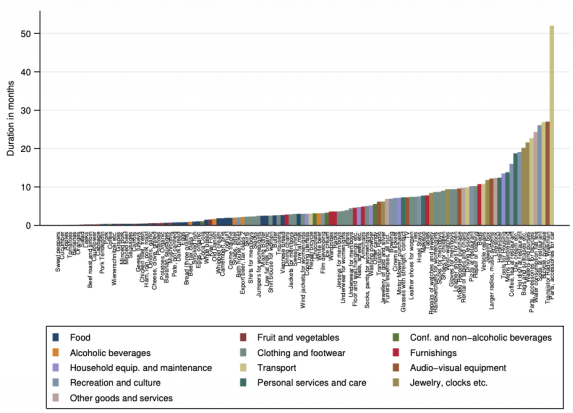


Figure 3.3: Estimated implied duration by good category (months).

The estimated durations reveal substantial heterogeneity across categories. Perishable goods such as food and vegetables display the shortest durations, with prices adjusting frequently within one to two months, reflecting their exposure to supply shocks, seasonality, and competitive retail markets. At the other extreme, durable goods such as jewellery exhibit considerably longer

durations, consistent with infrequent repricing in markets characterised by search frictions and product differentiation. This cross-sectional dispersion in adjustment speeds is precisely what makes the choice of aggregation method consequential.

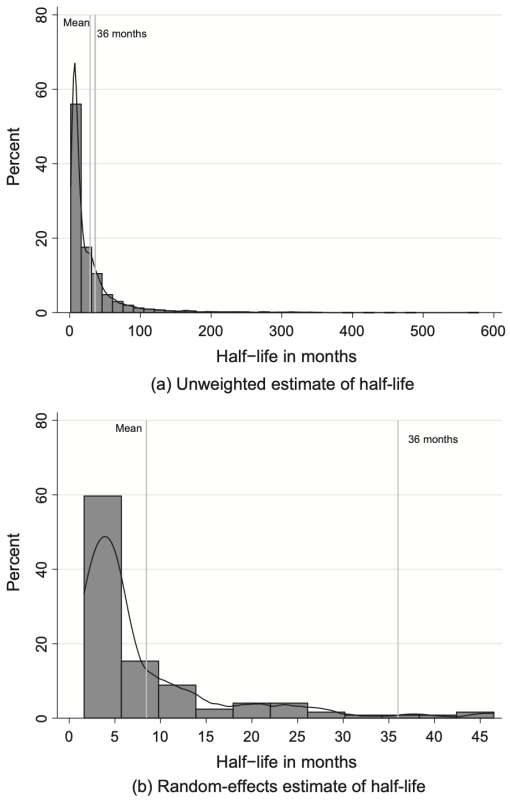


Figure 3.4: Distribution of estimated half-lives across good categories, weighted versus unweighted aggregation.

Aggregating estimated half-lives across all good categories using the standard **unweighted** method yields an average half-life of approximately **26 months**, broadly consistent with the lower end of Rogoff's (1996) consensus range. However, when aggregation is performed using a **precision-weighted** approach—assigning greater influence to estimates with lower standard errors—the average half-life collapses to approximately **8 months**. The threefold discrepancy between the two estimates is entirely attributable to aggregation bias: under a naive unweighted average, imprecisely estimated half-lives for durable goods receive the same weight as precisely estimated half-lives for frequently-repriced perishables, mechanically inflating the aggregate. Conclusions drawn using the standard approach in the literature may therefore substantially overestimate the true persistence of relative prices.

Heterogeneity Analysis

Authors also want to explain why half-lives differ across goods. The first test they run is to see whether distance plays a role in explaining half-life, therefore they run the following regression:

$$\hat{h}_{ij} = \beta \ln(dist_{ij}) + \rho\gamma_{ij} + u_{ij} + \varepsilon_{ij}$$

γ is the mean of price stickiness, measured as the frequency of price changes.

Distance effect. Across all specifications, geographical distance has a positive and statistically significant effect on persistence. In the baseline regression (Column 1), a one-percent increase in

Table 4. *Persistence, distance, and price stickiness regressions*

	(1)	(2)	(3)	(4)	(5)
$\ln(dist)$	0.214***	0.309***	0.297***	0.297***	0.313***
	(0.058)	(0.080)	(0.078)	(0.097)	(0.078)
$\ln(dist)^2$		0.122*	0.111	0.111	0.120*
		(0.070)	(0.069)	(0.069)	(0.068)
Frequency of price changes			0.400***	0.399**	
			(0.151)	(0.155)	
Frequency of price changes × $\ln(dist)$				0.002	
				(0.194)	
Frequency of price increases					−3.385***
					(0.789)
Frequency of price decreases					4.779***
					(0.905)
Constant	3.430***	3.367***	3.056***	3.057***	3.214***
	(0.046)	(0.058)	(0.121)	(0.125)	(0.126)
Marginal effect of distance at means		0.306***	0.295***	0.399***	0.310***
		(0.079)	(0.077)	(0.154)	(0.077)
Observations	8,343	8,343	8,302	8,302	8,302
R^2	0.04	0.05	0.14	0.13	0.16

Notes: Knapp–Hartung standard errors are shown within parenthesis below each estimate. The total effect of distance is measured at mean values of all covariates. ***, **, and * denote significance at the 1, 5, and 10 percent levels, respectively.

Figure 3.5

log distance increases the half-life by approximately 0.21 months. When including a quadratic term in log distance (Columns 2–5), the marginal effect of distance at mean values is around 0.30 months and remains significant at the 1% level. The positive coefficient on the interaction term implies that the impact of distance increases with geographical separation.

Price stickiness. Including the frequency of price changes (Column 3) shows that higher stickiness increases persistence: a one-percentage-point increase in the duration of price spells raises the half-life by approximately 0.4 months. This confirms that stickier prices imply slower convergence.

Asymmetric effects. When distinguishing between price increases and price decreases (Column 5), the effects are asymmetric:

- An increase in the frequency of price increases significantly reduces the half-life (coefficient −3.385).
- An increase in the frequency of price decreases significantly increases the half-life (coefficient 4.779).

The difference between these effects is statistically significant at the 1% level. Thus, more frequent price increases accelerate convergence, whereas more frequent price decreases slow it down.

Overall fit. The explanatory variables account for a modest share of variation in persistence (R^2 between 0.04 and 0.16), indicating that distance and price stickiness are important but not exhaustive determinants of half-life heterogeneity.

Main conclusion. Both geographical distance (a proxy for trade costs) and price stickiness significantly increase the persistence of relative prices. Moreover, the effect of stickiness is asymmetric across upward and downward price adjustments.

Adding a Border dummy in the regression does not lead to significant coefficients meaning that borders do not play any role following this setting.

We observe that the dummy variable for traded goods is negative and statistically significant, indicating that traded goods exhibit a lower half-life on average compared to non-traded goods. This result remains robust even when including an interaction term between the traded dummy and geographical distance.

Table 5. *Effect of borders, traded goods and relative income on the estimated half-life*

	(1)	(2)	(3)	(4)	(5)
$\ln(dist)$	0.323**	0.324**	−1.033**	−0.579	−0.602
	(0.129)	(0.130)	(0.451)	(0.574)	(0.575)
$\ln(dist)^2$	0.123*	0.124*	0.133*	1.056	1.019
	(0.073)	(0.073)	(0.072)	(0.730)	(0.731)
Frequency of price increases	−3.392***	−3.360***	−3.233***	−3.216***	−3.226***
	(0.791)	(0.796)	(0.786)	(0.788)	(0.788)
Frequency of price decreases	4.784***	5.002***	4.875***	4.851***	4.868***
	(0.907)	(0.915)	(0.904)	(0.906)	(0.906)
Border	0.015	0.030	0.028	0.020	0.026
	(0.159)	(0.160)	(0.158)	(0.158)	(0.158)
Traded		−0.887***	−0.890***	−0.445	−0.444
		(0.319)	(0.315)	(0.473)	(0.473)
$\ln(dist) \times$ Traded			1.385***	0.920	0.946*
			(0.441)	(0.573)	(0.573)
$\ln(dist)^2 \times$ Traded				−0.933	−0.904
				(0.734)	(0.734)
Income difference					0.033
					(0.035)
Constant	3.210***	3.998***	3.971***	3.536***	3.261***
	(0.144)	(0.314)	(0.310)	(0.465)	(0.547)
Contrasts of adjusted predictions:		0.887***	1.637***	3.693**	2.840***
non-traded versus traded goods		(0.319)	(0.400)	(1.666)	(1.062)
Observations	8,302	8,302	8,302	8,302	8,302
R^2	0.16	0.14	0.19	0.18	0.18

Notes: Knapp–Hartung standard errors are shown within parenthesis below each estimate. Contrasts of adjusted predictions are the differences between marginal effects for traded and non-traded goods computed at the average log distance. ***, **, and * denote significance at the 1, 5, and 10 percent levels, respectively.

Figure 3.6: Enter Caption

The coefficient associated with the interaction term is positive, implying that the effect of distance on persistence is stronger for traded goods. In other words, although traded goods generally converge faster, increasing geographical distance weakens this advantage by raising their estimated half-life.

Price stickiness is always significant in these regressions, suggesting that this variable is also an important factor influencing the rate of price convergence.

Conclusions

The central contribution of Bergman et al. (2018) is methodological: they demonstrate that the standard approach to aggregating half-life estimates across goods and regions produces systematically biased results. Specifically, computing simple unweighted averages of estimated half-lives is valid only under the knife-edge assumption that all individual estimates carry identical standard errors — a condition that is almost never satisfied in practice. When estimates differ in precision, as they inevitably do, the unweighted average conflates informative and uninformative estimates, introducing aggregation bias whose direction is theoretically ambiguous but is shown empirically to be large and positive.

Using Danish CPI micro-data covering 124 goods across 13 regions, the authors find that this bias is quantitatively severe. The standard unweighted method yields an average half-life of approximately **28 months**, whereas the precision-weighted random-effects estimator — which assigns influence to each estimate proportional to its statistical reliability — produces an average half-life of approximately **8 months**. Conventional methods therefore overestimate price persistence by a factor of three to four. Once estimation uncertainty is properly accounted for, the speed of relative price convergence is fully consistent with micro-level evidence from other countries and no longer constitutes a puzzle in its own right.

Beyond the aggregation correction, the paper identifies two structural determinants of the cross-sectional heterogeneity in half-lives. First, **geographical distance** increases persistence: regions that are farther apart exhibit slower price convergence, reflecting higher effective trade costs that limit arbitrage. Notably, this effect is asymmetric across good types — while traded goods generally converge faster than non-traded goods, the half-life of traded goods rises steeply with distance, narrowing the gap between the two categories as regions become more remote. Second, **price stickiness** plays a central and independent role: goods whose prices change less

frequently display greater persistence. Furthermore, the direction of price adjustment matters: more frequent price *increases* accelerate convergence, while more frequent price *decreases* slow it down, an asymmetry that points to underlying menu cost or pricing strategy heterogeneity. These results hold whether or not the aggregation bias correction is applied, confirming their robustness.

Taken together, the findings suggest that a substantial part of the PPP persistence puzzle is a statistical artefact stemming from improper aggregation, rather than evidence of deep or exotic economic frictions. Relative prices for individual goods adjust far more rapidly than the consensus literature has concluded, and PPP remains a credible long-run anchor for the real exchange rate once the analysis is conducted with appropriate econometric care.

3.3 Harrod-Balassa-Samuelson Model

The Harrod–Balassa–Samuelson (HBS) effect offers a structural explanation for why PPP may fail systematically, even in the long run, rather than as a transitory deviation. The core mechanism rests on a fundamental asymmetry between tradable and non-tradable sectors. In the tradable sector, the law of one price tends to hold: international competition equalises prices across countries. In the non-tradable sector, however, no such arbitrage force operates.

The key insight is that productivity growth is typically faster in the tradable sector, and faster in richer, more developed economies. Since wages are equalised across sectors within a country through labour mobility, higher tradable-sector productivity raises wages economy-wide. This wage increase is then passed through into the prices of non-tradables, whose productivity has not improved commensurately. The result is that **richer countries have systematically higher price levels**, not because of monetary misalignment, but because of structural productivity differentials. Crucially, this implies that the real exchange rate need not revert to PPP at all — it has a moving equilibrium determined by relative productivity growth. This mechanism therefore helps resolve part of the PPP puzzle: what looks like a slow or absent mean reversion in aggregate real exchange rates may simply reflect the HBS trend, rather than a failure of arbitrage or a measurement problem. Countries that grow faster than their trading partners will persistently exhibit an appreciating real exchange rate, entirely consistently with optimising behaviour and competitive goods markets.

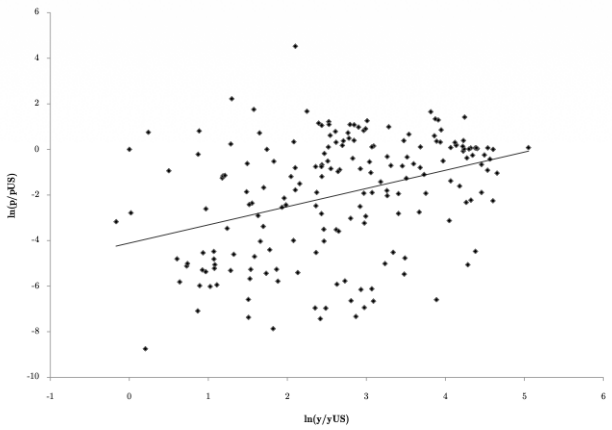


Figure 6: Log price level (vertical axis) against log real income per capita for year 2000, relative to the US. Source: Penn World tables.

Figure 3.7

In the figure above we show how there is positive relations between income differential $\ln(Y/Y^{usd})$ and price differential $\ln(P/P^{usd})$.

3.3.1 Assumptions

The model we present is a $2 \times 2 \times 2$ two sector, two country, two factors, where we introduce a Small Open Economy (SOE) where the RER is fully determined by the supply side and not by the demand side (no role for aggregate demand or Gov't).

1. The economy is a small open economy (SOE) with perfect capital mobility.

2. There are two sectors: a tradable sector (T), whose goods are priced in international markets, and a non-tradable sector (N), whose goods are priced in domestic markets.

3. There are two factors of production: capital (K) and labour (L).

4. The economy operates at full employment, such that

$$L = L_T + L_N,$$

and labour is supplied inelastically.

5. Capital is perfectly mobile across countries and across sectors.

6. Labour is immobile internationally but perfectly mobile across sectors domestically.

7. Output in both sectors is produced under constant returns to scale (CRS) using capital and labour in different proportions. The tradable sector (T) is assumed to be more capital-intensive than the non-tradable sector (N).

3.3.2 Model

We start our model trying to determine the relative price of P^N/P^T, where we assume $P_T = 1$ as numeraire and therefore P^N is also reporting the non traded goods relative price and representing the internal RER. We write below the Production Functions:

$$Y_T = A_T F(K_T, L_T), \tag{3.1}$$
$$Y_N = A_N H(K_N, L_N). \tag{3.2}$$

here we use *Hicks-neutral Total Factor Productivity* where both $A_{t,N}$ changes do not impact the capital labour ratio. We can define the above equations in intensive form so:

$$y_T \equiv \frac{Y_T}{L_T} = \frac{A_T F(K_T, L_T)}{L_T} = A_T F\left(\frac{K_T}{L_T}, 1\right) = A_T f(k_T),$$

$$y_N \equiv \frac{Y_N}{L_N} = \frac{A_N H(K_N, L_N)}{L_N} = A_N H\left(\frac{K_N}{L_N}, 1\right) = A_N h(k_N).$$

Now the firms for both T and N face a maximization problem where the former try to maximize the profit assuming the price $P^T = 1$ wile the second does the same accounting for P^N, hence we get:

$$\max_{k,L} \; A_T L_T f(k_T) - wL_T - \iota k_T L_T.$$

$$\max_{k,L} \; P_N A_N L_N h(k_N) - wL_N - \iota k_N L_N.$$

We show in the following box the derivation of the FOCs for the Traded goods (automatically, in the same way, we will get the one also for non traded).

SOLVING THE MAXIMIZATION PROBLEM
We recall that we are trying to find $\frac{\partial L^T}{\partial K_T}$ and we also know that $y_T = Y_T L_T$

$$\frac{\partial \mathcal{L}^T}{\partial K_T} := A_T L_T f'(k_T) \frac{1}{L_T} - i = 0 \tag{3.3}$$

the same we do w.r.t the L_T, so we get:

$$\frac{\partial \mathcal{L}^T}{\partial L_T} := A_T f(k_T) + A_T L_T f'(k_T) \left(-\frac{K_T}{L_T^2} \right) - w = 0 \tag{3.4}$$

Obtaining therefore:

$$A_T f'(k_T) = i$$

$$A_T[f(k_T) - k_T f'(k_T)] = w$$

After evaluating the FOCs we finally get:

RESULTS OF THE MAXIMIZATION PROBLEM

$$A_T f'(k_T) = \iota,$$

$$A_T \left[f(k_T) - k_T f'(k_T) \right] = w,$$

$$P_N A_N h'(k_N) = \iota,$$

$$P_N A_N \left[h(k_N) - k_N h'(k_N) \right] = w.$$

Note that wages and interest rates are equalized across sectors due to perfect factor mobility. In a small open economy, the interest rate r is determined in the world capital market. Consider the tradable sector with a Cobb–Douglas production function:

$$Y_T = A_T K_T^{\alpha_T} L_T^{1-\alpha_T}, \qquad 0 < \alpha_T < 1. \tag{3.5}$$

In intensive form:

$$y_T = A_T k_T^{\alpha_T}, \qquad k_T = \frac{K_T}{L_T}. \tag{3.6}$$

Under perfect competition, factor prices equal marginal products:

$$r = \frac{\partial Y_T}{\partial K_T} = \alpha_T A_T k_T^{\alpha_T - 1}, \tag{3.7}$$

$$w = \frac{\partial Y_T}{\partial L_T} = (1 - \alpha_T) A_T k_T^{\alpha_T}. \tag{3.8}$$

Solving the first equation for the capital–labour ratio:

$$k_T = \left(\frac{\alpha_T A_T}{r} \right)^{\frac{1}{1-\alpha_T}}. \tag{3.9}$$

Hence,

$$\frac{\partial k_T}{\partial r} < 0, \qquad \frac{\partial k_T}{\partial A_T} > 0. \tag{3.10}$$

Substituting into the wage equation:

$$w = (1 - \alpha_T) A_T \left(\frac{\alpha_T A_T}{r} \right)^{\frac{\alpha_T}{1-\alpha_T}}, \tag{3.11}$$

which implies

$$\frac{\partial w}{\partial r} < 0, \qquad \frac{\partial w}{\partial A_T} > 0. \tag{3.12}$$

Therefore, the world interest rate determines the capital–labour ratio in the tradable sector, which in turn determines the wage. An increase in r reduces k_T and w, whereas an increase in tradable-sector productivity A_T raises both.

Derivation of the log-differentiated CRS conditions

Now we move on and propose some algebraic manipulations in order to linearize our derived functions and express them in terms of growth rates, in order to track intertemporal dynamics. Under perfect competition and constant returns to scale (CRS), factor payments exhaust output in each sector. In intensive form, the tradable and non-tradable sectors satisfy:

$$A_T f(k_T) = r k_T + w, \tag{3.13}$$

$$P^N A_N h(k_N) = r k_N + w. \tag{3.14}$$

We use the notation $\dot{x} \equiv dx/dt$ and $\hat{x} \equiv \dot{x}/x$.

Tradable sector

Start from (3.13) and take logs:

$$\log A_T + \log f(k_T) = \log(r k_T + w). \tag{3.15}$$

Differentiate (3.15) with respect to time:

$$\frac{\dot{A}_T}{A_T} + \frac{f'(k_T)}{f(k_T)}\dot{k}_T = \frac{r\dot{k}_T + \dot{w}}{r k_T + w}. \tag{3.16}$$

Using (3.13) (i.e. $r k_T + w = A_T f(k_T)$), rewrite the RHS:

$$\frac{\dot{A}_T}{A_T} + \frac{f'(k_T)}{f(k_T)}\dot{k}_T = \frac{r\dot{k}_T + \dot{w}}{A_T f(k_T)}. \tag{3.17}$$

Now express $\dot{k}_T = k_T \hat{k}_T$ and $\dot{w} = w\hat{w}$, and note that $\dot{A}_T/A_T = \hat{A}_T$:

$$\hat{A}_T + \frac{f'(k_T)k_T}{f(k_T)}\hat{k}_T = \frac{r k_T}{A_T f(k_T)}\hat{k}_T + \frac{w}{A_T f(k_T)}\hat{w}. \tag{3.18}$$

Finally, use the marginal product condition for capital in the T sector, $r = A_T f'(k_T)$, which implies $r k_T/(A_T f(k_T)) = f'(k_T)k_T/f(k_T)$, so the $\hat{k}_T$ terms cancel in (3.18). Hence:

$$\hat{A}_T = \frac{w}{A_T f(k_T)}\,\hat{w}. \tag{3.19}$$

Define the labour income share in the tradable sector:

$$\beta_T \equiv \frac{w}{r k_T + w} = \frac{w}{A_T f(k_T)}. \tag{3.20}$$

Then (3.19) becomes:

$$\hat{A}_T = \beta_T \hat{w}. \tag{3.21}$$

Non-tradable sector

Proceed analogously from (3.14). Taking logs:

$$\log P^N + \log A_N + \log h(k_N) = \log(r k_N + w). \tag{3.22}$$

Differentiate with respect to time:

$$\hat{P}^N + \hat{A}_N + \frac{h'(k_N)}{h(k_N)}\dot{k}_N = \frac{r\dot{k}_N + \dot{w}}{r k_N + w}. \tag{3.23}$$

Using (3.14) (i.e. $rk_N + w = P^N A_N h(k_N)$) and writing $\dot{k}_N = k_N \hat{k}_N$, $\dot{w} = w\hat{w}$, we get:

$$\hat{P}^N + \hat{A}_N + \frac{h'(k_N)k_N}{h(k_N)}\hat{k}_N = \frac{rk_N}{P^N A_N h(k_N)}\hat{k}_N + \frac{w}{P^N A_N h(k_N)}\hat{w}. \tag{3.24}$$

Using the capital FOC in the N sector, $r = P^N A_N h'(k_N)$, implies $rk_N/(P^N A_N h(k_N)) = h'(k_N)k_N/h(k_N)$, so the $\hat{k}_N$ terms cancel, yielding:

$$\hat{P}^N + \hat{A}_N = \frac{w}{P^N A_N h(k_N)}\,\hat{w}. \tag{3.25}$$

Define the labour income share in the non-tradable sector:

$$\beta_N \equiv \frac{w}{rk_N + w} = \frac{w}{P^N A_N h(k_N)}. \tag{3.26}$$

Therefore,

$$\hat{P}^N + \hat{A}_N = \beta_N \hat{w}. \tag{3.27}$$

Explanation

From the equation:

$$\hat{P}^N = \frac{\beta_N}{\beta_T}\hat{A}_T - \hat{A}_N \tag{35}$$

Equation (35) shows that the internal relative price of non-traded goods, P^N, is determined by the productivity (TFP) differential between the tradable and non-tradable sectors, taking into account the labour income shares in each sector.

To understand the mechanism, suppose that TFP increases in the tradable sector ($A_T \uparrow$), while A_N remains unchanged. Since the world interest rate r is exogenously given in a small open economy with perfect capital mobility, the marginal product of capital must remain equal to r. An increase in A_T therefore raises the marginal product of labour in the tradable sector, increasing the demand for labour and pushing up the equilibrium wage w.

Because labour is perfectly mobile across sectors domestically, the wage must be equalized across T and N. Hence, the wage increase originating in the tradable sector also applies to the non-tradable sector.

In the non-tradable sector, however, productivity has not changed. Since the interest rate r is fixed and common to both sectors, firms in the N sector face higher labour costs without a corresponding increase in productivity. To satisfy the zero-profit condition under perfect competition, the price of non-traded goods P^N must therefore increase.

Thus, higher productivity in the tradable sector leads to an increase in the relative price of non-traded goods. More generally, the larger the productivity differential ($A_T - A_N$), the higher will be P^N. Even accounting for the same productivity growth in both sectors, we generally assume $\beta_N > \beta_T$ since non tradable sector is commonly more labour-intensive. Assuming an overall increase in technology, L will increase, so w will increase; however, since the non-tradable sector is more labour-intensive, this will put more pressure on wages in the N sector, which is absorbed by prices, so P_N/P_T will increase.

3.3.3 HBS Effect and the real exchange rate

Since we have derived how P_N evolves with the differential in productivity growth, we can now explain the reason why rich countries generally display a higher price level than poor countries **Penn-effect**. Since our assumption that *traded goods are priced in the international market* means that the PPP hold for that type of goods, hence assuming equal weight for the goods we would have:

$$q = \underbrace{s + p^{T*} - p^{T}}_{=0} + \alpha[(p^{N*} - p^{T*}) - (\underset{P^N=1 \implies p^N=0}{p^N} - P^T)]$$

we can rearrange Equation (35) to obtain:

$$\hat{q} = \alpha \left[\frac{\beta_N}{\beta_T} (\hat{A}^*_T - \hat{A}_T) - (\hat{A}^*_N - \hat{A}_N) \right] \tag{36}$$

Equation (36) enables us to determine whether a country's currency is overvalued or undervalued.; therefore, rich countries have a high propensity to produce tradable goods, hence letting the relative difference in productivity growth in traded goods to pull $Q \uparrow$, implying an overvalued currency. Generally, rich countries experience high growth in the tradable sector, leading to a rise in wages; this also induces non-tradable wages to increase, since $w = W_N = W_T$. Therefore, since A_N did not grow, the increase in wages in the N sector drains into P_N.

3.3.4 Empirical Evidence

Overall, the empirical literature offers moderate support for the predictions of the Harrod–Balassa–Samuelson (HBS) model. However, establishing whether the underlying mechanism operates through relative productivity differentials remains difficult, with results varying across countries, time periods, and measurement specifications. A key challenge concerns the measurement of productivity and relative prices, since neither is directly observed; researchers must impose *a priori* classifications of tradable (T) and non-tradable (N) sectors.

Asea and Mendoza (1994). An early test of the HBS model cast within a two-country intertemporal general equilibrium framework with explicit tradable and non-tradable sectors. Using the Hodrick–Prescott filter to extract low-frequency real exchange rate (RER) movements for 14 OECD countries, the authors find that labor productivity differentials explain the low-frequency component of relative price movements, providing support for the HBS mechanism.

Canzoneri, Cumby and Diba (1999). Using a panel of 13 OECD countries and applying panel unit-root and panel cointegration techniques, Canzoneri et al. test for a long-run equilibrium relationship between internal relative prices and relative labor productivity in the tradable sector, finding supportive evidence. However, their test of purchasing power parity (PPP) for tradables yields mixed results: no support is found when the US dollar serves as the reference currency, whereas support emerges when the Deutsche Mark (DM) is used instead.

Recent literature on emerging economies. A growing strand of the literature examines whether the HBS model accounts for RER dynamics in emerging markets, with particular emphasis on European transition economies (post-socialist countries). These economies underwent a characteristic two-phase RER trajectory: an initial sharp depreciation followed by a sustained appreciation. The structural transformation of their productive sectors makes productivity-driven RER movements especially relevant in this context.

A formal test of the role of productivity as a determinant of the internal relative price of non-tradables is provided by Lederš and Mihailović. Constructing a database of 21 emerging

and developed countries, they measure relative prices and relative productivity across N and T sectors using data from the Groningen Growth and Development Centre (GGDC). Relative labour productivity is defined as the ratio of the weighted-average labour productivity of the traded sector to that of the non-traded sector, and relative prices are constructed using the Value Added Deflator. The baseline regression is:

$$\ln\left(\frac{P_{it}^N}{P_{it}^T}\right) = \alpha_i + \gamma_t + \beta \ln\left(\frac{R_{it}^T}{R_{it}^N}\right) + \varepsilon_{it}, \tag{3.28}$$

where α_i denotes country fixed effects, γ_t is a time dummy included to absorb cross-sectional correlation, R_{it}^T/R_{it}^N is the relative labour productivity of the traded to the non-traded sector, and ε_{it} is the error term.

Prior to estimation, the authors test for unit roots in the series. Standard ADF tests fail to reject the null of a unit root, confirming non-stationarity. Estimation proceeds via a modified OLS estimator designed to correct for cross-equation error correlation and endogeneity bias. The average estimated slope is $\hat{\beta}^{\text{avg}} > 0$ and approximately equal to 0.25, consistent with the HBS prediction. Individual country coefficients are displayed in Figure 3.8: while some are negative, the common slope is positive, in line with the theory.

Table 3: Panel "Harrod-Balassa-Samuelson" estimates. 1974-2004.

	No common time dummy		Common time dummy	
Country	$\hat{\beta}$	t-ratio	$\hat{\beta}$	t-ratio
Hong Kong	1.04	8.01	0.87	4.75
India	0.21	2.50	0.57	2.65
Japan	0.55	17.71	-1.31	-3.61
Korea	1.10	13.27	0.97	3.7
Malaysia	-0.05	-0.67	0.12	2.01
Philippines	0.59	1.56	-0.12	-0.99
Singapore	0.41	4.74	0.28	0.54
Thailand	0.62	6.27	0.09	0.4
Argentina	0.87	1.87	2.63	2.34
Bolivia	0.38	3.09	0.12	1.36
Chile	0.10	0.77	-0.19	-0.54
Colombia	0.07	0.12	-0.08	-0.72
Costa Rica	0.81	8.48	0.74	4.84
Mexico	0.72	3.87	0.47	1.07
Denmark	0.10	2.91	-0.66	-1.79
Spain	0.91	10.35	0.32	2.31
France	0.73	15.43	-0.15	-1.24
Netherlands	-0.29	-2.80	-0.03	-1.85
Sweden	0.72	31.42	0.42	5.11
UK	1.06	5.59	-0.15	-0.61
US	0.96	16.58	0.26	1.21
Panel	0.55	32.96	0.25	4.57

Notes: Fully Modified OLS panel estimate. Relative price of nontradables on relative tradable productivity. See text for data sources.

Figure 3.8: Country-specific and average slope coefficients from the Lederš–Mihailović regression (3.28).

Nevertheless, cointegration tests fail to reject the null hypothesis of a unit root in the residuals of (3.28), undermining the interpretation of a stable long-run relationship and weakening support for the HBS hypothesis.

A further challenge to the model emerges from the time variation in estimated slope coefficients across studies: rather than the constant Penn effect implied by the standard HBS framework, the data suggest that the productivity–relative price relationship shifts over time. This instability has motivated the introduction of a *time-varying* HBS specification, in which the strength of the Balassa–Samuelson effect is allowed to evolve. The secular increase in the estimated effect is consistent with rising international capital mobility, which tightens the link between traded-sector productivity and factor prices across countries.

Chapter 4

Open Economy and Exchange Rate Dynamics

4.1 Mundell-Fleming Model

The **Mundell-Fleming Model** is an extension of the open-economy closed-economy with a sticky-price Keynesian IS-LM model. This model also enables the accounting for stock variables, thereby bridging flow open-economy macro-models with stock open-economy models and broadening the capacity to explain economic interactions in an open market economy.

This model also allows us to study the effectiveness of both *monetary policy* and *fiscal policy* under different exchange rate regimes, in particular under *credible peg* and *float* regimes. Finally, the model introduces capital mobility as an important component that will be exploited for the examination of the interactions between the concepts of internal and external balance.

4.1.1 Assumptions

Before diving into the model's details, we introduce a set of assumptions that will help us.

- **Small Open Economy** so all foreign variables are exogenous.
- **Fixed prices in short run**
- **Perfect Capital mobility** implies flexible asset prices.
- **Taxes and savings** rise with income
- **No trade restrictions**
- **BoP depends only on income and exchange rate**
- **Demand for money depends on income and interest rate**
- **Monetary policy operates through Open Market Operations**
- **Fiscal Policy Operates via changes in government expenditures on home goods financed by debt accumulation or decumulation**
- **The economy operates under either *fixed exchange rate or pegged* (credible if there is a true commitment by the monetary authorities) or *float exchange rate*.**

4.1.2 Model

Model in Log-Levels

The MFM consists of three equilibrium conditions in three markets, namely: *(i) goods and services*, *(ii) money* and *Assets*. In the exposition of the model, all variables are expressed in log terms except for the interest rate, which is already expressed in percentage terms.

We introduce here an **Open-Economy IS curve** which describes the equilibrium in the goods market. Here we have:

$$\underbrace{y}_{supply} = \underbrace{\gamma y - \sigma i + g + \delta(s + p^* - p)}_{\text{Total domand of goods}}$$

Here γy is the propensity to consume of individuals and $\gamma \in (0, 1)$. This is also referred to as the domestic-absorption channel. Regarding the real exchange rate, we have that if $q \uparrow$ then we have real depreciation, meaning that domestic goods are cheaper with respect to foreign goods (under the Marshall-Lerner condition). Finally we can write:

$$IS : i = \frac{1}{\sigma} g + \frac{\delta}{\sigma}(s + p^* - p) + \frac{\gamma - 1}{\sigma} y \tag{4.1}$$

since$\sigma > 0$ and $\gamma \in (0, 1)$ then $\frac{\gamma-1}{\sigma} < 0$.

We now move on to the second main component of our model.

We now set up the model for the**Money Demand**, also defined as LM. In fact, we have:

$$LM : i = -\frac{1}{\lambda}(m - p) + \frac{\phi}{\lambda} y \tag{4.2}$$

Here ϕ is the income elasticity of money, while λ is the interest semi-elasticity of money demand, which in our case are both positive.

Finally we get to the last equation needed in our system.

We introduce the **Capital Market Equilibrium** or FF curve, by assuming that the UIP hold and that $E_t[s_{t+1}] = s_t$, therefore:

$$FF : i = i^* \tag{4.3}$$

With perfect capital mobility, this curve is perfectly horizontal. The log-level Mundell–Fleming model can be rewritten in terms of *percentage changes* (total differentials around the long-run equilibrium). Totally differentiating the IS and LM equations yields the compact system:

$$dy = \frac{\delta}{1 - \gamma}\, ds - \frac{\sigma}{1 - \gamma}\, di^* + \frac{1}{1 - \gamma}\, dg, \tag{4.4}$$

$$dm = \frac{\phi\delta}{1 - \gamma}\, ds - \left(\lambda + \frac{\phi\sigma}{1 - \gamma}\right) di^* + \frac{\phi}{1 - \gamma}\, dg. \tag{4.5}$$

This system expresses endogenous percentage changes in output (dy) and real money balances (dm) as functions of exogenous policy and external variables.

Policy regimes:

- **Peg ($ds = 0$):** monetary policy (dm) is endogenous, fiscal policy (dg) is the active instrument.
- **Float ($dm = 0$):** the exchange rate (ds) adjusts endogenously, monetary policy is the active instrument.

Thus, the model can be analysed through a linear two-equation system in percentage changes.

4.1.3 Output Stabilisation of Fiscal Policy

Now we can analyse the model by exploiting the fact that we might impose different exchange rate regimes.

Credible Peg

Under credible Peg, the Central Bank commits to maintain the exchange rate fixed, and therefore, to do so, we assume the CB to operate through market operations. Here if the nominal exchange rate is fixed, then $ds = 0$. Here, an expansionary fiscal policy would imply $dg > 0$, (assuming no change in the foreign exchange rate) where dg plays a role in both (4.4) and (4.5), in particular we have:

$$dy = \frac{1}{1-\gamma} dg > 0, \quad dm = \frac{\phi}{1-\gamma} > 0$$

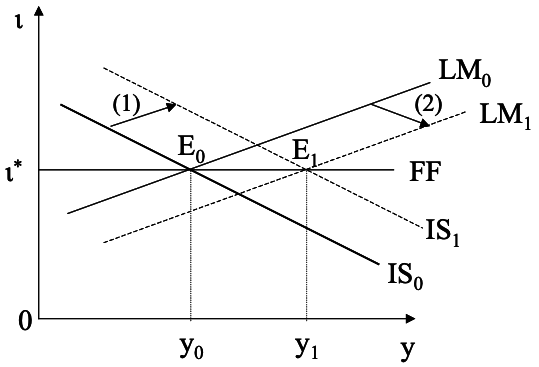


Figure 4.1

Given an increase in government spending, we have that the IS curve shifts upwards, therefore, we have pressure of excess demand that stimulates output $dg > 0 \implies dy > 0$.The increase in income leads to an increase in demand for money $dy \uparrow \implies L(Y,i) \uparrow$ assuming that the money supply is fixed $\frac{\bar{M}}{S} = L(Y,i)$ to rebalance, we need the interest rate to increase. An interest rate above the world rate would lead investors to buy domestic assets, increasing capital flows and putting upward pressure on the exchange rate $E^{h/f}$ *downarrow*. This would prompt the CB to offset the purchase, thereby increasing the supply of domestic currency, $M \uparrow$.

Floating Exchange regime

Assuming a floating exchange rate regime, the central bank does not intervene to return the exchange rate to a fixed level; therefore, we assume $dm = di^* = 0$. We instead assume that $ds \neq 0$. Now we assume $dg > 0$, hence $dy = \frac{1}{1-\gamma} dg$, however, the fiscal policy operates also through an indirect channel, which is the one through the exchange rate, therefore:

$$\frac{dy}{ds}\frac{ds}{dg} = \frac{\delta}{1-\gamma}(-\frac{1}{\delta}) = -\frac{1}{1-\gamma} < 0$$

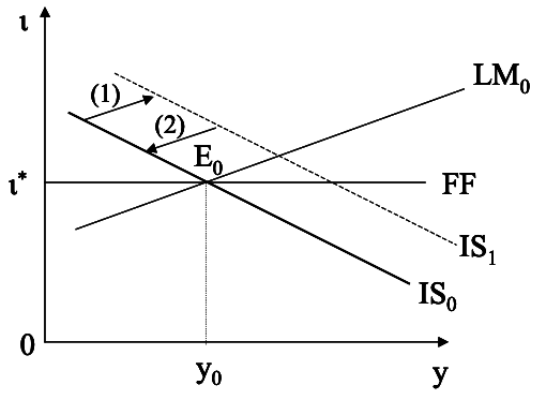


Figure 4.2

The ds/dg derives from the second equation of dm where we impose $dm = di^* = 0$. Combining the two effects, we obtain a total effect on output of 0. So what happens is that since $dg > 0$ then there is an increase in income and upward pressure on the IS curve, here the Nominal exchange rate moves to maintain the equilibrium, this makes the money demand to increase, but since we assume fixed money supply it means that the internal interest rate must rise, therefore leading to an appreciation of the exchange rate and therefore we reduce export, while import rises so aggregate demand falls back.

Summarising the results above, in the original Mundell-Fleming model under perfect capital mobility:

1. Fiscal policy is *effective* as a stabilisation tool under a *fixed* exchange rate regime;
2. but completely *ineffective* under a *flexible* exchange rate regime.

4.1.4 Output stabilisation properties of Monetary Policy

We now turn to monetary policy effectiveness under different exchange rate regimes.

Credible Peg

Under a credible peg, we have that $ds = 0$, we also assume that $di^* = dg = 0$ since we are in a SoE and the Government does not apply any fiscal policy. Thus, by looking at the percentage changes around the long-run equilibrium levels, we have $dy = dm = 0$. We start from the following:

$$\underbrace{DC + NFA^{CB}}_{assets} = \underbrace{MB}_{liability}$$

An expansionary monetary policy consists of an open market purchase of domestic bonds by the central bank. The excess demand for bonds raises their price and lowers the domestic interest rate.

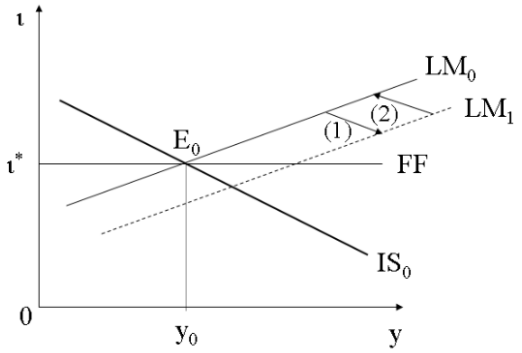


Figure 4.3

Under a flexible exchange rate regime, the monetary expansion ($dm > 0$) shifts the LM curve outward and, from equation (43), implies a depreciation of the exchange rate:

$$ds = \frac{1-\gamma}{\phi\delta}\, dm > 0.$$

The fall in the domestic interest rate induces investors to purchase foreign bonds, generating capital outflows. Under a fixed exchange rate regime, the central bank must defend the currency by selling foreign reserves. As a result, domestic credit increases while foreign reserves decrease by the same amount, leaving the monetary base unchanged.

4.1.5 Floating Exchange Rate

Now, in this case, we have that $ds \neq 0$ since the exchange rate can move. Here, we consider $dm > 0$, so we have a shift upward of the LM curve that will have an effect on depreciating the exchange rate. This depreciation leads to expenditure switching, with exports increasing and thereby expanding domestic output.

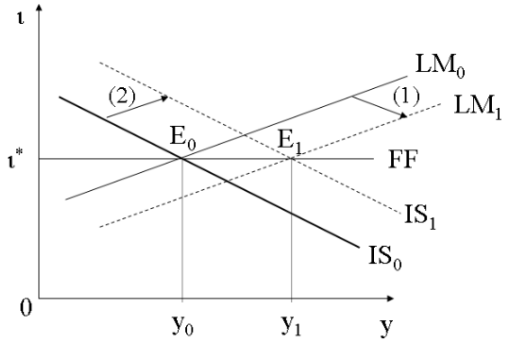


Figure 4.4

So here the effects is given by:

$$\frac{dy}{ds} = \frac{\delta}{1-\gamma} > 0, \quad \frac{dy}{ds}\frac{ds}{dm} = \frac{\delta}{1-\gamma}\frac{1-\gamma}{\phi\gamma} = \frac{1}{\phi} > 0$$

In the original Mundell–Fleming model with perfect capital mobility:

- Monetary policy is ineffective under a fixed exchange rate regime.
- Monetary policy is effective under a flexible exchange rate regime.

4.2 Flexible Price Monetary Model

The monetary approach to the balance of payments (BoP), developed in the 1960s during the Bretton Woods era, views external imbalances as fundamentally monetary phenomena. In this framework, BoP deficits and surpluses arise from disequilibria in the money market: an excess supply of money generates reserve outflows under fixed exchange rates or exchange rate depreciation under flexible rates. **This perspective provides the long-run monetary foundation of the Mundell–Fleming model.**

4.2.1 Assumptions

The main assumptions under this model are the following:

- **UIP and PPP** hold continuously.
- **Flexible Prices**
- **BoP adjustment are a monetary phenomenon**
- **Stable Money demand function**
- **Production and employment full level** so income is fixed and no output-inflation trade-off.

4.2.2 Model

We now assume the demand function for money as follows:

$$\frac{M_t^d}{P_t} = Y_t^{\phi} e^{-\lambda i_t} e^{\varepsilon_t} \tag{4.6}$$

so by taking the logs we obtain:

$$m_t^s - p_t = \phi y_t - \lambda i_t + \varepsilon_t$$

We than consider $\phi \in (0,1)$ (income elasticity of money) and $\theta > 0$ (interest semielasticity of money) and $\varepsilon_t \sim iid(0, \sigma^2)$. We ca now write the two money demand function for the two countries:

$$(a) : m_t - p_t = \phi y_t - \lambda i_t$$
$$(b) : m_t^* - p_t^* = \phi y_t^* - \lambda i_t^*$$

under UIP and risk neutrality, we have:

$$i_t - i_t^* = E_t[s_{t+1}] - s_t$$

and PPP implies that:

$$s_t = p_t - p_t^*$$

Solving for the price levels in (13) and (14) and substituting into (16) yields

$$s_t = m_t - \phi y_t + \lambda i_t - \left(m_t^* - \phi y_t^* + \lambda i_t^*\right).$$

Define economic fundamentals as

$$f_t \equiv (m_t - m_t^*) - \phi(y_t - y_t^*).$$

Here if the relative money supply increase then we would have increase in domestic prices through the money market equilibrium, this would push lower interest rate, with capital outflow and consequential depreciation. Instead higher domestic activity strengthen the currency.

Then the exchange rate can be written compactly as

$$s_t = f_t + \lambda(i_t - i_t^*) = f_t + \lambda\big(E_t[s_{t+1}] - s_t\big),$$

showing that the exchange rate depends on relative monetary fundamentals and expected future depreciation. Here the dynamics are captured by the forward NER, hence we can solve this by proceding with forward substitution and get:

$$s_t = \underbrace{\gamma \sum_{i=0}^{K} (1-\gamma)^i E[f_{t+i}]}_{\text{forward looking solution}} + \underbrace{(1-\gamma)^{k+1} E[s_{t+k+1}]}_{\text{terminal complementary function}}$$

As $k \to \infty$ we need the complementar function to vanish therefore we need $(1-\gamma) < 1$ so to rule out bubbles. So we have the following:

$$s_t = \underbrace{\gamma \sum_{i=0}^{\infty} (1-\gamma)^i E[f_{t+i}]}_{\text{Discounte future fundamentals}}$$

4.2.3 Rational Bubbles

So far, we have seen how the flexible price monetary model delivered a solution made of two main components (i) *Particular solution* and (ii) *Complementary solution*. By assuming the transversality condition holds, we rule out the possibility of bubbles arising and dominating the system's dynamics. We can therefore now introduce the bubble component, which we assume to be generated by:

$$s_t^b = \frac{1}{1-\gamma} s_{t-1}^b + \varepsilon_t$$

where the coefficients are greater than one, implying explosive behaviour. Adding this bubble component to the fundamental solution we obtain:

$$s_t = s_t^f + s_t^b$$

Now, assuming how the bubble is generated, we can get what we have obtained above to the fundamental solution such that:

$$\lim_{k\to\infty} (1-\gamma)^k E_t[s_{t+k}] = \underbrace{\lim_{k\to\infty} (1-\gamma)^k E_t\Big[s_{t+k}^f\Big]}_{=0} + \underbrace{\lim_{k\to\infty} (1-\gamma)^k E_t\Big[s_{t+k}^b\Big]}_{=s_t^b} = s_t^b \neq 0. \tag{4.7}$$

Now, to understand why the behaviour is explosive, we could in principle solve s_t^b forward and obtain:

$$s_t^n = (1-\gamma)^k E_t[s_{t+k}^b]$$

Even if $s_t = s_t^f + s_t^b$ is unstable, it is still a valid solution. To see that we could do as follows:

$$s_t = \gamma f_t + (1-\gamma) E_t[s_t]$$
$$s_t^f + s_t^b = \gamma f_t + (1-\gamma) E_t[s_t^f + s_t^b]$$

knowing that $f_t = \frac{1}{\gamma}(s_t - (1-\gamma)E_t[s_t] =$ and $s_t ? s_t^f + s_t^b$ we can plug it in the above equation and get:

$$s_t^f + s_t^b = \gamma \underbrace{\frac{s_t^f + s_t^b - (1-\gamma)E_t\left[s_{t+1}^f + s_{t+1}^b\right]}{\gamma}}_{=f_t,} + (1-\gamma)E_t\left[s_{t+1}^f\right] + (1-\gamma)E_t\left[s_{t+1}^b\right] \tag{4.8}$$

$$s_t^f + s_t^b = s_t^f + s_t^b \tag{4.9}$$

Despite the seemingly paradoxical terminology, rational bubbles may still arise within a framework where agents behave rationally. In this context, rationality implies that agents are aware of the structure of the model and are able to form expectations accordingly. Although such assumptions may appear strong, the monetary model with rational expectations provides an important benchmark for understanding exchange rate dynamics in a forward-looking setting.

The presence of bubbles is not merely a theoretical possibility. A large empirical literature has attempted to detect bubbles in financial markets, particularly in foreign exchange markets. Early contributions include the variance bounds tests proposed by (Shiller 1981), which compare the volatility of asset prices to the volatility implied by their fundamental values. Empirical applications of this methodology, such as (Huang 1981), have investigated the presence of bubbles in exchange rates, including the Deutsche mark. More broadly, these studies suggest that deviations of prices from fundamentals may occasionally reflect speculative components consistent with rational bubble dynamics.

Therefore, although the assumption of fully rational agents may not perfectly reflect real-world behavior, the model remains a useful analytical benchmark for interpreting episodes in which asset prices, including exchange rates, deviate persistently from their fundamental values.

4.3 Dornbusch Overshooting Model

The model we present enables to connect and introduce flexible and sticky prices by considering prices to be sticky (NK assumption) in the short run and allowing them to be flexible in the long run. Wage and price rigidity have long been recognized as central features in macroeconomics, dating back to [**?**]. In the Keynesian tradition, such rigidities imply that monetary policy can have real effects in the short run because prices and wages adjust slowly. However, in the long run, once prices fully adjust, money growth only leads to proportional changes in the price level, reflecting money homogeneity and long-run monetary neutrality. More recently, nominal rigidities have been micro-founded by the New Keynesian "menu costs" literature, which shows how even small costs of adjusting prices can generate persistent price stickiness.

We recall that Dornbush's model is not microfounded, as he assumes explicit functional forms that are not derived.

The model presented here has been developed to account for and explain some main stylised facts:

1. **Large exchange rate volatility** with respect to the fundamentals m, y.
2. **differential speed of adjustment** of assets versus goods market.
3. **Effects of monetary expansion**: A monetary expansion first causes a currency depreciation, then during adjustment the exchange rate may appreciate as prices rise, and the exchange rate itself transmits monetary policy to inflation and output.

4.3.1 Assumptions

We now introduce the set of assumptions under which the model is constructed:

1. **SOE**: Where the world interest rate is i^* (asset market) and is exogenous so is is p^* (goods market).

2. **Domestic and Foreign goods are imperfect substitutes**

3. **Differential of adjustment** between goods (slow) and assets (immediate).

4. **Perfect Foresight** so agents predict future exactly.

Capital Market Structure and Equilibrium

Here we assume that domestic and foreign assets are perfect substitutes given a proper premium, so under risk neutrality we assume:

$$i = i^* + x$$

where $x = E[\dot{s}]$ and as $x > 0$ depreciation of domestic currency.

Formation on the exchange rate are given by:

$$x = \theta(\bar{s} - s)$$

where agents distinguish between long and short run, so the expected NER is proportional to the discrepancy between long-run exchange rate and s.

Money Market Structure and Equilibrium

We assume our money demand as:

$$m - p = \phi y - \lambda i$$

Now we assume that in the long run $s = \bar{s}$ and $p = \bar{p}$ and $i = i^*$ therefore:

$$\bar{p} = m - \phi y + \lambda i^* \tag{1}$$

Inserting this to the previous equation we get:

$$s = \bar{s} - \frac{1}{\lambda\theta}(p - \bar{p}) \tag{2}$$

Solving for p we obtain the asset market equilibrium QQ:

$$QQ : p = \bar{p} - \lambda\theta s + \lambda\theta\bar{s} \tag{3}$$

where we can interpret it as follows: Given the long-run values of $\bar{s}$ and $\bar{p}$, the spot exchange rate s can be determined as a function of the deviation of the current price level from its equilibrium value (2). For a given price level p, the domestic interest rate i and the interest rate differential $i - i^*$ are determined endogenously. Hence, an increase in the price level ($p \uparrow$) implies an increase in the domestic interest rate ($i \uparrow$). This generates a capital inflow into the country, which leads to an appreciation of the domestic currency and therefore a decrease in the spot exchange rate ($s \downarrow$).

According to the UIP condition, the exchange rate must adjust so that the expected depreciation of the currency equals the interest rate differential. Therefore, the appreciation of the spot exchange rate must be sufficiently large so that agents expect a future depreciation that exactly offsets the higher domestic interest rate. In other words, the exchange rate appreciates beyond its long-run equilibrium level so that the anticipated depreciation satisfies the UIP condition.

Goods Market Structure and Equilibrium

Here we assume a normalization of foreign log price such that $p^* = 0$ therefore the relative price of domestic goods is given by $q = s - p$, hence

$$\ln D = u + \delta(s - p) + \gamma y - \sigma i$$

Inflation is then defined as the excess demand therefore the difference between the aggregate demand and output up to some coefficient, so:

$$\dot{p} = \pi(\ln D - y)$$

therefore

$$\dot{p} = \pi[u + \delta(s - p) + (\gamma - 1)y - \sigma i]$$

Assuming that in the long run inflation is zero and all the variables converge to their steady state level $s = \bar{s}, i = i^*$, then:

$$\bar{s} = \bar{p} + \frac{1}{\delta}[\sigma i^* + (1 - \gamma)y - u]$$

From the long–run equilibrium exchange rate

$$\bar{s} = \bar{p} + \frac{1}{\delta}[\sigma i^* + (1 - \gamma)y - u],$$

we can write

$$\bar{s} - s = \bar{p} - s + \frac{1}{\delta}[\sigma i^* + (1 - \gamma)y - u],$$

so that

$$s = (s - \bar{s}) + \bar{p} + \frac{1}{\delta}[\sigma i^* + (1 - \gamma)y - u].$$

Substituting this expression for s into the price adjustment equation

$$\dot{p} = \pi\left[u + \delta(s - p) + (\gamma - 1)y - \sigma i\right],$$

and rearranging gives

$$\dot{p} = \pi\left[\delta(s - \bar{s}) + \delta(\bar{p} - p) + \sigma(i^* - i)\right].$$

Using UIP $i - i^* = \frac{1}{\lambda}(s - \bar{s})$, we obtain

$$\dot{p} = \pi\left[(\delta + \sigma\theta)(s - \bar{s}) + \delta(\bar{p} - p)\right].$$

Finally, substituting $s - \bar{s} = -\frac{1}{\lambda\theta}(p - \bar{p})$,

$$\dot{p} = -\pi\left(\frac{\delta + \sigma\theta}{\theta\lambda} + \delta\right)(p - \bar{p}) = -\nu(p - \bar{p}),$$

where

$$\nu = \pi\left(\frac{\delta + \sigma\theta}{\theta\lambda} + \delta\right).$$

The differential equation

$$\dot{p} = -\nu(p - \bar{p})$$

has the solution

$$p(t) - \bar{p} = (p_0 - \bar{p})e^{-\nu t},$$

or equivalently

$$p(t) = \bar{p} + (p_0 - \bar{p})e^{-\nu t}.$$

Hence, prices converge exponentially to the long-run equilibrium level $\bar{p}$ at speed ν.

Substituting this result into the exchange rate equation gives the time path of the nominal exchange rate

$$s(t) = \bar{s} - \frac{1}{\lambda\theta}(p_0 - \bar{p})e^{-\nu t} = \bar{s} + (s_0 - \bar{s})e^{-\nu t}.$$

Thus, the exchange rate also converges to its long-run equilibrium $\bar{s}$. If initially $p_0 < \bar{p}$, then $s_0 > \bar{s}$, meaning the exchange rate is temporarily above its long-run level and subsequently appreciates as it returns to equilibrium (exchange rate overshooting).

4.3.2 Model Equilibrium and Transition path

The figure below shows three important relations: (i) QQ schedule asset and money market equilibrium(with continuous adjustment), (i) Long run PPP under normalization $s = p$, and finally (iii) $\dot{p}$, which shows combinations of interest rate and and interest rates along which the good market is in equilibrium. Points above $\dot{p}$ presents excess supply, while below excess demand.

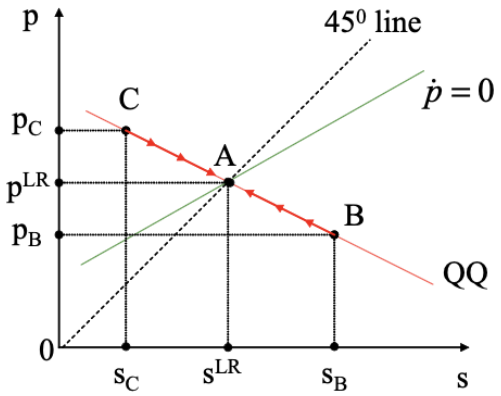


Figure 4.5

To find $\dot{p}$ schedule we use the following derivation

$$0 = \pi\left[u + \delta(s - p) + (\gamma - 1)y - \sigma\frac{m - p - \phi y}{-\lambda}\right] \tag{4.10}$$

$$\frac{\lambda\delta + \sigma}{\lambda}p = u + \delta s + \frac{\sigma}{\lambda}m + \left[(\gamma - 1) - \frac{\phi\sigma}{\lambda}\right]y \tag{4.11}$$

$$p = \frac{\lambda}{\lambda\delta + \sigma}u + \frac{\lambda\delta}{\lambda\delta + \sigma}s + \frac{\sigma}{\lambda\delta + \sigma}m + \frac{\lambda(\gamma - 1) - \phi\sigma}{\lambda\delta + \sigma}y \tag{4.12}$$

Since UIP and money market equilibrium hold continuously, the economy is *always* located on the asset–market equilibrium schedule (the QQ curve). In contrast, goods market equilibrium is only achieved in the long run because domestic prices are sticky and adjust gradually over time. Nevertheless, conditions in the goods market are crucial for driving the economy from a short-run disequilibrium toward its long-run equilibrium.

In the Dornbusch model two main adjustment mechanisms operate:

- A depreciation of the domestic currency ($s \uparrow$) lowers the relative price of domestic goods ($(s-p) \uparrow$), making domestic output cheaper relative to foreign goods and generating excess demand for domestic output.
- As excess demand emerges, domestic prices gradually increase ($p \uparrow$). The rise in prices reduces real money balances and increases domestic interest rates, dampening aggregate demand and gradually restoring goods market equilibrium.

Now we recall from perfect foresight assumptions that exchange rates expected depreciation x depends on $\theta(s-\bar{s})$ where θ is the rate for the perfect foresight exchange rate path. Furthermore v is the rate at which prices and exchange rate converge in the long run equilibrium determined by the model. So it must be that $\theta = v$, given: Starting from

$$\theta = \pi \left(\frac{\delta + \sigma\theta}{\theta\lambda} + \delta \right)$$

multiply both sides by θ:

$$\theta^2 = \pi \left(\frac{\delta + \sigma\theta}{\lambda} + \delta\theta \right)$$

Expand the right-hand side:

$$\theta^2 = \frac{\pi\delta}{\lambda} + \frac{\pi\sigma}{\lambda}\theta + \pi\delta\theta$$

Rearranging terms gives the quadratic equation

$$\theta^2 - \theta \left(\frac{\pi\sigma}{\lambda} + \pi\delta \right) - \frac{\pi\delta}{\lambda} = 0$$

whose stable (positive) root is

$$\tilde{\theta} = \frac{\pi \left(\frac{\sigma}{\lambda} + \delta \right)}{2} + \sqrt{\left(\frac{\pi \left(\frac{\sigma}{\lambda} + \delta \right)}{2} \right)^2 + \frac{\pi\delta}{\lambda}}.$$

Therefore the economy's convergence rate depends on different parameters. The economy converges faster (given π) if:

- the lower the interest rate response to the money demand λ, the higher the responds of goods demand σ and the higher the elasticity of demand for domestic output δ.

4.3.3 Overshooting Dynamics

Assume a positive and permanent monetary shock $m \uparrow$. From the long–run price equation

$$\bar{p} = m - \phi y + \lambda i^*$$

it follows that the long–run price level increases and therefore the asset–market schedule QQ shifts outward to QQ'.

Since prices are predetermined in the short run, the increase in money supply raises real money balances, which must be absorbed through a fall in the domestic nominal interest rate $i \downarrow$ in order to restore money market equilibrium. The lower interest rate implies, through UIP, an expectation of a future depreciation of the domestic currency.

At the current nominal exchange rate, both the fall in the domestic interest rate and the expectation of future depreciation reduce the attractiveness of domestic assets. This generates capital outflows and leads to an immediate depreciation of the spot exchange rate $s \uparrow$. The depreciation must be large enough to create an expectation of future appreciation that exactly offsets the lower domestic interest rate.

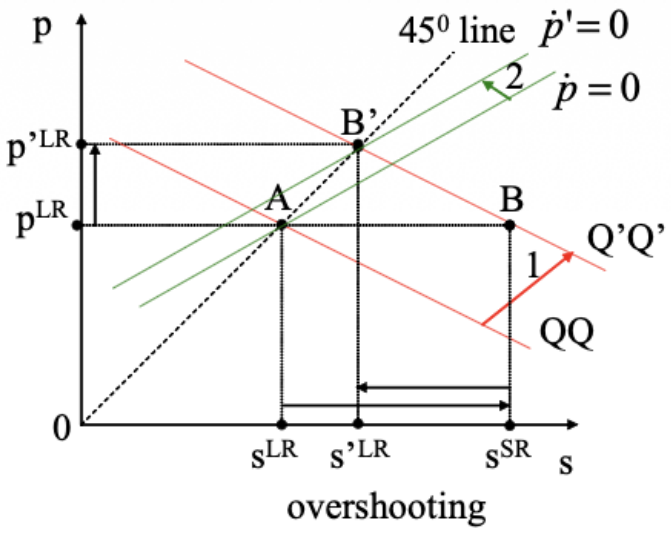


Figure 4.6

$$m \uparrow \Longrightarrow i \downarrow \Longrightarrow UIP \Longrightarrow s \uparrow (depreciation)$$

In B, we are in an excess demand situation, which is given by a low interest rate and an undervalued currency, resulting in more exports. This will lead prices to rise, gradually reducing the real money balance and reversing the currency depreciation into appreciation in the long-run equilibrium until $s = \bar{s}$. Here we see the classical dynamics of the **exchange rate overshooting**. Basically, we have a two-stage effect, in particular, we have:

1. Given we start at A, we assume a positive and permanent monetary policy shock $m \uparrow$. With sticky prices in the short run, the increase in the nominal money supply raises real money balances, which leads to a decrease in the domestic interest rate $i \downarrow$. This generates capital outflows toward foreign assets. According to uncovered interest parity, the exchange rate must immediately depreciate so that agents expect a future appreciation. Therefore the spot exchange rate jumps above its new long–run level, moving the economy to point B.

2. At point B, the exchange rate is highly depreciated and domestic goods become relatively cheaper than foreign goods. This generates excess demand in the goods market, which gradually pushes prices upward $p \uparrow$. As prices increase, real money balances decrease (since m is constant), which leads to a gradual increase in the interest rate. The rise in the domestic interest rate attracts capital inflows and induces an appreciation of the exchange rate. The economy therefore converges toward the new long–run equilibrium at point B'.

The monetary expansion shifts the QQ schedule to $Q'Q'$ because, for any given price level, the increase in money supply lowers the domestic interest rate and therefore requires a more depreciated exchange rate to satisfy uncovered interest parity. At the same time, the long–run equilibrium price level increases, which shifts the $\dot{p} = 0$ locus upward.

Analytically, we have that $\dot{p}$ is proportional to the excess demand given by $\ln D - \ln Y$, since the excess demand is finite, and also the inflation is therefore there are no jumps. p is instantaneously fixed together with y and i^* is constant. Considering therefore, an unanticipated monetary policy expansion $dm > 0$ will produce $di < 0$, recall that:

$$\frac{di}{dm} = -\frac{1}{\lambda} < 0$$

since $m \uparrow \implies dp_{LR} \uparrow$, so we have that in the long run $dm = d\bar{p} = d\bar{s}$, we now recall:

$$dp - dm = -\phi dy + \lambda di^* + \theta\lambda(d\bar{s} - ds)$$

in the short run $dp = 0$ so we get:

$$\frac{ds}{dm} = 1 + \frac{1}{\lambda\theta} > 1$$

meaning that the effect of an unexpected monetary inflation is creating an effect on the exchange rate that is more than proportionate, we also recall that $d\bar{s} = dm$ so:

$$ds = \left(1 + \frac{1}{\lambda\theta}\right) d\bar{s}$$

so we have derived the exchange rate overshooting analytically.
Following Dornbush, the short-run dynamics are entirely dominated by the asset market, more specifically, capital mobility and expectation formation. By considering then the Variance of that we can find that:

$$Var(ds) = \left(1 + \frac{1}{\lambda\theta}\right)^2 Var(dm)$$

4.4 Taylor Rule and Exchange Rate (Engle and West 2006)

In this paper the authors explore the relationship between Taylor rules and the real exchange rate. In particular, before this contribution it was difficult to reconcile the empirical evidence that exchange rates behave approximately as random walks with the theoretical view that exchange rates should depend on macroeconomic fundamentals. Engel and West show that these two facts are not necessarily inconsistent. Even if the exchange rate depends on fundamentals, it may still display dynamics that are close to a random walk when the exchange rate is determined as the present value of expected future fundamentals.

Before this paper there was already a long tradition studying the relationship between interest rates and exchange rates, both in theoretical and empirical work. This link arises naturally because interest rate differentials affect international capital flows, which in turn influence exchange rate movements.

Part of the literature also extended monetary policy rules by incorporating exchange rates directly into the Taylor rule. For example, in regimes where exchange rate stability is important—such as fixed or managed exchange rate systems—central banks may react to fluctuations in the exchange rate s_t when setting the policy interest rate. Empirical research on this relationship has employed several approaches, including VAR models, tests of interest rate parity conditions, and calibrated general equilibrium sticky-price models. While theory predicts a strong connection between interest rates and exchange rates, empirical results have often been mixed and sensitive to the currencies considered, the sample period, and the econometric methodology used.

4.4.1 Model

The model proposed is a SOE version of a New Keynesian Model, where we assume domestic and foreign country, that produce goods to be consumed and traded and households hold both domestic and foreign bonds.

The main research question is **What is the effect of monetary policy on the real exchange rate**?

Following the paper households are assumed to behave as in a close economy and are assumed to spend a share of their income on domestic goods and the rest in foreign goods therefore:

$$\pi_t^h = (1-\alpha)\pi_t^h + \alpha\pi_t^f$$

Similarly to a closed economy model we assume firms producing goods under monopolisticallly competitive setting, with calvo price stickiness, this implies NKPH for both home and foreign country:

$$\pi_{d,t} = \beta E_t \pi_{d,t+1} + \kappa y_t^h + u_{c,t}^h \tag{4.13}$$

$$\pi_t^* = \beta E_t \pi_{t+1}^* + \kappa y_t^* + u_{c,t}^* \tag{4.14}$$

Combining these two Phillips curves we obtain

$$\dot{\pi}_t = \beta E_t \dot{\pi}_{t+1} + \kappa y_t + u_{c,t}. \tag{4.15}$$

Here the variables express differences between the two countries where

$$\dot{\pi}_t \equiv \pi_{d,t} - \pi_t^*, \qquad y_t = y_t^h - y_t^*, \qquad u_{c,t} \equiv u_{c,t}^h - u_{c,t}^*.$$

We assume that the output gap is related to the real exchange rate and to an exogenous output shock. In particular, defining the real exchange rate as

$$q_t = s_t - p_t = s_t - \left(p_t^h - p_t^*\right),$$

the output gap is given by

$$y_t = \theta q_t + u_{y,t}, \qquad \theta > 0. \tag{4.16}$$

This relation can be interpreted as an IS curve. As also the current literature does we insert the exchange rate component directly into the taylor rule by adding deviations of actual rela exchange rate to its steady state. Now the important assumption here is that the foreign country (U.S) uses the classic taylor rule while domestic introduce the RER in it. U.S is considered big so the FED does not face problems related to the exchange rate therefore it is not a priority:

$$\begin{aligned} \text{Foreign:} \quad & i_t^* = \gamma_\pi E_t[\pi_{t+1}^*] + \gamma_y y_t^* + u_{m,t}^* \\ \text{Home:} \quad & i_t^h = \gamma_q q_t + \gamma_\pi E_t[\pi_{t+1}^h] + \gamma_y y_t^h + u_{m,t}^h \end{aligned}$$

we now consider i_t as the differential of the home and foreign interest rate therefore:

$$i_t = \gamma_q q_t + \gamma_\pi E_t\left(\pi_{t+1}^h - \pi_{t+1}^*\right) + \gamma_y \left(y_t^h - y_t^*\right) + u_{m,t}^h - u_{m,t}^*.$$

Start from the Taylor rule for the interest rate differential

$$i_t = \gamma_q q_t + \gamma_\pi E_t(\pi_{t+1}^h - \pi_{t+1}^*) + \gamma_y(y_t^h - y_t^*) + (u_{m,t}^h - u_{m,t}^*)$$

Define

$$y_t = y_t^h - y_t^*, \qquad u_{m,t} = u_{m,t}^h - u_{m,t}^*$$

so that

$$i_t = \gamma_q q_t + \gamma_\pi E_t(\pi_{t+1}^h - \pi_{t+1}^*) + \gamma_y y_t + u_{m,t}$$

Assume UIP holds

$$i_t = i_t^h - i_t^* = E_t s_{t+1} - s_t$$

Using the definition of the real exchange rate

$$q_t = s_t + p_t^* - p_t^h$$

it follows that

$$E_t s_{t+1} - s_t = E_t q_{t+1} - q_t + E_t(\pi_{t+1}^h - \pi_{t+1}^*)$$

Substituting into the Taylor rule gives

$$E_t q_{t+1} = (1 + \gamma_q) q_t + (\gamma_\pi - 1) E_t(\pi_{t+1}^h - \pi_{t+1}^*) + \gamma_y y_t + u_{m,t}$$

Solving for q_t and defining $b = \frac{1}{1+\gamma_q}$ yields

$$q_t = b E_t q_{t+1} + b(1 - \gamma_\pi) E_t(\pi_{t+1}^h - \pi_{t+1}^*) - b\gamma_y y_t - b u_{m,t}.$$

We can now solve the difference equation and obtain:

$$q_t = b \sum_{j=0}^{\infty} b^j E_t \left[\pi_{t+j+1} - \gamma_\pi \left(p_{t+j+1} - p_{t+j}\right) - \gamma_y y_{t+j} - u_{m,t+j}\right] + \lim_{j \to \infty} E_t \, b^j q_{t+j}. \tag{4.17}$$

Assuming the transversality condition we assume last term to decay to zero since $b \in (0,1)$. Finally we can say that adding risk premium to UIP won't change this equation if we assume that $u_{m,t}$ includes the shock. However we recall it moves in inverse direction with respect to monetary policy. Finally our model ends up with 4 equations:

$$i_t = \gamma_q q_t + \gamma_\pi E_t \pi_{t+1} + \gamma_y y_t + u_{m,t} \quad \text{(Tayl. rule)} \tag{4.18}$$
$$i_t = E_t s_{t+1} - s_t \quad \text{(UIP)} \tag{4.19}$$
$$q_t = s_t + p_t^* - p_t \quad \text{(RER)} \tag{4.20}$$
$$q_t = bE_t q_{t+1} + b(1-\gamma_\pi)E_t \pi_{t+1} - b\gamma_y y_t - bu_{m,t} \quad \text{(For.RER)} \tag{4.21}$$

Consider a positive cost–push shock that increases domestic inflation. According to the Taylor rule, since $\gamma_\pi > 1$, the central bank reacts by increasing the nominal interest rate so that $i_t > i_t^*$.

Using UIP,

$$i_t = E_t s_{t+1} - s_t,$$

a higher domestic interest rate implies that the difference between the expected future exchange rate and the current exchange rate must increase. This can occur either because the expected future exchange rate increases or because the current exchange rate falls. Following the standard argument also discussed in the Mundell–Fleming model, we assume that the adjustment occurs through a fall in the current exchange rate, implying an appreciation of the domestic currency.

Since the real exchange rate is defined as $q_t = s_t + p_t^* - p_t$, the appreciation of the nominal exchange rate implies that the real exchange rate also falls.

From the aggregate demand relation

$$y_t = \theta q_t + u_{y,t},$$

a decrease in q_t leads to a decline in output. The intuition is that the appreciation of the currency makes domestic goods relatively more expensive compared to foreign goods, reducing external demand. Moreover, the increase in the real interest rate also reduces consumption, reinforcing the negative effect on output.

4.4.2 Taylor Rules and Exchange Rate Dynamics

These slides present a simple open-economy framework that links monetary policy, exchange rates, and macroeconomic dynamics. The objective is to show how monetary policy shocks and other disturbances affect the exchange rate and the broader economy.

The model combines four key relationships. First, monetary policy is described by a Taylor rule in which the central bank sets the nominal interest rate in response to inflation, output, and possibly the real exchange rate:

$$i_t = \gamma_q q_t + \gamma_\pi E_t \pi_{t+1} + \gamma_y y_t + u_{m,t}.$$

Second, the exchange rate is determined in financial markets through the uncovered interest parity (UIP) condition:

$$i_t = E_t s_{t+1} - s_t.$$

This relationship implies that interest rate differentials determine expected exchange rate movements.

Third, aggregate demand depends on the real exchange rate:

$$y_t = \theta q_t + u_{y,t},$$

so that a depreciation of the domestic currency increases demand for domestic goods and raises output.

Finally, inflation is described by a Phillips curve in which inflation depends on expected future inflation and the level of economic activity:

$$\pi_t = \beta E_t \pi_{t+1} + \kappa y_t.$$

Together, these equations form a simple New Keynesian open-economy system in which monetary policy affects exchange rates through the interest rate, and exchange rate movements feed back into output and inflation.

The framework can then be used to analyze the effects of different shocks. For example, a cost-push shock increases inflation, prompting the central bank to raise the interest rate through the Taylor rule. Higher interest rates lead to an appreciation of the currency through UIP, which reduces output by making domestic goods more expensive relative to foreign goods. Conversely, a monetary policy expansion lowers the interest rate, leading to a depreciation of the currency and an increase in output.

This structure illustrates how monetary policy, financial markets, and macroeconomic variables interact in determining exchange rate dynamics.

4.5 Empirical Evidence

These lecture notes cover the empirical evidence on exchange rate determination, drawing on a series of influential studies spanning several decades. The lectures begin by testing the flexible price and sticky price monetary models against data, reviewing early work by Frenkel (1980) and the long-sample cointegration approach of Rapach and Wohar (2002). Attention then turns to the forecasting literature, starting with the landmark Meese and Rogoff (1983) finding that structural models cannot outperform a simple random walk out of sample — a result that has shaped the field ever since. The notes survey the broader forecasting literature through Rossi (2013), consider whether survey-based expectations improve on model-based forecasts, and examine how economic policy uncertainty affects exchange rate expectations using the framework of Beckmann and Czudaj (2017). The lectures conclude with Engel and West (2005), who offer a theoretical reconciliation of the disconnect puzzle: if at least one fundamental contains a unit root and the discount factor is near unity, standard present-value exchange rate models imply that exchange rates should behave approximately as random walks — so the Meese-Rogoff result is a prediction of the models, not evidence against them.

We can now consider a model that incorporates flexible price and sticky prices features, and evaluate it, so:

$$s = \alpha_0 + a_1(m - m^*) + a_2(y - y^*) + a_3(i - i^*) + a_4(\pi^e - \pi^{e*}) + u$$

here:

- flexible prices: $a_1 = 1, a_2 < 0$ and either $a_3 > 0, a_4 = 0$ or $a_3 = 0, a_4 > 0$. Here, the variables attached to a_3 and a_4 express the same Fisher expression; therefore, one should be 0.
- In the sticky prices, we assume that prices do not change in the short run, so inflation should not change; therefore, $a_4 = 0$ and all the other terms should stay.

4.5.1 Frenkel 1980

In his paper, Frenkel tests for the flexible price monetary model by using the interwar period 21-25 Monthly data on Mark/pound, Franc/Pound and Dollar/Pound. In this period, the author used the hyperinflation as a natural experiment for anormal condition, under the new flexible regime after the gold standard was abandoned.

He approaches and addresses the analysis under the monetary approach to the exchange rate, assuming CIP to hold and therefore formulating the following regression:

$$s_t = \alpha_0 + a_1(m_t - m_t^*) + a_2(y_t - y_t^*) + a_3 \underbrace{(f_t - s_t)}_{i - i^*} + u_t$$

using the hypothesis that $a_1 = 1, a_2 < 0, a_3 > 0$. The empirical results show that the coefficient associated with the money supply is close to unity and carries the predicted sign, in line with the implications of the monetary model of the exchange rate. The parameters associated with output are statistically insignificant, suggesting that real activity plays a limited role in explaining exchange rate movements within this framework.

By contrast, the forward premium is found to be statistically significant and its magnitude is close to the estimated interest elasticity of money demand. Moreover, the empirical relationship appears to be considerably stronger during episodes of hyperinflation, whereas the results are weaker during more stable monetary periods. These findings are consistent with the evidence presented in [**?**], where the monetary model performs better in high-inflation environments.

TABLE 3—EXCHANGE RATE EQUATIONS: MONTHLY DATA
(Standard Errors in Parentheses)

Dependent Variable: $ln\,S_t$	Constant	$ln\,M$	$ln\,M^*$	$ln\,y$	$ln\,y^*$	π	S.E.	R^2	D.W.	ρ
Mark/Pound	−6.030	.970				3.886	.340	.99	2.56	.89
(Feb. 1921–Aug. 1923)	(1.696)	(.092)				(1.131)				
Franc/Pound	.001	.999	−.972	.188	.926	3.914	.069	.92	1.86	1.00
(Feb. 1921–May 1925)	(.010)	(.099)	(.099)	(.281)	(.520)	(.970)				
Franc/Dollar	.006	.995	−.995	.225	−.369	3.971	.075	.86	1.81	1.00
(Feb. 1921–May 1925)	(.011)	(.099)	(.100)	(.327)	(.370)	(.974)				

Note: The Mark/Pound exchange rate equation was estimated using two-stage least squares following Fair's method with lagged values of the dependent and independent variables, a constant, time, and time squared as instruments. ρ is the final value of the autocorrelation coefficient; an iterative Cochrane-Orcutt transformation was employed. R^2 was computed as $1 - var(\hat{u}_t)/var(ln\,S_t)$. The Franc/Pound and the Franc/Dollar exchange rate equations were estimated in first difference form using the Theil-Goldberger mixed-estimation procedure with stochastic restrictions.

Figure 4.7

It might be the case that under hyperinflation, the prices result in being very flexible. Usually, we reject flexible prices. However, also using sticky prices, it supports earlier studies but not after Bretton Woods....what might be the problem? It could arise from many sources:

- **Econometric Model**
- **Sample too short Exchange Rate model is incorrect**

4.5.2 Rapach and Wohar 2002

A prominent response to the weak short-run empirical performance of the monetary model is to extend the time horizon of the analysis. Rapach and Wohar argue that what matters for detecting long-run equilibrium relationships is the *length* of the sample period rather than the frequency of observations. To exploit this, they construct a dataset of annual observations spanning 1880–1995 for 14 countries, providing a time span substantially longer than those used in the post-Bretton Woods literature.

Theoretical specification. Their empirical specification is grounded in the flexible-price monetary model:

$$s = \alpha_0 + a_1(m - m^*) + a_2(y - y^*) + a_3(i - i^*) + u. \tag{4.22}$$

Since the analysis focuses on the long-run equilibrium, the interest rate differential can be eliminated by appealing to uncovered interest parity. In the steady state, the expected change in the exchange rate is zero on average regardless of whether s is stationary or $I(1)$, so that $i = i^*$ and the interest differential drops out. The long-run specification therefore reduces to

$$s = \alpha_0 + a_1(m - m^*) + a_2(y - y^*) + u, \tag{4.23}$$

which captures the equilibrium relationship between the nominal exchange rate and its monetary fundamentals. The model predicts that s moves proportionally with relative money supplies ($a_1 = 1$) and inversely with relative real output ($a_2 < 0$), the latter reflecting the standard money demand channel.

Cointegration interpretation. Equation (4.23) can be interpreted as a potential cointegrating vector. If s, $(m - m^*)$, and $(y - y^*)$ are each integrated of order one but the disturbance u is stationary, the linear combination is $I(0)$ and the variables are said to be *cointegrated*: although each series may drift independently in the short run, they are bound together by a stable long-run attractor. Testing the monetary model therefore amounts to two distinct questions: first, whether a cointegrating relationship among these variables exists at all; and second, whether the estimated coefficients are consistent with the theoretical restrictions $a_1 = 1$ and $a_2 < 0$.

Unit root pre-testing. Cointegration requires that all variables entering the relationship share the same order of integration. Accordingly, the analysis begins by applying standard unit root tests to s, $(m-m^*)$, and $(y-y^*)$ to verify that each series is $I(1)$. Only if this precondition is satisfied does it make sense to proceed to cointegration testing.

Table 2
Cointegrating coefficient estimates, $e_t = \beta_0 + \beta_1(m_t^* - m_t) + \beta_2(y_t^* - y_t)$

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
	OLS estimates		FM-OLS estimates		DOLS estimates			JOH-ML estimates		
Country	β_1	β_2	β_1	β_2	β_1	β_2	SW-Wald[a]	β_1	β_2	JOH-χ^2 [b]
Australia	0.50	−0.14	0.62	−0.21	0.45	−0.19	2.59	0.45	−0.41	2.32
(1880–1995)	(0.09)	(0.21)	(0.29)	(0.63)	(0.35)	(0.73)		(0.30)	(0.63)	
Belgium	0.90	−1.10	1.04	−1.06	1.00	−1.01	0.01	0.93	−1.06	0.06
(1880–1989)	(0.09)	(0.08)	(0.38)	(0.34)	(0.25)	(0.23)		(0.25)	(0.23)	
Canada	0.17	0.04	0.23	0.05	0.13	0.10	179**	0.20	0.08	14.04**
(1880–1995)	(0.03)	(0.04)	(0.13)	(0.15)	(0.08)	(0.08)		(0.08)	(0.08)	
Finland	1.05	–	1.07	–	1.01	–	0.01	0.82	–	3.10†
(1911–1995)	(0.03)		(0.13)		(0.07)			(0.11)		
France (trend)	1.03	−1.34	1.09	−1.06	1.03	−1.16	0.28	1.03	−0.09	2.15
(1880–1989)	(0.04)	(0.10)	(0.11)	(0.25)	(0.10)	(0.31)		(0.14)	(0.54)	
Italy	0.99	−0.93	0.95	−1.20	0.96	−1.34	2.55	0.98	−1.87	11.36**
(1880–1995)	(0.01)	(0.09)	(0.03)	(0.20)	(0.03)	(0.39)		(0.02)	(0.25)	
Portugal	1.13	–	1.09	–	1.07	–	1.31	0.94	–	0.47
(1890–1995)	(0.04)		(0.09)		(0.06)			(0.09)		
Spain	0.88	−1.21	0.83	−1.27	0.86	−1.29	11.94**	0.85	−1.05	10.16**
(1901–1995)	(0.03)	(0.09)	(0.06)	(0.22)	(0.05)	(0.16)		(0.04)	(0.14)	
Switzerland (trend)	0.45	−0.79	1.46	−2.08	0.86	−1.30	1.54	1.59	−2.42	17.77**
(1880–1995)	(0.15)	(0.15)	(0.55)	(0.53)	(0.46)	(0.43)		(0.52)	(0.61)	
United Kingdom	0.41	−0.97	0.39	−0.85	0.45	−0.99	22.24**	0.31	−0.71	10.42**
(1880–1995)	(0.04)	(0.04)	(0.14)	(0.16)	(0.19)	(0.24)		(0.13)	(0.16)	

†,*,** indicate significance at the 10, 5, and 1 percent levels, respectively; for Finland and Portugal, β_2 is constrained to be equal to zero; standard errors for the coefficient estimates are given in parentheses; (trend) indicates that a linear trend is included in the cointegrating relationship.

[a] Stock and Watson (1993) one-sided (upper-tail) test of $H_0: \beta_1 = 1$, $\beta_2 = -1$; 10, 5, and 1 percent critical values for a $\chi^2(2)$ equal 4.61, 5.99, and 9.21, respectively; for Finland and Portugal, the test is for $H_0: \beta_1 = 1$; for Finland and Portugal, the 10, 5, and 1 percent critical values for a $\chi^2(1)$ equal 2.71, 3.84, and 6.64, respectively.

[b] Johansen (1991) one-sided (upper-tail) test of $H_0: \beta_1 = 1$, $\beta_2 = -1$; 10, 5, and 1 percent critical values for a $\chi^2(2)$ equal 4.61, 5.99, and 9.21, respectively; for Finland and Portugal, the test is for $H_0: \beta_1 = 1$; for Finland and Portugal, the 10, 5, and 1 percent critical values for a $\chi^2(1)$ equal 2.71, 3.84, and 6.64, respectively.

Figure 4.8: Unit root test results and cointegration estimates for the flexible-price monetary model. Columns (8)–(11) report tests of the theoretically implied cointegrating vectors.

Empirical results. The estimates provide partial but meaningful support for the monetary model. The coefficient on relative money supply, $\hat{a}_1$, is positive and statistically significant across all specifications, and in several cases close to the theoretically implied value of unity. The coefficient on relative output, $\hat{a}_2$, carries the predicted negative sign in every case and is statistically significant for 6 of the 10 countries considered, consistent with the money demand mechanism. Columns (8) to (11) report cointegration tests against the theoretically restricted vectors; evidence of a stable long-run relationship between the exchange rate and its fundamentals is found for several countries in the sample.

Taken together, the results suggest that the monetary model's long-run predictions are broadly consistent with the data once a sufficiently long time horizon is considered. This finding holds under both the flexible-price and sticky-price parameterisations of the model, pointing to the importance of sample length as a key determinant of the power of exchange rate tests.

4.5.3 The random Walk Hypothesis

Another approach is to assess the ability of a structural model to produce good predictions and therefore generate accurate forecasts. In the literature this is typically studied through the out-of-sample forecasting performance of exchange rate models over different horizons. An influential contribution is the work of Meese and Rogoff (1983), who showed that neither the flexible-price nor the sticky-price monetary models outperform a simple random walk in forecasting exchange rates.

The empirical specification is extended by combining several theoretical exchange rate models into a single synthesis regression. In addition to the monetary fundamentals of the basic model, the specification incorporates elements from the real interest rate model of Frankel and from the Hooper–Morton framework. The Frankel model allows for persistent inflation differentials across countries, while the Hooper–Morton model introduces the possibility of shifts in the long–run equilibrium real exchange rate and the presence of a risk premium.

The resulting synthesis regression can be written as

$$s = \alpha_0 + a_1(m - m^*) + a_2(y - y^*) + a_3(i - i^*) + a_4(\pi^e - \pi^{e*}) + a_5 tb + a_6 tb^* + u. \tag{4.24}$$

In this specification the nominal exchange rate depends on monetary fundamentals, interest rate differentials, expected inflation differentials, and external balance indicators. The variables tb and tb^* denote the domestic and foreign trade balances, which proxy cumulative current account imbalances and capture long–run external pressures on the exchange rate.

Different theoretical models correspond to restrictions on the coefficients of this general equation. The flexible–price monetary model imposes the restrictions discussed in the basic monetary specification, while combining the sticky–price model with the real interest rate model implies $a_5 = a_6 = 0$, excluding trade balance effects. In contrast, the Hooper–Morton model predicts that improvements in the domestic current account lead to an appreciation of the domestic currency ($a_5 < 0$), whereas improvements in the foreign current account imply a depreciation ($a_6 > 0$). Finally, an additional specification combines covered and uncovered interest parity, suggesting that the forward exchange rate contains information about the future nominal exchange rate.

Table 1: The Meese-Rogoff (1983) Forecasting Results

Ex. rate	Model: / Horizon	Random walk	Forward rate	Univariate AR	VAR	Monetary model	Dornbusch model	Complete eq. (45)
\$/DM	1	3.72	3.20	3.51	5.40	3.17	3.65	3.50
	6	8.17	9.03	12.40	11.83	9.64	12.03	9.95
	12	12.98	12.60	22.53	15.06	16.12	18.87	15.69
\$/yen	1	3.68	3.72	4.46	7.76	4.11	4.40	4.20
	6	11.58	11.93	22.04	18.90	13.38	13.94	11.94
	12	18.31	18.95	52.18	22.98	18.55	20.41	19.20
\$/£	1	2.56	2.67	2.79	5.56	2.82	2.90	3.03
	6	6.45	7.23	7.27	12.97	8.90	8.88	9.08
	12	9.96	11.62	13.35	21.28	14.62	13.66	14.57
trade-	1	1.99	n.a.	2.72	4.10	2.40	2.50	2.74
weighted	6	6.09	n.a.	6.82	8.91	7.07	6.49	7.11
\$ NEER	12	8.65	14.24	11.14	10.96	11.40	9.80	10.35

Source: Meese and Rogoff (1983), Table 1, p. 13, slightly shortened by the authors. The table reports the RMSE for the full forecasting period, November 1976 through June 1981, approximately in percentage terms. The three structural models are estimated using and IV estimator to correct for first-order serial correlation. The measure of money is narrow money (M1-B) and the long-term interest rate differential serves as the proxy for the long-run expected inflation differential in the two sticky-price models.

Figure 4.9

Table reports the root mean squared forecast errors (RMSE) for several exchange rate models across different forecasting horizons (1, 6, and 12 months) and currency pairs. The models considered include a random walk benchmark, the forward rate model, a univariate autoregressive model, a VAR, and several structural exchange rate models such as the monetary model, the Dornbusch sticky-price model, and the complete synthesis regression.

The main result is that the random walk model performs at least as well as, and often better than, the structural exchange rate models. In many cases the RMSE of the random walk is smaller than that of the theoretically motivated models, particularly at longer horizons. This finding implies that macroeconomic fundamentals—such as money supply, output, interest rate differentials, or trade balance variables—do not provide systematically better short-run forecasts of exchange rates than a naïve benchmark that simply assumes the exchange rate follows a random walk.

This result, first documented by Meese and Rogoff (1983), became known as the *exchange rate forecasting puzzle*. Despite strong theoretical foundations, structural exchange rate models fail

to outperform a simple statistical benchmark in out-of-sample forecasting exercises, highlighting the difficulty of predicting exchange rate movements in the short run.

In this application, there are many econometric errors and methodological problems. They, in fact use actual fundamentals instead of their forecasts and then use fundamentals that are non-stationary. Using an extension of this papers that accounts for that we show that fundamentals are actually useful in predicting the exchange rate in the long-term.

4.5.4 Long Run Predictions

Using the CIP and solving forward, assuming the transversality condition to hold, we get that:

$$s_t - f_t = \sum_{j=0}^{k} (1-\gamma)^j E_t[\Delta f_{t+j+1}]$$

If the fundamentals are $I(1)$, then by taking the difference we would have $\Delta f_t \sim I(0)$, therefore the textbf discounted sum of the RHS should be stationary. If the RHS is stationary, then it has to be also the LHS, so s_t, f_t must be cointegrated. We can write the error correction model as:

$$s_{t+1} - s_t = \beta(s_t - f_t) + \varepsilon_{t+1}$$

This hold for any period k, we recall that fundamentals are given as:

$$f_t = (m_t - m_t^*) - \phi(y_t - y_t^*)$$

if fundamentals are importantly increasing with k, then it should be that β strengthen its explanatory power.

Table 2: Long-Horizon Exchange Rate Forecasts

Country Horizon	Canada	Japan	UK
1	0.998	0.981	1.000
6	1.014	0.955	0.976
12	1.033	0.939	0.938
16	1.009	0.847	0.889

Source: Authors' own calculations of Theil's $U = RMSE^f/RMSE^{RW}$. Money is M1 and output is GDP. Data from IMF's IFS database for 1974:1–2001:3.

Figure 4.10: Enter Caption

We see here the ratio $RMSE^f/RMSE^{RW}$, meaning that predictions of this regression are better than RW when it is less than 1. Here we see that for Japan and the UK, as k increases, the RMSE of the model estimated decreases.

	Canadian dollar			Swiss Franc		
horizon	β	R^2	OUT/RW	β	R^2	OUT/RW
1	0.040	0.059	0.998	0.074	0.051	0.997
4	0.155	0.179	1.119	0.285	0.180	0.981
8	0.349	0.351	1.145	0.568	0.336	0.917
12	0.438	0.336	1.436	0.837	0.538	0.738
16	0.450	0.254	1.699	1.086	0.771	0.411
	Deutsche mark			Yen		
1	0.035	0.015	1.015	0.047	0.020	0.988
4	0.205	0.104	1.037	0.263	0.125	0.928
8	0.554	0.265	1.002	0.575	0.301	0.819
12	0.966	0.527	0.796	0.945	0.532	0.712
16	1.324	0.762	0.524	1.273	0.694	0.574

Figure 4.11: Enter Caption

4.5.5 Rossi "Exchange Rate Predictability" Review on Predictability

This article provides an in-depth analysis of exchange rate predictability by presenting and comparing a range of empirical methodologies, highlighting their respective strengths and limitations. The central questions addressed are **whether exchange rates are predictable** and, if so, **which economic variables contain predictive power**.

The analysis is framed around the well-known *Meese and Rogoff puzzle*, which shows that standard structural exchange rate models consistently fail to outperform a simple random walk in out-of-sample forecasting. The paper revisits this puzzle by examining alternative modeling approaches and assessing whether incorporating different sets of fundamentals or econometric techniques can improve predictive performance. As analyzed by the authors exchange rate predictability depends on various elements: (i) choice of predictors, (ii) forecast horizon, (iii) sample period, (iv) model and (v) forecast evaluation method. In particular each component here can affect the final results on predictability.

The main results found in the paper are:

- **Choice of predictors**: The empirical evidence provides limited support for traditional economic models in forecasting exchange rates. While some models may outperform the random walk in-sample—typically conditional on specific countries or time periods—there is no systematic out-of-sample superiority of standard macroeconomic predictors. Among the available models, Uncovered Interest Rate Parity (UIRP) shows some predictive ability at short horizons, whereas monetary models display limited success at longer horizons. More recent contributions highlight that Taylor rule fundamentals and net foreign asset positions exhibit relatively more promising out-of-sample forecasting performance, although their predictive power remains unstable over time.

- **Model choice**: The empirical evidence suggests that nonlinear specifications are generally the least successful in forecasting exchange rates. Linear models tend to perform relatively better, particularly error correction models (ECM) and panel ECM specifications, which exhibit some predictive ability at longer horizons. However, their performance critically depends on the choice of predictors, with Taylor rule fundamentals and net foreign asset positions showing comparatively stronger results. Models allowing for regime changes can occasionally improve forecasting performance, but their success is highly sample-dependent and does not provide robust, systematic gains.

- **Data**: The treatment of the data plays a crucial role in determining forecasting performance. Data transformations—such as detrending, filtering, and seasonal adjustment—can substantially affect predictive ability and may explain why identical models yield different results across studies. In addition, the use of revised versus real-time data, as well as the calibration of cointegration relationships, can significantly influence the apparent success of economic models. Overall, differences in data handling contribute to the lack of robustness in exchange rate predictability results.

- **Forecast evaluation**: The choice of the forecast evaluation method plays a crucial role in assessing exchange rate predictability. In particular, the selection of the benchmark model is fundamental: using an inappropriate benchmark may overstate the predictive ability of economic models, while the random walk without drift represents the most challenging and appropriate benchmark in practice.

  The forecast horizon is also key. Monetary fundamentals typically exhibit no predictive ability at short horizons (e.g., one month or one quarter), although some evidence suggests that, within ECM specifications, they may outperform the random walk at longer horizons (three to four years).

More generally, both the evaluation method and the choice of the forecast sample substantially influence results. For instance, UIRP may outperform the random walk under specific testing procedures (e.g., Clark-West rather than Diebold-Mariano-West), but its performance is unstable over time and has deteriorated in recent decades. Similarly, the forecasting ability of panel ECM models is not robust to variations in the estimation window or the out-of-sample evaluation period.

Overall, the empirical evidence suggests that exchange rate predictability is weak, model-dependent, and highly sensitive to methodological choices, with no approach delivering robust and consistent forecasting gains over the random walk benchmark.

4.5.6 Exchange Rate Expectations and Economic Policy Uncertainty

This paper analyzes whether economic policy uncertainty can be incorporated into exchange rate models, with a particular focus on its role in shaping exchange rate expectations. The main finding is that forecast errors are strongly affected by policy uncertainty, indicating that such uncertainty is not fully accounted for by market participants.

The role of expectations is central for policymakers, as they represent a key transmission channel of economic policy. However, the exchange rate literature has largely overlooked this dimension. Most empirical studies focus on realized exchange rates s_t, whereas standard theoretical frameworks—such as Uncovered Interest Parity (UIP)—are formulated in terms of expected future exchange rates $E_t[s_{t+1}]$.

Even when survey-based expectation data are available, they are often misused, typically serving as predictors of future exchange rates rather than being modeled as the dependent variable. This paper instead treats expectations as the object of interest and investigates how they are influenced by policy uncertainty.

We start from a general exchange rate representation given by:

$$s_t = (1-b)\sum_{j=0}^{\infty} b^j \mathbb{E}_t\left[f_{1,t+j} + u_{1,t+j}\right] + b\sum_{j=0}^{\infty} b^j \mathbb{E}_t\left[f_{2,t+j} + u_{2,t+j}\right]. \tag{4.25}$$

The terms $f_{1,t+j}$ and $f_{2,t+j}$ denote the systematic components, typically associated with macroeconomic fundamentals, while $u_{1,t+j}$ and $u_{2,t+j}$ represent unsystematic components, capturing shocks or innovations.

The key idea is that economic policy uncertainty may affect both the systematic and the unsystematic components of exchange rate dynamics. This approach is related to the *news* literature, where exchange rate movements are driven by revisions in expectations about future fundamentals, as well as to the microstructure literature, which emphasizes the role of information and order flow.

A crucial empirical challenge is the availability of appropriate data. In particular, the analysis requires survey-based measures of exchange rate expectations and reliable indicators of policy uncertainty.

The underlying idea is that exchange rate expectations are driven by different macroeconomic fundamentals at different points in time, often referred to as "scapegoats". Economic policy uncertainty may amplify fluctuations in expectations by affecting both the systematic components (fundamentals) and the unsystematic components (shocks).

Empirically, we distinguish between expected exchange rates and forecast errors, defined as $s_{t+1} - E_t[s_{t+1}]$, and analyze how policy uncertainty influences both. Controlling for a standard set of macroeconomic fundamentals within a VAR framework, we isolate the additional effect of uncertainty.

From a political economy perspective, policy uncertainty reflects imperfect information about policymakers' preferences and future actions. As a result, uncertainty affects expectation formation and may lead to systematic forecast errors. Different dimensions of uncertainty (overall,

monetary, fiscal) can have distinct effects across horizons, influencing both short-run dynamics and long-run exchange rate behavior.

After getting the forecast residuals they estimate a VAR (2) model where:

$$Y_t = [\underbrace{EPU_t}_{news}, \underbrace{\varepsilon_{t+h}}_{for.error}, \underbrace{\tilde{r}_t}_{\text{int. diffe}}, \underbrace{\tilde{m}_t}_{\text{mone-supp diffe}}, \underbrace{\tilde{y}_t}_{\text{out diffe}}]$$

The uncertainty measure display high correlation, therefore they run a different analysis on the different type of policy uncertainty.

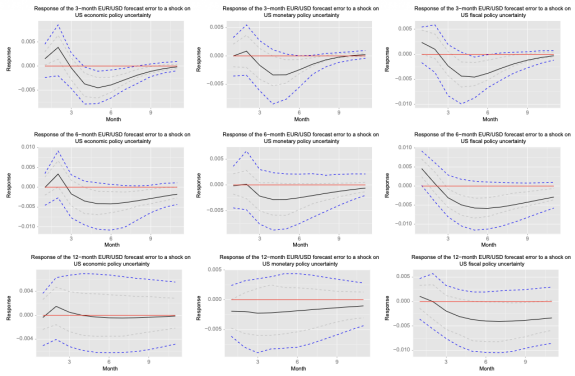


Fig. 4. Response of the EUR/USD forecast error to a shock on US economic policy uncertainty. The plots show the reaction of the EUR/USD forecast error $(s_{t+h}-E_t s_{t+h})$ for $h=3,6,12$) to a shock of the US news-based economic policy uncertainty index and its monetary and fiscal counterparts. The first row gives the response of the forecast error at a three month horizon, the second at a six month horizon, and the third at a twelve month horizon. The first column gives the reaction to a shock of US economic policy uncertainty, the second to US monetary policy uncertainty, and the third to US fiscal policy uncertainty. The reaction shown in units of the response variable is represented by the solid black line and the corresponding confidence bands by dashed lines (the 95% level in blue and the 68% in grey), which represent the quantiles of the estimated bootstrapped impulse response coefficients (see Efron and Tibshirani, 1993, for details). The solid red line displays the zero line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Figure 4.12: Enter Caption

Policy uncertainty has limited and mostly insignificant effects on long-horizon forecast errors, consistent with the dominance of fundamentals in explaining long-run exchange rate dynamics. In the short run, effects are somewhat stronger but generally weak and short-lived. A notable exception is the JPY/USD, where forecast errors significantly decrease following increases in policy uncertainty, suggesting that markets may anticipate policy responses more effectively in this case.

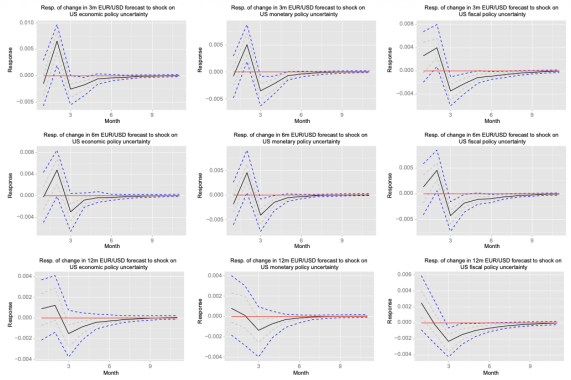


Fig. 8. Response of the change in EUR/USD forecast to a shock on US economic policy uncertainty. The plots show the reaction of the change of the EUR/USD forecast ($\Delta E_t(s_{t+h})$ for $h=3,6,12$) to a shock of the US news-based economic policy uncertainty index and its monetary and fiscal counterparts. The first row gives the response of the forecast error at a three month horizon, the second at a six month horizon, and the third at a twelve month horizon. The first column gives the reaction to a shock of US economic policy uncertainty, the second to US monetary policy uncertainty, and the third to US fiscal policy uncertainty. The reaction shown in units of the response variable is represented by the solid black line and the corresponding confidence bands by dashed lines (the 95% level in blue and the 68% in grey), which represent the quantiles of the estimated bootstrapped impulse response coefficients (see (Efron and Tibshirani, 1993), for details). The solid red line displays the zero line. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Figure 4.13: monetary policy uncertainty

When turning to exchange rate expectations, the impact of policy uncertainty is weaker and less systematic compared to forecast errors. While some short-run positive effects are observed, they are rarely robust at conventional significance levels. Overall, the evidence suggests that market participants do not fully incorporate policy uncertainty into their expectations, leading

to larger forecast errors during periods of high uncertainty. The Japanese yen again stands out as an exception, being consistently expected to appreciate under higher uncertainty, in line with its role as a safe-haven currency.

Table 4
Granger causality analysis.

			EUR/USD			JPY/USD			CAD/USD			GBP/USD		
	h		EPU	MPU	FPU	EPU	MPU	FPU	EPU	MPU	FPU	EPU	MPU	FPU
e_{t+h}	3	(a)	0.0334	0.3172	0.0367	0.9922	0.1744	0.1162	0.0303	0.0566	0.3589	0.2549	0.1143	0.5040
		(b)	0.9418	0.0035	0.0002	0.6829	0.3692	0.7942	0.9375	0.9049	0.6756	0.8481	0.1803	0.2170
	6	(a)	0.0567	0.0020	0.0281	0.3723	0.0498	0.6250	0.0286	0.2667	0.6888	0.1104	0.4145	0.4104
		(b)	0.9785	0.2004	0.2758	0.2995	0.5428	0.0651	0.8599	0.0104	0.1514	0.5391	0.2818	0.4186
	12	(a)	0.6186	0.0734	0.2827	0.3399	0.7859	0.5322	0.0623	0.8379	0.7840	0.1837	0.8437	0.7208
		(b)	0.6171	0.0558	0.2853	0.8978	0.0024	0.3958	0.9579	0.0889	0.1271	0.8398	0.7471	0.3793
$\Delta E_t(s_{t+h})$	3	(a)	0.0008	0.0021	0.0032	0.8922	0.9399	0.2143	0.0127	0.6688	0.3317	0.0346	0.0635	0.1593
		(b)	0.3121	0.7595	0.3907	0.1583	0.1333	0.1936	0.0975	0.0933	0.8274	0.9513	0.4235	0.2678
	6	(a)	0.0468	0.0236	0.0044	0.7519	0.8407	0.3307	0.0237	0.5047	0.2240	0.1146	0.0641	0.3231
		(b)	0.3900	0.2996	0.3471	0.1625	0.0687	0.0814	0.0842	0.0311	0.6640	0.9335	0.5823	0.2770
	12	(a)	0.3177	0.4777	0.0888	0.7071	0.9125	0.5294	0.2263	0.7330	0.3996	0.1599	0.0492	0.4281
		(b)	0.2053	0.9489	0.7179	0.1922	0.2666	0.0759	0.0873	0.0553	0.7780	0.8679	0.3368	0.3157

Note: The table reports p-values for Granger causality tests. The following two null hypotheses have been tested: (a) H_0: US policy uncertainty (EPU, MPU or FPU) does not Granger-cause e_{t+h} and $\Delta E_t(s_{t+h})$, respectively, (b) H_0: e_{t+h} and $\Delta E_t(s_{t+h})$, respectively, does not Granger-cause US policy uncertainty (EPU, MPU or FPU). EPU represents US economic policy uncertainty, MPU denotes US monetary policy uncertainty, FPU gives US fiscal policy uncertainty. e_{t+h} represents the forecast error at horizon h, and $\Delta E_t(s_{t+h})$ denotes the first difference of exchange rate expectations at horizon h.

Figure 4.14

Granger causality results reveal a bi-directional relationship between policy uncertainty and forecast errors, although the strength and significance vary across currencies and horizons. Policy uncertainty Granger-causes forecast errors in several cases, particularly for the EUR/USD and CAD/USD at medium horizons, while no significant effects are found for the GBP/USD.

Conversely, forecast errors also Granger-cause policy uncertainty for most currencies, suggesting that unexpected exchange rate movements may trigger policy responses or increase perceived uncertainty. This reverse causality is generally stronger and more widespread than the forward effect.

Regarding expectations, policy uncertainty exhibits some predictive power, especially for the EUR/USD and GBP/USD at short horizons. However, there is little evidence that expectations Granger-cause uncertainty. Overall, the results suggest that while uncertainty influences expectations and forecast errors, feedback effects from market outcomes to policy uncertainty are also important.

Uncertainty important when market participants form their expectations, in particular in the short-term. Some evidence suggesting long-term effects. Difficult to interpret causality results. Only US uncertainty in the model, country-specific uncertainty may also be important. Interesting paper with potentially important results.

4.5.7 Engle and West (2005)

The paper discussed here addresses the long-standing puzzle in international economics where floating exchange rates are well approximated by a random walk and are difficult to predict using macroeconomic fundamentals. The puzzle they want to solve is the following: **Exchange rates seem to be disconnected from macroeconomic fundamentals**. Following the authors' exchange rate can manifest near random walk behavior if:

- **Fundamentals are persistent**: are I(1) processes or ate least one fundamental has a unit root.
- $\beta \approx 1$: The discount factor in standard exchange rate models is near unity.

Here, if the discount factor is very close to unity, permanent shocks become increasingly important. To validate the link between exchange rates and fundamentals, the author shifts the focus to predicting fundamentals from exchange rates, thereby reversing causality. What they will find is that exchange rates help predict fundamentals, indicating that they contain information about macroeconomic variables, especially nominal ones.

Starting from the flexible price model we can write what follows:

$$s_t = (m_t - m_t^*) - \kappa(y_t - y_t^*) + \theta(i_t - i_t^*)$$

assuming the UIP holds $i_t - i_t^* = \Delta s_{t+1}^e + \rho_t$ and not considering y_t then we get:

$$s_t = m_t + \theta(\Delta s_{t+1}^e + \rho_t)$$

we can then write $b = \theta/(1-\theta)$ therefore the discount factor is related to interest semi-elasticity of money demand hence we get (be aware of annualization, since it must be corrected if data is annualised, otherwise multiply by 400):

$$s_t = (1-b)m_t + b\rho_t + bs_{t+1}^e, \qquad \text{solve diff. eq with no bubbles}$$
$$s_t = (1-b)\sum_{j=0}^{\infty} b^j \mathbb{E}_t m_{t+j} \; + \; b\sum_{j=0}^{\infty} b^j \mathbb{E}_t p_t$$

We notice that such a formula holds for many other asset prices, not only exchange rates such that:

$$s_t = (1-b)\sum_{j=0}^{\infty} b^j \mathbb{E}_t \left(a_1' x_{t+j}\right) + b\sum_{j=0}^{\infty} b^j \mathbb{E}_t \left(a_2' x_{t+j}\right)$$

where x_t is an $n \times 1$ vector of fundamentals, b is the discount factor, and a_1 and a_2 are $n \times 1$ vectors of parameters.

The authors then proceed by generating synthetic data and verifying numerical results. In particular, they consider a model where fundamentals are $x_t \sim I(1)$, so they then consider the first difference in the exchange rate and evaluate the correlation over time and with fundamentals given different levels of b.

TABLE 1
POPULATION AUTOCORRELATIONS AND CROSS CORRELATIONS OF Δs_t

				Correlation of Δs_t with:					
	b (1)	φ_1 (2)	φ (3)	Δs_{t-1} (4)	Δs_{t-2} (5)	Δs_{t-3} (6)	Δx_{t-1} (7)	Δx_{t-2} (8)	Δx_{t-3} (9)
1.	.50	1.0	.3	.15	.05	.01	.16	.05	.01
2.			.5	.27	.14	.07	.28	.14	.07
3.			.8	.52	.42	.34	.56	.44	.36
4.	.90	1.0	.3	.03	.01	.00	.03	.01	.00
5.			.5	.05	.03	.01	.06	.03	.01
6.			.8	.09	.07	.06	.13	.11	.09
7.	.95	1.0	.3	.02	.01	.00	.02	.01	.00
8.			.5	.03	.01	.01	.03	.01	.01
9.			.8	.04	.04	.03	.07	.05	.04
10.	.90	.90	.5	.04	−.01	−.03	.02	−.03	−.05
11.	.90	.95	.5	.05	.01	−.01	.04	−.00	−.02
12.	.95	.95	.5	.02	−.00	−.01	.01	−.02	−.03
13.	.95	.99	.5	.02	.01	.00	.03	.01	−.00

NOTE.—The model is $s_t = (1-b)\sum_{i=0}^{\infty} b^i E_t x_{t+i}$ or $s_t = b\sum_{i=0}^{\infty} b^i E_t x_{t+i}$. The scalar variable x_t follows an AR(2) process with autoregressive roots φ_1 and φ. When $\varphi_1 = 1.0$, $\Delta x_t \sim$ AR(1) with parameter φ. The correlations in cols. 4–9 were computed analytically. If $\varphi_1 = 1.0$, as in rows 1–9, then in the limit, as $b \to 1$, each of these correlations approaches zero.

Figure 4.15: Enter Caption

They see that as $b \to 1$ then predictability shrinks. Now to prove the exchange rate is a random walk, we want to express it as a present value, look at its change and decompose it into a predictable part and a surprise.

1. **Step 1**

   Note that

$$\begin{aligned}
(1-b)E_t \sum_{i=0}^{\infty} b^i f_{t+i} &= (1-b)E_t \left(f_t + bf_{t+1} + b^2 f_{t+2} + \cdots\right) \\
&= E_t \left(f_t - bf_t + bf_{t+1} + b^2 f_{t+2} + \cdots - b^2 f_{t+1} - b^3 f_{t+2} - \cdots\right) \\
&= f_t + E_t \sum_{i=1}^{\infty} b^i (f_{t+i} - f_t) \\
&= f_t + E_t \sum_{i=1}^{\infty} b^i \Delta f_{t+i}.
\end{aligned}$$

This implies that

$$s_t - f_t = E_t \sum_{i=1}^{\infty} b^i \Delta f_{t+i}.$$

From the previous slides we know that

$$s_t = (1-b) f_t + b E_t s_{t+1},$$

which can be rewritten (subtract bs_t from both sides and rearrange) as

$$(1-b) s_t = (1-b) f_t + b E_t s_{t+1} - b s_t.$$

Using the previous result $s_t - f_t = E_t \sum_{i=1}^{\infty} b^i \Delta f_{t+i}$, we obtain

$$E_t \Delta s_{t+1} = \frac{1-b}{b} E_t \sum_{i=1}^{\infty} b^i \Delta f_{t+i}.$$

Interpretation. The spread $s_t - f_t$ equals the discounted sum of expected future changes in f_t. Consequently, expected changes in s_t are proportional to the same discounted expectations, implying that the dynamics of s_t are fully driven by expectations about future movements of f_t.

2. **Step 2**

   Lead

   $$s_t = (1-b) E_t \sum_{i=0}^{\infty} b^i f_{t+i}$$

   one period and take expectation with respect to information available at time t:

   $$E_t s_{t+1} = (1-b) E_t \sum_{i=0}^{\infty} b^i f_{t+1+i}.$$

   The actual exchange rate at time $t+1$, using information available at time $t+1$, is

   $$s_{t+1} = (1-b) E_{t+1} \sum_{i=0}^{\infty} b^i f_{t+1+i}.$$

   Implying that the forecast error is

   $$\begin{aligned} s_{t+1} - E_t s_{t+1} &= (1-b) \sum_{i=0}^{\infty} b^i \left(E_{t+1} f_{t+1+i} - E_t f_{t+1+i} \right) \\ &= \frac{1-b}{b} \sum_{i=1}^{\infty} b^i \left(E_{t+1} f_{t+i} - E_t f_{t+i} \right), \end{aligned}$$

   since

   $$\sum_{i=0}^{\infty} b^i f_{t+1+i} = \frac{1}{b} \sum_{i=1}^{\infty} b^i f_{t+i}.$$

3. **Step 3**

   Combine the results from Step 1 and Step 2 to obtain

$$\Delta s_{t+1} = \frac{1-b}{b} E_t \sum_{i=1}^{\infty} b^i \Delta f_{t+i} + \frac{1-b}{b} \sum_{i=1}^{\infty} b^i \left(E_{t+1} f_{t+i} - E_t f_{t+i} \right).$$

   Interpretation. The actual depreciation rate depends on:

   - expected future changes in fundamentals, and
   - revisions in expectations (forecast errors) of future fundamentals.

Beveridge-Nelson decomposition

The authors move on to show their theorem, assuming a DGP as a Vector Moving Average:

$$\Delta f_t = \Theta(L)\varepsilon_t = \sum_{j=0}^{\infty} \Theta_j \varepsilon_{t-j}$$

where $\Theta_0 = I$ and $\Theta(L) = I - \Theta_1 L + \Theta_2 L^2 + \Theta_3 L^3 + \cdots$ is a polynomial in the lag operator L (i.e., $Lf_t = f_{t-1}$), and ε_t are the forecast errors.

- In the presence of a unit root in f_t, we can rewrite the VMA model as

  $$\Delta f_t = \Theta(1)\varepsilon_t + \Theta^*(L)\varepsilon_t$$

  where

  $$\Theta(1) = I + \sum_{j=1}^{\infty} \Theta_j \quad \text{and} \quad \Theta^*(L) = \Theta(L) - \Theta(1).$$

  Note: If f_t contains a unit root, then $\Theta(1) \neq 0$.

- The decomposition above is called the Beveridge–Nelson decomposition. Its aim is to divide a non-stationary time series into a stochastic trend ($\Theta(1)$) and a stationary cyclical component ($\Theta^*(L)$).

- We can then define:

  $$\Delta f_t = \Delta f_t^T + \Delta f_t^C.$$

  Where T stands for trend and C for cyclical component.

Now, with this in mind, we want to compute forecast errors of fundamentals. Let $\mathbb{P}[\Delta f_{t+i} \mid \mathcal{F}_t]$ denote the projection of future fundamentals Δf_{t+i} conditional on the information set at time t,

$$\mathcal{F}_t = \{\Delta f_t, \Delta f_{t-1}, \dots\}.$$

By definition,

$$\mathbb{E}_t[\Delta f_{t+i}] = \mathbb{P}[\Delta f_{t+i} \mid \mathcal{F}_t].$$

One-step-ahead forecast under the VMA representation.

Using $\Delta f_t = \Theta(L)\varepsilon_t$, we compute:

$$\mathbb{E}_t[\Delta f_{t+1}] = \Theta(L)\mathbb{E}_t[\varepsilon_{t+1}].$$

Since $\mathbb{E}_t[\varepsilon_{t+1}] = 0$ and $\mathbb{E}_t[\varepsilon_t] = \varepsilon_t$, we obtain:

$$\begin{aligned}\mathbb{E}_t[\Delta f_{t+1}] &= (I + \Theta_1 L + \Theta_2 L^2 + \cdots)\mathbb{E}_t[\varepsilon_{t+1}] \\ &= 0 + \Theta_1\varepsilon_t + \Theta_2\varepsilon_{t-1} + \cdots \\ &= (\Theta_1 + \Theta_2 L + \cdots)\varepsilon_t.\end{aligned}$$

Implication for Beveridge–Nelson projections.

Using the Beveridge–Nelson decomposition, forecasts of future fundamentals reduce to projections based on current and past shocks. In particular, the key objects are:

$$\mathbb{E}_{t+1}[f_{t+i}] - \mathbb{E}_t[f_{t+i}] \quad \text{and} \quad \mathbb{E}_t[\Delta f_{t+i}],$$

which depend only on the innovation ε_t and the impulse response matrices $\{\Theta_j\}$. Using the Beveridge–Nelson decomposition,

$$f_t = f_t^C + f_t^T,$$

we can write the revision in expectations as

$$\mathbb{E}_{t+1}f_{t+i} - \mathbb{E}_t f_{t+i} = \left(\mathbb{E}_{t+1}f^C_{t+i} - \mathbb{E}_t f^C_{t+i}\right) + \left(\mathbb{E}_{t+1}f^T_{t+i} - \mathbb{E}_t f^T_{t+i}\right).$$

The key simplification comes from the properties of the trend component:

- **Martingale property of the trend:** The permanent component satisfies

$$\mathbb{E}_t f^T_{t+i} = f^T_t,$$

so the best forecast of the future trend is the current trend.

- **Forecast revision of the trend:** New information at time $t+1$ updates the trend only through the innovation,

$$\mathbb{E}_{t+1}f^T_{t+i} - \mathbb{E}_t f^T_{t+i} = \Delta f^T_{t+1}.$$

Therefore,

$$\mathbb{E}_{t+1}f_{t+i} - \mathbb{E}_t f_{t+i} = \mathbb{E}_{t+1}f^C_{t+i} - \mathbb{E}_t f^C_{t+i} + \Delta f^T_{t+1}.$$

Next, consider expected changes:

$$\mathbb{E}_t[\Delta f_{t+i}] = \mathbb{E}_t[\Delta f^C_{t+i}] + \mathbb{E}_t[\Delta f^T_{t+i}].$$

- **Unpredictability of the trend innovation:** The change in the trend satisfies

$$\Delta f^T_{t+1} = \text{innovation at } t+1,$$

hence

$$\mathbb{E}_t[\Delta f^T_{t+i}] = 0.$$

Thus,

$$\mathbb{E}_t[\Delta f_{t+i}] = \mathbb{E}_t[\Delta f^C_{t+i}].$$

Interpretation: All predictable movements come from the cyclical component, while the trend evolves only through unexpected shocks. Now we find that the change in the exchange rate is given by:

$$\Delta s_{t+1} = \frac{1-b}{b}\sum_{j=1}^{\infty} b^j \mathbb{P}\left[\Delta f^C_{t+j} \mid F_t\right] + \frac{1-b}{b}\sum_{j=1}^{\infty} b^j \left(\mathbb{P}\left[f^C_{t+j} \mid F_{t+1}\right] - \mathbb{P}\left[f^C_{t+j} \mid F_t\right]\right) + \Delta f^T_{t+1}.$$

The **first two terms** on the RHS show the contribution of the cyclical component of the fundamentals. Considering b close to one, we can say that the two terms have minor effects, while the real effect is well approximated by the trend $\Delta f_{t+1} = \Theta(1)\varepsilon_{t+1}$. So if fundamentals have at least one unit root, then Δs_t has a unit root. **Here we show that if we have high persistence, the model does predict unit root effects** so the unpredictability is an implication of the model when the fundamentals have a unit root.

If fundamentals have a unit root ($I(1)$), then they contain *permanent shocks*. As the discount factor $b \to 1$, the exchange rate change satisfies:

$$\Delta s_{t+1} \approx \text{innovation in the trend}$$

Hence:

- Levels (s_t) are highly persistent (non-stationary).
- Differences (Δs_t) are driven only by new shocks.
- Therefore, Δs_t is approximately white noise $\Rightarrow$ low (zero) autocorrelation.

Conclusion: Unpredictability of exchange rate changes is a direct implication of unit root fundamentals, not a contradiction of the model.

Empirical Analysis

They propose an empirical analysis in which they study six currencies over the period from 1974 to 2001, considering four fundamentals (money, output, prices, and interest rates [US relative to the foreign country]). The first step is therefore to test for a unit root, rejecting it for any fundamental. Assuming no cointegration in the data, they test for Granger non-causality in bivariate VAR models.

TABLE 3
BIVARIATE GRANGER CAUSALITY TESTS, DIFFERENT MEASURES OF Δf_t
FULL SAMPLE: 1974:1–2001:3

	Canada	France	Germany	Italy	Japan	United Kingdom
	A. Rejections at 1% (***), 5% (**), and 10% (*) Levels of H_0: Δs_t Fails to Cause Δf_t					
1. $\Delta(m - m^*)$		*		**	**	
2. $\Delta(p - p^*)$			***	***	***	
3. $i - i^*$		**			**	
4. $\Delta(i - i^*)$		**			***	
5. $\Delta(m - m^*) - \Delta(y - y^*)$		*		*		
6. $\Delta(y - y^*)$						
	B. Rejections at 1% (***), 5% (**), and 10% (*) Levels of H_0: Δf_t Fails to Cause Δs_t					
1. $\Delta(m - m^*)$						
2. $\Delta(p - p^*)$	*					
3. $i - i^*$					**	
4. $\Delta(i - i^*)$						
5. $\Delta(m - m^*) - \Delta(y - y^*)$						
6. $\Delta(y - y^*)$						

NOTE.—See the notes to earlier tables for variable definitions. Statistics are computed from fourth-order bivariate VARs in $(\Delta s_t, \Delta f_t)'$. Because four observations were lost to initial conditions, the sample generally is 1975:2–2001:3, with exceptions as indicated in the note to table 2.

Figure 4.16: Enter Caption

In the table above, we observe that we reject the null of non-causality between the exchange rate and fundamentals, meaning that, in most cases, the exchange rate helps predict fundamentals. In panel B, they examine whether fundamentals help predict exchange rates, but report that all coefficients are Granger non-causal with respect to the exchange rate.

4.5.8 Summary of Theoretical and Empirical Results

Engel and West (2005) provide a dual-layered resolution to the Meese-Rogoff puzzle by combining analytical proofs with new empirical evidence.

Theoretical Results: The Near-Random Walk Theorem. The authors prove that in a standard rational expectations present-value model, an exchange rate (s_t) will manifest near-random walk behavior if: (1) the underlying economic fundamentals (f_t) are highly persistent, specifically $I(1)$, and (2) the discount factor (b) applied to future fundamentals is near unity. Analytically, as $b \to 1$, the model places increasing weight on long-term expectations, causing the variance of the exchange rate change to be dominated by unpredictable innovations (news)

TABLE 4
BIVARIATE GRANGER CAUSALITY TESTS, DIFFERENT MEASURES OF Δf_t, EARLY PART OF SAMPLE: 1974:1–1990:2

	Canada	France	Germany	Italy	Japan	United Kingdom
	A. Rejections at 1% (***), 5% (**), and 10% (*) Levels of H_0: Δs_t Fails to Cause Δf_t					
1. $\Delta(m - m^*)$		**		*		
2. $\Delta(p - p^*)$			**	***	**	
3. $i - i^*$		***				*
4. $\Delta(i - i^*)$		***			**	**
5. $\Delta(m - m^*) - \Delta(y - y^*)$		**		**		
6. $\Delta(y - y^*)$					**	
	B. Rejections at 1% (***), 5% (**), and 10% (*) Levels of H_0: Δf_t Fails to Cause Δs_t					
1. $\Delta(m - m^*)$						
2. $\Delta(p - p^*)$	*		*			
3. $i - i^*$						
4. $\Delta(i - i^*)$						
5. $\Delta(m - m^*) - \Delta(y - y^*)$						
6. $\Delta(y - y^*)$						

NOTE.—See the notes to the earlier tables for variable definitions.

TABLE 5
BIVARIATE GRANGER CAUSALITY TESTS, DIFFERENT MEASURES OF Δf_t, LATER PART OF SAMPLE: 1990:3–2001:3

	Canada	France	Germany	Italy	Japan	United Kingdom
	A. Rejections at 1% (***), 5% (**), and 10% (*) Levels of H_0: Δs_t Fails to Cause Δf_t					
1. $\Delta(m - m^*)$		**			***	
2. $\Delta(p - p^*)$	*	***	*			
3. $i - i^*$			*		**	**
4. $\Delta(i - i^*)$			**		**	**
5. $\Delta(m - m^*) - \Delta(y - y^*)$		*				**
6. $\Delta(y - y^*)$					*	
	B. Rejections at 1% (***), 5% (**), and 10% (*) Levels of H_0: Δf_t Fails to Cause Δs_t					
1. $\Delta(m - m^*)$			**		**	
2. $\Delta(p - p^*)$		***		**		
3. $i - i^*$					***	
4. $\Delta(i - i^*)$		**			***	
5. $\Delta(m - m^*) - \Delta(y - y^*)$			**		**	
6. $\Delta(y - y^*)$		*				

NOTE.—See the notes to the earlier tables for variable definitions.

Figure 4.17: Enter Caption

rather than past data. Numerical simulations confirm that with a realistic discount factor of $b \geq 0.9$, the autocorrelation of exchange rate changes becomes statistically negligible, explaining why fundamentals fail to forecast exchange rates in short horizons.

Empirical Results: Exchange Rates as Leading Indicators. Shifting the empirical focus from exchange rate predictability to the information content of prices, the authors use G7 data (1974–2001) to demonstrate that the link between variables exists in the opposite direction. Their Granger causality tests show that exchange rates possess significant power to predict future fundamentals, particularly nominal variables such as inflation and money growth. Furthermore, they demonstrate that exchange rates act as efficient aggregators of information; incorporating the exchange rate into a forecasting information set significantly improves the correlation between the model's predicted present value and the actual exchange rate. This suggests that while exchange rates appear to move randomly, they are correctly functioning as forward-looking asset prices that incorporate expectations of future macroeconomic conditions.

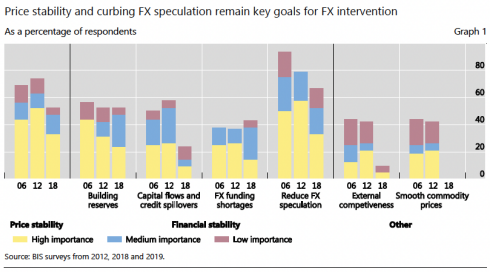
Chapter 5

Central Bank Intervention

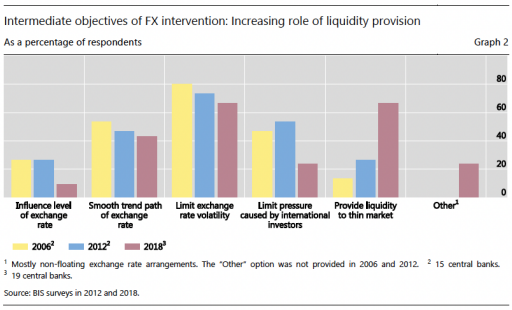
5.1 Introduction

The monetary authorities buy or sell foreign exchange to influence exchange rates. But is that the only reason behind central bank intervention? The three main goals for a Central Bank are:

- **Price stability Financial Stability**: building reserves, capital flow and credit spillovers, reduce speculation on FX.
- **Others**: competitiveness, smooth commodity prices.
- **Defend Fixed Exchange Rate**



(a) First image caption



(b) Second image caption

Figure 5.1: Overall caption for both images

- **Conceptual Distinction:** The authors distinguish between *goals* (the ultimate purpose, such as price stability) and *intermediate objectives* (operational targets, such as smoothing the exchange rate path).
- **Primary Goals:** Key motivations for intervention include maintaining price stability, curbing excessive speculation, and ensuring financial stability by accumulating reserves or alleviating funding shortages.
- **Operational Objectives:** Intermediate objectives are set to operationalize these goals, with restricting exchange rate volatility and providing liquidity to thin markets being identified as the most critical targets.
- **Regional Trends:** The survey data indicates an increasing role for liquidity provision and a relative decline in intervening for external competitiveness compared to previous years.

5.1.1 Central Bank Balance Sheet

We now start analysing the Central Bank Balance Sheet to understand what the components are that the Central Bank uses for its own policies. We start by reporting the balance sheet, which is organised into two main sections: **Assets** and **Liabilities**.

Table 5.1: Central Bank Balance Sheet

Assets	**Liabilities**
Official international reserves	Deposits of private financial institutions (Bank reserves)
Domestic credit	Currency in circulation
• Government bonds	Other
• Loans to domestic financial institutions	
Other	

Notes: Domestic credit affects money supply through open market operations. The sum of currency in circulation and bank reserves is defined as the **monetary base** (or **base money**).

Once the main components of the balance sheet have been identified, we can distinguish between the two benchmark forms of foreign exchange intervention: **non-sterilised** and **sterilised** interventions.

- **Non-Sterilised Intervention.** A non-sterilised intervention occurs when the central bank conducts foreign exchange operations without offsetting them through domestic asset transactions, thereby allowing the monetary base to change.

  Consider the case in which the Federal Reserve aims to depreciate the dollar relative to the yen. To achieve this, the central bank purchases yen from a commercial bank. On the balance sheet, this implies an increase in foreign reserves on the asset side and a corresponding increase in deposits of financial institutions (i.e., reserves) on the liability side. Formally,

  $$\Delta MB = \Delta R > 0,$$

  where reserves R are a component of the monetary base. Hence, the monetary base expands one-for-one with the increase in reserves.

- **Sterilised Intervention.** A sterilised intervention combines a foreign exchange operation with an offsetting domestic open market operation, leaving the monetary base unchanged.

  Following the previous example, after purchasing foreign currency (which increases reserves), the central bank sells domestic government bonds to absorb liquidity. This reduces reserves by the same amount, so that

  $$\Delta MB = 0.$$

  In effect, the central bank alters the composition of its balance sheet by replacing domestic assets with foreign assets, without changing the overall supply of base money.

Non-sterilised interventions affect the monetary base and are therefore equivalent to conventional monetary policy actions. As a result, there is broad consensus regarding their effectiveness.

In contrast, sterilised interventions leave the monetary base unchanged, and their effectiveness is more controversial. Their potential impact operates through several channels:

- **Portfolio balance channel**

- **Signalling channel**

- **Order flow channel**

5.2 Portfolio Balance Model

We now introduce the *portfolio balance model*, originally developed by **McKinnon** in the 1970s and 1980s as a framework to analyse the joint determination of interest rates and exchange rates in the presence of portfolio reallocations by both private agents and central banks.

The model focuses on short-run dynamics and therefore operates under the standard assumption of price and output rigidity, i.e., prices and real output are taken as fixed. Within this setting, exchange rate movements are driven by changes in the relative supplies of and demands for domestic and foreign assets.

FIGURE: The operation of the portfolio balance model

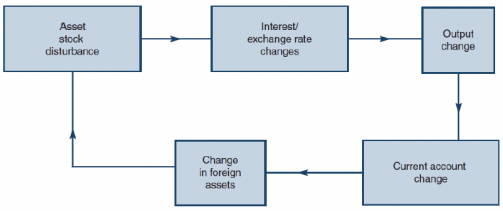


Figure 5.2

In the short run the $\Delta CA \approx 0$ in order to isolate pure financial effects, which we assume to produce immediate effects

$$CA = NX + NI + NT$$

(where NX Net exports; NI Net income from abroad (interest, dividends, wages; NT Net transfers (remittances, aid)).

Assumptions In the following model, we impose the following assumptions which are:

- **Short Run**: prices and output fixed.
- **Small Open Economy**
- **3 Assets**: domestic bonds, foreign bonds and money.
- **Free capital mobility**
- **Not perfect substitutability** between domestic and foreign bonds. This would allow us to analyse the model's rebalancing procedure for exchange rates and interest rates.

Important identities Here, we formulate important identities which will be fundamental to our analysis.

- **Private Wealth**: Total private wealth is given by all the assets held by the private component as follows:

$$W = M + \underbrace{B_P}_{\text{domestic bonds}} + \underbrace{SF_P}_{\text{for. bonds converted}}$$

Portfolio sheet of private

- **Domestic Money**: s equal to the authorities holding domestic and foreign bonds:

$$M = B_a + SR$$

(remember the asset liability scheme) **Portfolio sheet of central bank**

- **Domestic Bonds** are assumed to be fixed (since determined by fiscal policy) and to be held both by private and authorities:

$$\bar{B} = B_a + B_p$$

- **Foreign Bonds** are not fixed since foreign is big economy:

$$F = F_p + R$$

Therefore, we can open up the elements and then obtain:

$$W = B + SF$$

Private wealth is the total holdings of money and foreign assets, converted into domestic currency.

Now we want to characterise the functions of domestic bonds, foreign bonds, and holding money, in particular we have:

- $M = m(r, \mathbb{E}\dot{s}, W, Y), \quad m_r < 0,\ m_{\mathbb{E}\dot{s}} < 0,\ m_W > 0,\ m_Y > 0$ if there is an increase in s so a depreciation then we would have that the opportunity cost of holding money decreases so agents buy foreign assets so they switch money to foreign assets letting M decrease.

- $B_p = b(r, \mathbb{E}\dot{s}, W, Y), \quad b_r > 0,\ b_{\mathbb{E}\dot{s}} < 0,\ b_W > 0,\ b_Y < 0$ as $Y \uparrow$ increase transaction motive and therefore we decrease the bonds relative to money.

- $SF_p = f(r, \mathbb{E}\dot{s}, W, Y), \quad f_r < 0,\ f_{\mathbb{E}\dot{s}} > 0,\ f_W > 0,\ f_Y < 0$

Now we write the equation for the current account, assuming it is unchanged and equal to 0. We get therefore:

$$CA = \frac{dF}{dt} = \dot{F}_p = T(\frac{S}{P}, Y) + r^*(F_p + R)$$

The Portfolio Constraint and Total Differentiation in the Portfolio Balance Model The wealth identity

$$W \equiv M(r, E\dot{s}, W, Y) + B_p(r, E\dot{s}, W, Y) + SF_p(r, E\dot{s}, W, Y)$$

is an *accounting identity* that holds for all values of its arguments. Since each asset demand depends on $(r, E\dot{s}, W, Y)$, any change in these variables must preserve the equality. This is captured by *total differentiation*.

Total Differentiation Taking the total differential of the identity:

$$dW = dM + dB_p + d(SF_p),$$

and applying the chain rule:

$$dM = M_r dr + M_{E\dot{s}} dE\dot{s} + M_W dW + M_Y dY,$$

with analogous expressions for dB_p and $d(SF_p)$. Substituting:

$$dW = (M_r + B_r + F_r)dr + (M_{E\dot{s}} + B_{E\dot{s}} + F_{E\dot{s}})dE\dot{s} + (M_W + B_W + F_W)dW + (M_Y + B_Y + F_Y)dY.$$

Rearranging:

$$\big[1 - (M_W + B_W + F_W)\big]dW = (M_r + B_r + F_r)dr + (M_{E\dot{s}} + B_{E\dot{s}} + F_{E\dot{s}})dE\dot{s} + (M_Y + B_Y + F_Y)dY.$$

Since the identity must hold for arbitrary variations, coefficients must satisfy consistency conditions.

Step 1: Differentiation with respect to Wealth W Holding $(r, E\dot{s}, Y)$ fixed, we isolate the dW terms:

$$1 = M_W + B_W + F_W,$$

or

$$\boxed{m_W + b_W + f_W = 1}.$$

Interpretation: A marginal increase in wealth must be fully allocated across all assets. This is the **portfolio constraint on the margin**.

Step 2: Differentiation with respect to the Interest Rate r Holding $(W, E\dot{s}, Y)$ fixed implies $dW = 0$, $dE\dot{s} = 0$, $dY = 0$. Then:

$$0 = M_r + B_r + F_r,$$

or

$$\boxed{m_r + b_r + f_r = 0}.$$

Interpretation: A change in r redistributes portfolio shares without affecting total wealth. Hence, increases in one asset must be offset by decreases in others.

Step 3: Differentiation with respect to Expected Depreciation $E\dot{s}$ Holding (W, r, Y) fixed implies:

$$0 = M_{E\dot{s}} + B_{E\dot{s}} + F_{E\dot{s}},$$

or

$$\boxed{m_{E\dot{s}} + b_{E\dot{s}} + f_{E\dot{s}} = 0}.$$

Interpretation: Expected depreciation reallocates wealth across assets without changing its total level.

Economic Implications These conditions imply:

$$b_r = -(m_r + f_r) > 0, \qquad f_{E\dot{s}} = -(m_{E\dot{s}} + b_{E\dot{s}}) > 0,$$

showing that each asset responds most strongly to its own return. This reflects **gross substitutability**: assets compete for portfolio shares, and changes in returns induce reallocations across them.

5.2.1 Equilibrium In Portfolio Balance Model

Since the main argument of this model is to analyse how interest rates and exchange rates vary with different portfolio allocations, we would like a graphical representation on the interest-exchange rate plane. Here we assume that in equilibrium, all components do not change, hence:

$$d\mathbb{E}\dot{s} = dM = dB_p = dF_p = 0$$

Now we can take the total differentiation of $W = M + B_p + SF_p$ so we obtain:

$$dW = dM + \underbrace{dB_p}_{=0} + F_p dS + S \underbrace{dF_p}_{=0}$$

Notice that variations with respect to output Y are omitted, as Y is assumed fixed in the short run.

Money Market Schedule Starting from the money demand function

$$M = m(r, \mathbb{E}\dot{s}, W, Y),$$

its total differential is given by

$$dM = m_r\, dr + m_W\, dW + m_{E\dot{s}}\, d\mathbb{E}\dot{s}.$$

In equilibrium, we assume $\mathbb{E}\dot{s}$ is fixed, so $d\mathbb{E}\dot{s} = 0$. Moreover, using the wealth identity,

$$dW = F_p\, dS.$$

Imposing money market equilibrium ($dM = 0$), we obtain

$$0 = m_r\, dr + m_W F_p\, dS.$$

Rearranging:

$$\frac{dS}{dr} = -\frac{m_r}{m_W F_p} > 0,$$

since $m_r < 0$ and $m_W > 0$.
Interpretation: An increase in the interest rate reduces money demand, requiring an exchange rate depreciation (increase in S) to restore equilibrium via an increase in wealth.

Domestic Bond Market Schedule Consider the domestic bond demand function

$$B_p = b(r, \mathbb{E}\dot{s}, W, Y).$$

Its total differential is

$$dB_p = b_r\, dr + b_W\, dW + b_{E\dot{s}}\, d\mathbb{E}\dot{s}.$$

Again, holding $\mathbb{E}\dot{s}$ fixed and using $dW = F_p\, dS$, we have

$$dB_p = b_r\, dr + b_W F_p\, dS.$$

Imposing equilibrium in the bond market ($dB_p = 0$), we obtain

$$0 = b_r\, dr + b_W F_p\, dS,$$

which implies

$$\frac{dS}{dr} = -\frac{b_r}{b_W F_p} < 0,$$

since $b_r > 0$ and $b_W > 0$.
Interpretation: An increase in the interest rate raises the demand for domestic bonds. To restore equilibrium, the exchange rate must appreciate (decrease in S), reducing wealth and thus bond demand.

Foreign Bond Market Schedule Finally, consider the foreign bond demand function

$$SF_p = f(r, \mathbb{E}\dot{s}, W, Y).$$

Taking the total differential:

$$F_p\, dS = f_r\, dr + f_W\, dW + f_{E\dot{s}}\, d\mathbb{E}\dot{s}.$$

Holding $\mathbb{E}\dot{s}$ fixed and substituting $dW = F_p\, dS$:

$$F_p\, dS = f_r\, dr + f_W F_p\, dS.$$

Rearranging:

$$(1 - f_W)F_p\, dS = f_r\, dr,$$

so that

$$\frac{dS}{dr} = \frac{f_r}{(1 - f_W)F_p} < 0,$$

since $f_r < 0$ and $0 < f_W < 1$.
Interpretation: An increase in the interest rate reduces the demand for foreign bonds. To restore equilibrium, the exchange rate must appreciate, reducing the domestic value of foreign assets.

Now, assuming that the slope of the bond market schedule is steeper than the one of the foreign bond $-\frac{b_r}{b_W} < \frac{f_r}{1-f_W}$, we get a **STABLE EQUILIBRIUM** not otherwise.

FIGURE: Equilibrium in the portfolio balance model.

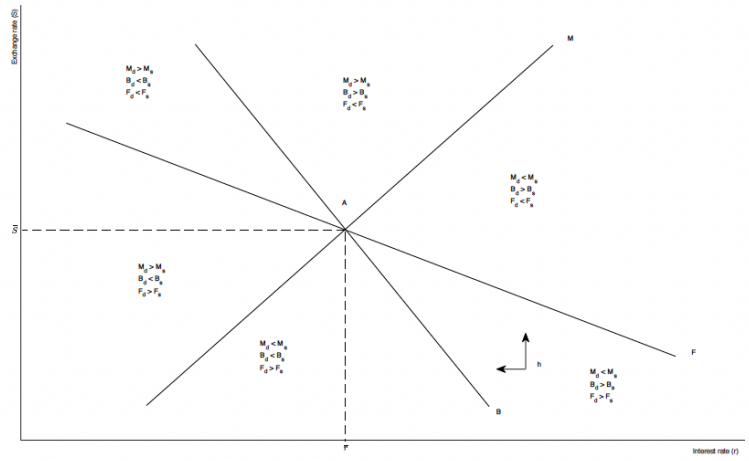


Figure 5.3: Enter Caption

- **Money schedule**: we see here that for a given interest rate a depreciation would mean that foreign assets appreciate, therefore $W \uparrow = M + B_p + SF_p \uparrow$ which implies that $M_d \uparrow = (r, E\dot{s}, W \uparrow, Y) > M_s \ fixed.$
- **Bond schedule** for the same reason, we have if we have a depreciation, then foreign assets are worth more, so more wealth $W \uparrow$ hence $\uparrow B_p = B_p(r, E\dot{s}, \uparrow W, Y)$.
- **Foreign Bond Schedule**: In this case considering same level of exchange rate an increase in the interest rate would make the internal bonds more competitive so there would be a shift from foreign bonds to domestic bonds causing $f_d < f_s$.

We can now solve the portfolio balance model, written as follows:

$$dB_p = b_r\, dr + b_w\, dW \tag{5.1}$$
$$dM = m_r\, dr + m_w\, dW \tag{5.2}$$
$$dF_p = f_r\, dr + f_w\, dW - F_p\, dS \tag{5.3}$$

using algebra notation we get:

$$\begin{bmatrix} dB_p \\ S\, dF_p \\ dM \end{bmatrix} = \begin{bmatrix} b_r & b_w & 0 \\ f_r & f_w & -F_p \\ m_r & m_w & 0 \end{bmatrix} \begin{bmatrix} dr \\ dW \\ dS \end{bmatrix}$$

and by inverting the matrix we can solve for the endogenous variable so:

$$\begin{bmatrix} dr \\ dW \\ dS \end{bmatrix} = \begin{bmatrix} \dfrac{m_w}{b_r m_w - b_w m_r} & 0 & -\dfrac{b_w}{b_r m_w - b_w m_r} \\ -\dfrac{m_r}{b_r m_w - b_w m_r} & 0 & \dfrac{b_r}{b_r m_w - b_w m_r} \\ \dfrac{f_r m_w - f_w m_r}{F_p\,(b_r m_w - b_w m_r)} & -\dfrac{1}{F_p} & \dfrac{b_r f_w - b_w f_r}{F_p\,(b_r m_w - b_w m_r)} \end{bmatrix} \begin{bmatrix} dB_p \\ S\, dF_p \\ dM \end{bmatrix}$$

Open Market Operation

In an open market operation, the central bank purchases domestic bonds from the private sector and injects money into the economy. This implies

$$dM = -dB_p > 0, \qquad S\,dF_p = 0,$$

since the central bank increases its holdings of domestic bonds $dB_a > 0$, and given the bond supply identity $\bar{B} = B_a + B_p$, it follows that $dB_p < 0$, i.e., the private sector holds fewer domestic bonds. No foreign bonds are involved.

FIGURE: The effects of an open market operation.

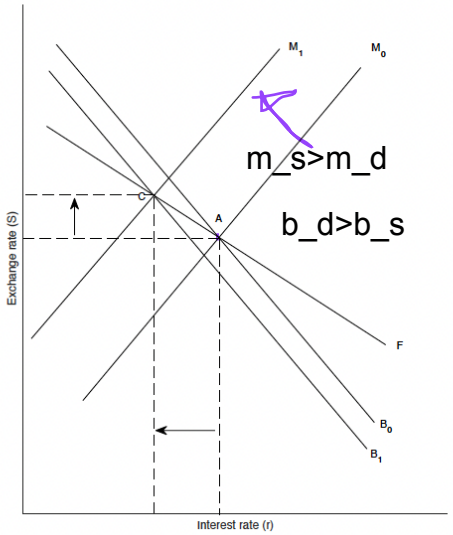


Figure 5.4: Effects of an open market operation in the Portfolio Balance Model.

Consider the initial equilibrium at point A, where all three asset markets clear. The operation disturbs two of the three equilibrium schedules simultaneously. In the money market, the increase in money supply generates an excess supply of money at the initial (r, S):

$$M_s > M_d.$$

This shifts the M-curve to the left, from M_0 to M_1: at any given exchange rate S, equilibrium in the money market now requires a lower interest rate than before. In the domestic bond market, the central bank has absorbed part of the bond supply, reducing private sector holdings. This creates excess demand for domestic bonds at the initial (r, S):

$$B_d > B_s,$$

which shifts the B-curve downward, from B_0 to B_1: at any given S, equilibrium in the bond market now requires a lower r. The F-curve is unaffected, since $dF_p = 0$. Agents rebalance their portfolios in response. Excess money holdings are allocated across both domestic and foreign assets according to relative returns. In the domestic bond market, the combined effect of increased demand for bonds (from excess money) and reduced bond supply drives bond prices up and the interest rate sharply down. At the same time, part of the excess liquidity flows into foreign assets, increasing the demand for foreign currency and generating a depreciation $S \uparrow$. The new equilibrium is the intersection of M_1, B_1 and F, characterized by a **large fall in** r and a **small depreciation of the exchange rate**. The interest rate effect dominates because both the M-curve and B-curve shift in the same direction, creating strong downward pressure on r.

The depreciation is small because only the fraction of excess liquidity directed toward foreign assets — rather than the total money injection — drives the exchange rate, reflecting portfolio rebalancing across all assets rather than a complete shift away from domestic bonds.

Analytical Solution The operation implies $dM = -dB_p > 0$ and $S\,dF_p = 0$. Substituting into the matrix solution:

Interest rate:

$$dr = -\frac{F_p(m_w + b_w)}{D}\,dM < 0$$

since $m_w + b_w = 1 - f_w > 0$, $F_p > 0$, $D > 0$.

Wealth:

$$dW = \frac{-f_r F_p}{D}\,dM > 0$$

where we used Walras's Law: $m_r + b_r = -f_r > 0$.

Exchange rate:

$$dS = \frac{-f_r(m_W + b_W) + f_W(m_r + b_r)}{b_r m_W - b_W m_r} dM$$

we recall that $b_r + m_r = -f_r$ and $m_W + b_W + f_W = 1$ so we can get:

$$dS = \frac{-f_r(1 - f_W) - f_W f_r}{b_r m_W - b_W m_r} dM > 0$$

since $f_r < 0$, confirming a depreciation of the domestic currency.

The interest rate effect is large since both m_w and b_w appear in the numerator, reflecting simultaneous pressure from the money and bond markets. The depreciation is small since only f_r drives it, capturing the fraction of excess liquidity directed toward foreign assets.

5.2.2 Non-sterilized Foreign Exchange Intervention

In a non-sterilized foreign exchange operation, the central bank purchases foreign assets (foreign bonds or foreign currency) from the private sector and injects domestic money into the economy. This implies

$$dM > 0, \qquad dR > 0.$$

so basically we have that:

$$dM = -SdF_p = SdR$$

meaning that the monetary authorities buy foreign assets up to the value of the money supply injected into the market, increasing their reserves.

FIGURE: The effects of a non–sterilized foreign exchange operation.

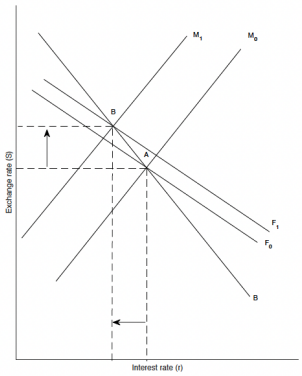


Figure 5.5

Assume the economy is initially in equilibrium. At the initial interest rate and exchange rate, the increase in money supply generates an excess supply of money:

$$M_s > M_d.$$

Agents, therefore, rebalance their portfolios by reducing excess money holdings and increasing their demand for both domestic and foreign assets. In particular, the demand for foreign bonds increases:

$$F_d > F_s.$$

This creates excess demand for foreign assets, which requires an increase in the exchange rate (a depreciation of the domestic currency) to restore equilibrium. The depreciation raises the domestic currency value of foreign assets, helping to satisfy the higher demand.

At the same time, part of the portfolio adjustment takes place through the domestic bond market, where excess demand for bonds pushes up their price and lowers the interest rate:

$$r \downarrow .$$

The new equilibrium is therefore characterized by a lower interest rate and a higher exchange rate.

5.2.3 Sterilized Foreign Exchange Intervention

A sterilized intervention is a two-stage operation in which the central bank first buys foreign bonds — injecting money exactly as in a non-sterilized intervention — and then immediately sells domestic bonds to drain the money it just created, leaving the money supply unchanged.

1. **Stage 1 — FX leg:** The central bank buys foreign bonds from the private sector and pays with money:
$$dM = -S\,dF_p = S\,dR > 0.$$
2. **Stage 2 — Sterilization leg:** The central bank sells domestic bonds to the private sector, withdrawing the money just created:
$$-dM = dB_p > 0.$$

The two stages cancel in the money market, so the net effect is:

$$dM = 0, \qquad S\,dF_p = -dB_p < 0,$$

meaning that only the *composition* of private portfolios changes: the private sector holds **more domestic bonds and fewer foreign bonds** than before.

FIGURE: The effects of a sterilized foreign exchange operation.

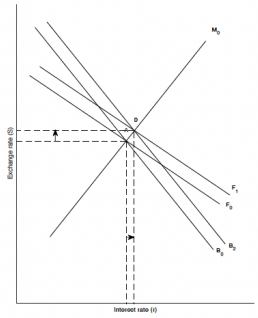


Figure 5.6: Effects of a sterilized foreign exchange operation in the Portfolio Balance Model.

This portfolio rebalancing activates two simultaneous forces. First, the reduction in foreign bond holdings creates excess demand for foreign assets. Since the remaining foreign bonds in private portfolios must be held willingly, the exchange rate must rise to increase their domestic currency value — the domestic currency **depreciates**, $S \uparrow$. The F-curve therefore shifts upward from F_0 to F_1.

Second, the increase in domestic bond supply creates excess supply in the domestic bond market. Bond prices fall and, since $r = C/P_B$, the interest rate **rises**, $r \uparrow$. The B-curve shifts upward from B_0 to B_1. The M-curve is unaffected since $dM = 0$.

The new equilibrium at the intersection of B_1, F_1 and M is thus characterized by a **higher interest rate and a depreciated exchange rate**. Crucially, both effects are smaller than under a non-sterilized intervention, since the money channel is shut down by construction and only the portfolio rebalancing channel operates.

	Δr	ΔS
Open market operation ($\Delta M = -\Delta B_p$)	$-$ (large)	$+$ (small)
Non-sterilized foreign exchange operation ($\Delta M = -S\Delta F_p$)	$-$ (small)	$+$ (large)
Sterilized foreign exchange operation ($-S\Delta F_p = \Delta B_p$)	$+$	$+$

5.2.4 Bond-Financed Fiscal Expansion

We study the effects of fiscal policy on exchange rates and interest rates within the portfolio balance model. The government can finance an increase in expenditure in two ways.

- **Borrowing from the Central Bank (Money-Financed Deficit):** Both money and wealth increase by the amount of the deficit:

$$dM = dW = d\text{DF} > 0, \qquad dB_p = 0, \qquad dF_p = 0.$$

The money injection creates an excess supply of money, shifting the M-curve to the left. At the same time, the rise in wealth generates excess demand for both domestic and foreign bonds: wealthier agents wish to hold more of every asset. The B-curve shifts downward (excess demand for domestic bonds lowers r) and the F-curve shifts upward (excess demand for foreign bonds depreciates S). The M-curve also shifts leftward due to the excess money supply.

FIGURE: Effects of a money-financed government budget deficit.

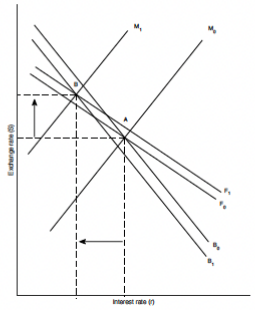


Figure 5.7: Effects of a money-financed government budget deficit.

The total effect is unambiguous: the interest rate **falls** and the exchange rate **depreciates**. All three curves shift in the same direction, making both effects large.

- **Borrowing from the Public (Bond-Financed Deficit):** Domestic bonds and wealth increase by the amount of the deficit:

$$dB_p = dW = d\text{DF} > 0, \qquad dM = 0, \qquad dF_p = 0.$$

The money supply is completely unchanged. Two simultaneous forces are activated.

First, the increase in bond supply creates an **excess supply of domestic bonds** at the initial (r, S). Bond prices fall and the interest rate rises. This shifts the B-curve upward.

Second, the increase in wealth generates excess demand for both money and foreign bonds ($m_W > 0$, $f_W > 0$). The excess demand for money — with supply unchanged — requires r to rise to clear the money market, shifting the M-curve rightward. The excess demand for foreign bonds puts upward pressure on S, shifting the F-curve upward.

The interest rate effect is **unambiguous**: both the bond supply increase and the wealth-driven money demand increase push r upward. However, the exchange rate effect is **ambiguous**, as two competing forces act on S:

– **Wealth effect** ($f_W > 0$): higher wealth raises demand for foreign bonds ⇒ upward pressure on S (depreciation).
– **Substitution effect** ($f_r < 0$): the rise in r makes domestic bonds more attractive, crowding out demand for foreign bonds ⇒ downward pressure on S (appreciation).

FIGURE: Effects of a money-financed government budget deficit.

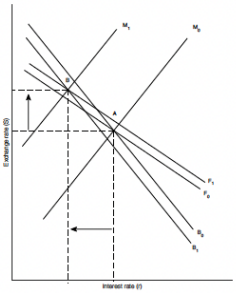
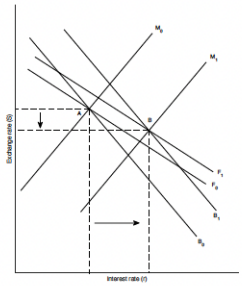


FIGURE: Effects of a bond-financed government budget deficit.



(a) First image (b) Second image

Figure 5.8: Overall caption

The outcome depends on the relative magnitude of f_r:

– If f_r is **small** (wealth effect dominates): the F-curve shifts up substantially and the exchange rate **depreciates**, $S \uparrow$.
– If f_r is **large** (substitution effect dominates): the F-curve barely shifts and the exchange rate may **appreciate**, $S \downarrow$.

The total effect is therefore: $\Delta r > 0$ always, and ΔS is **positive or negative** depending on which effect dominates.

Summary. The key distinction between the two financing methods is the following. Money financing injects liquidity, lowering r and unambiguously depreciating the currency. Bond financing instead raises r unambiguously — by flooding the bond market and increasing money demand — but leaves the exchange rate direction theoretically indeterminate, depending on whether the wealth effect on foreign bond demand dominates the substitution effect driven by the higher interest rate.

5.2.5 Risk Premium, Imperfect and perfect substitutability

So far, we have assumed that foreign and domestic bonds are imperfect substitutes, and we have assumed that the change in expected future changes in the exchange rates is zero, so static expectations. Using UIP we have:

$$r - r^* = \mathbb{E}\dot{s} + \rho$$

taking the first difference we get:

$$dr - dr^* = d\mathbb{E}\dot{s} + d\rho$$

Assuming SOE, then $dr^* = 0$ and so also $d\mathbb{E}\dot{s} = 0$, therefore we get that the changes in the interest rate are completely driven by the changes in the risk premium.

Now, assuming perfect substituability and therefore ruling out the risk premium, we have that changes in the domestic rate are completely tied to changes in the foreign rate, so the domestic and foreign rates behave equally; therefore, their schedules are the same.

In this case, if we consider a sterilised policy, M remains unchanged, and so does the B/F schedule (it does not create any disequilibrium since the bonds are the same).

Figure 8: Perfect substitutability of domestic and foreign bonds and expansionary monetary policy.

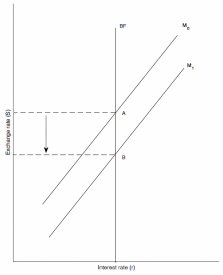


Figure 5.9: Enter Caption

5.3 Long-Run Portfolio Balance Model

We now extend the portfolio balance framework to the long run by relaxing the restriction that the current account must be zero. The law of motion for the net foreign asset position of the private sector is

$$CA_t \;=\; \dot{F}_{p,t} \;=\; \frac{dF_{p,t}}{dt} \;=\; T\Big(\tfrac{S_t}{P_t},\, Y_t\Big) \;+\; r^*\big(F_{p,t} + R_t\big), \tag{5.4}$$

where

- $F_{p,t}$ is the stock of net foreign assets held by the private sector;
- $T\left(\frac{S_t}{P_t}, Y_t\right)$ is the trade balance, a function of the real exchange rate S_t/P_t and domestic income Y_t;
- r^* is the (exogenous) world real interest rate;
- R_t denotes the stock of official reserves held by the central bank.

The first term captures net exports; the second is net investment income earned on the country's net foreign asset position $F_{p,t} + R_t$.

Sign restrictions and Marshall–Lerner. Under the Marshall–Lerner condition, a real depreciation improves the trade balance:

$$\frac{\partial T}{\partial (S/P)} > 0, \qquad \frac{\partial T}{\partial Y} < 0. \tag{5.5}$$

A rise in S/P makes domestic goods relatively cheaper, raising exports and compressing imports; higher domestic income raises imports and worsens T.

Economic interpretation. Equation (5.4) makes explicit that the current account has two components: a flow component (the trade balance) and a stock-driven component (interest income on the net foreign asset position). When $CA_t > 0$, domestic savings exceed domestic investment and the economy is a net lender to the rest of the world, so $\dot{F}_{p,t} > 0$. This accumulation feeds back into the asset-market block of the model: a larger supply of foreign bonds in private portfolios requires an adjustment in relative asset prices (and thus in S_t and i_t) to restore portfolio equilibrium.

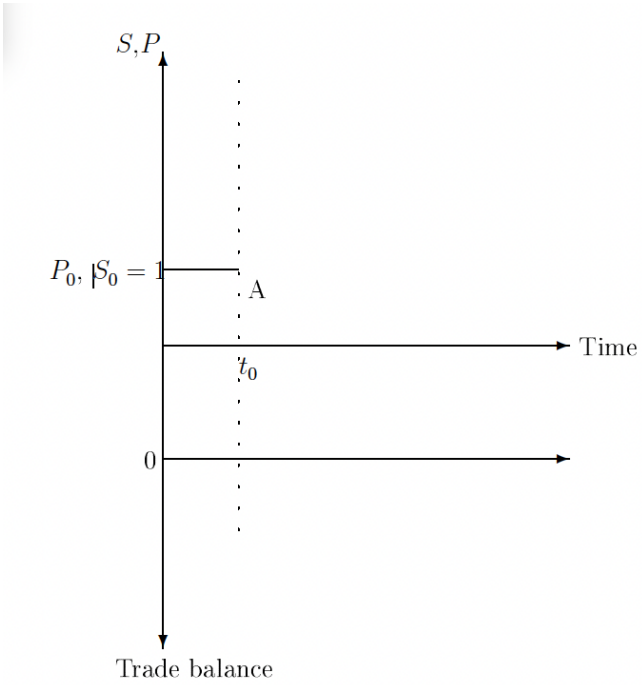


Figure 5.10: Initial long-run equilibrium with $S_0 = P_0 = 1$ and $CA_0 = T_0 = 0$.

Dynamics after a money-financed fiscal expansion

Consider an economy starting from full equilibrium: the three asset markets clear, net foreign investment income is zero, and the trade balance is zero, so that $CA_{t_0^-} = 0$. We normalize $S_0 = P_0 = 1$.

At time t_0 the government finances a budget deficit by printing money (an open-market purchase of domestic bonds). The short-run portfolio-balance analysis implies:

(i) The domestic interest rate falls, $i \downarrow$.

(ii) The exchange rate depreciates on impact, $S_1 > S_0$ (jump from A to C).

(iii) With prices sticky in the very short run, $S_1/P_0 > 1$: the real exchange rate depreciates, so by (5.5) the trade balance improves, $T_1 > 0$ (movement from F to G in Figure 5.11; note the downward-pointing arrow).

Because $T_1 > 0$ and, initially, $r^* F_{p,t_0} = 0$, equation (5.4) implies $\dot{F}_{p,t} > 0$: the country accumulates net foreign assets. The resulting excess supply of foreign bonds in private portfolios induces households to rebalance toward domestic bonds, which *appreciates* the currency gradually (S falls from S_1 toward S_2, movement from C to E). The appreciation erodes competitiveness, so the trade balance deteriorates (movement from G to H).

Meanwhile, the money-supply increase gradually feeds into prices: P rises from P_0 to the new long-run level $P_1 > P_0$.

The new long-run equilibrium (point E). In the long run, the real exchange rate returns to its initial (neutral) level, but at a *different* nominal configuration:

$$\frac{S_2}{P_1} < \frac{S_1}{P_0}, \qquad \text{with } S_2 < S_1 \text{ and } P_1 > P_0. \tag{5.6}$$

At point E two conditions must hold simultaneously:

(a) $CA_t = 0$ (no further accumulation of foreign assets),

(b) $F_p^{\text{new}} > F_{p,0}$, so $r^* F_p^{\text{new}} > 0$.

From (5.4), $CA = 0$ with $r^* F_p > 0$ requires

$$T\left(\frac{S_2}{P_1}, Y\right) \;=\; -\,r^* F_p^{\text{new}} \;<\; 0, \tag{5.7}$$

i.e. a *trade-balance deficit* in the long run financed by positive net interest income from abroad. Hence the new long-run real exchange rate must be *more appreciated* than the initial one (the country can "afford" to be less competitive because it earns rents on its foreign assets).

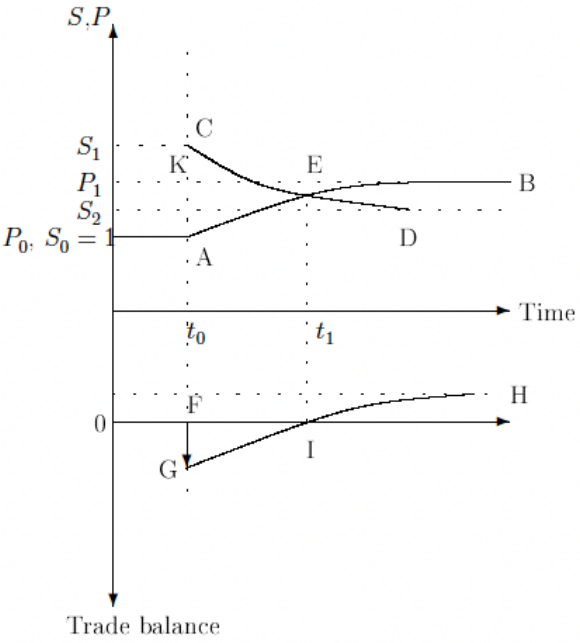


Figure 5.11: Full dynamics: jump depreciation, gradual appreciation via portfolio rebalancing, and long-run price-level adjustment. The path of S crosses the path of P, producing a *real* overshooting that is distinct from the Dornbusch (sticky-price) mechanism.

Overshooting without sticky prices. The model delivers exchange-rate overshooting ($S_1 > S_2$) as in Dornbusch, but the economic driver is different:

- In **Dornbusch**, overshooting arises because prices are *sticky*: the money-market equilibrium requires a large impact jump in S to generate the expected appreciation consistent with UIP.
- In the **portfolio-balance model**, overshooting arises from the *stock–flow interaction* between the current account and foreign-asset accumulation: even if prices adjusted instantaneously, the initial trade surplus would induce an accumulation of foreign assets, which then forces the currency to appreciate gradually toward the new long-run level.

Overshooting in this framework is therefore a *portfolio* phenomenon, not a *price-rigidity* phenomenon.

Summary of the new steady state. Relative to the initial equilibrium, the long-run effects of a money-financed budget deficit are:

Variable	Long-run change
Nominal exchange rate S	depreciated ($S_2 > S_0$)
Price level P	higher ($P_1 > P_0$)
Real exchange rate S/P	appreciated relative to t_0
Nominal interest rate i	lower
Trade balance T	deficit
Net foreign assets F_p	higher
Current account CA	zero

5.4 Signaling Channel

In the following section, we elaborate on the idea that official intervention serves as a signal of future monetary policy by providing the foreign exchange market with new relevant information.

Given the idea that **the exchange rate is a function of expected future discounted fundamentals**, signals affect expectations and therefore affect exchange rates.

Model Setup

We make the following baseline assumptions:

- **The central bank possesses superior information and can credibly reveal it to the market by intervening in the foreign exchange market.**
- **Two-country model consistent with the standard flexible-price monetary model.**
- **Perfect Credibility of the Monetary Policy**

Monetary Model

We specify the domestic and foreign money demand functions:

$$\text{DOM:} \quad m_t - p_t = \alpha_1 y - \alpha_2 r_t \tag{5.8}$$

$$\text{FOR:} \quad m_t^* - p_t^* = \alpha_1 y^* - \alpha_2 r_t^* \tag{5.9}$$

where we assume symmetric parameters across countries (lower-case letters denote logs throughout).

We impose two standard conditions:

- **Uncovered Interest Parity (UIP):** $r_t = r_t^* + E_t[s_{t+1}] - s_t$
- **Purchasing Power Parity (PPP):** $p_t = p_t^* + s_t$

Under these assumptions households hold only their own currency, and UIP eliminates portfolio-balance effects—so *sterilised intervention has no direct effect on asset prices.* We also assume rational expectations, so $s_{t+1}^e = E_t[s_{t+1}]$.

Solving for the Exchange Rate

Step 1: Solve for the price level. From (5.8):

$$p_t = m_t - \alpha_1 y + \alpha_2 r_t$$

Step 2: Substitute PPP. Inserting PPP ($p_t = p_t^* + s_t$) and subtracting the foreign counterpart:

$$s_t = (m_t - m_t^*) - \alpha_1 (y - y^*) + \alpha_2 \big(E_t[s_{t+1}] - s_t \big)$$

Step 3: Rearrange. Collecting s_t terms on the left-hand side:

$$(1 + \alpha_2)\, s_t = \underbrace{(m_t - m_t^*) - \alpha_1 (y - y^*)}_{\equiv f_t} + \alpha_2\, E_t[s_{t+1}]$$

Dividing through by $(1 + \alpha_2)$:

$$s_t = \frac{1}{1 + \alpha_2}\, f_t + \frac{\alpha_2}{1 + \alpha_2}\, E_t[s_{t+1}]$$

where $f_t \equiv (m_t - m_t^*) - \alpha_1 (y - y^*)$ collects current fundamentals.

Step 4: Forward iteration (repeated substitution). Apply the same logic to $E_t[s_{t+1}]$, then to $E_t[s_{t+2}]$, and so on. Letting $\lambda \equiv \frac{\alpha_2}{1+\alpha_2} \in (0,1)$, after n iterations:

$$s_t = \frac{1}{1+\alpha_2} \sum_{k=0}^{n} \lambda^k E_t[f_{t+k}] + \lambda^{n+1} E_t[s_{t+n+1}]$$

Imposing the **no-bubbles (transversality) condition**:

$$\lim_{n\to\infty} \lambda^{n+1} E_t[s_{t+n+1}] = 0$$

yields the *fundamental solution*:

$$\boxed{S_t = \frac{1}{1+\alpha_2} \sum_{k=0}^{\infty} \left(\frac{\alpha_2}{1+\alpha_2}\right)^k E_t\big(M_{t+k} - M^*_{t+k} - \alpha_1(Y - Y^*)\big)} \tag{5.10}$$

Interpretation

Equation (5.10) expresses the current exchange rate as a *present discounted value* of expected future monetary fundamentals, with discount factor $\frac{\alpha_2}{1+\alpha_2}$. The key implication for the signaling channel follows immediately: **any intervention that revises market expectations about future money supplies** (M_{t+k}, M^*_{t+k}) **or output differentials will move** S_t **today**, even if the intervention is fully sterilised and leaves current money supplies unchanged.

5.4.1 No Intervention of Monetary Authority

Under the assumption that the money supply is constant over time, so $M_{0,t} = M_0$ and $M^*_{0,t} = M^*_0$ we have that: the term $E_t(M_0 - M^*_0 - \alpha_1(Y - Y^*)) = C$ constant, therefore the multiplier $(\frac{\alpha_2}{1+\alpha_2})^k$ for $k \to \infty$ is given by $\frac{1+\alpha_2}{1}$, so we obtain $S_0 = M_0 - M^*_0 + \alpha_1(Y - Y^*)$

5.4.2 Intervention of the Monetary Authority in Period 2

We now assume that the **central bank intervenes in period 1 solely to signal that monetary policy will change in period 2**. Crucially, no change in the money supply occurs in period 1, so current fundamentals remain unchanged. The announced policy at $t = 2$ is:

$$M_2 = M_0 + \gamma I, \qquad M^*_2 = M^*_0 + \gamma I^*$$

where $\gamma > 0$ governs the sensitivity of future monetary policy to the intervention signal. A positive domestic intervention $I > 0$ means the domestic central bank *buys* foreign currency, while $I^* > 0$ means the foreign central bank *sells* foreign currency.

Step 1: Exchange rate at $t = 2$. From $t = 2$ onwards monetary policy is held constant at M_2, M^*_2, so the exchange rate collapses to the static monetary solution (the geometric sum evaluates to unity):

$$S_2 = M_2 - M^*_2 - \alpha_1(Y - Y^*)$$

Substituting $M_2 = M_0 + \gamma I$ and $M^*_2 = M^*_0 + \gamma I^*$, and recalling that $S_0 = M_0 - M^*_0 - \alpha_1(Y - Y^*)$, we obtain:

$$S_2 = \underbrace{M_0 - M^*_0 - \alpha_1(Y - Y^*)}_{= S_0} + \gamma(I - I^*) = S_0 + \gamma(I - I^*)$$

Taking the rational expectation formed at $t = 1$ (the signal is perfectly credible):

$$E_1[S_2] = S_0 + \gamma(I - I^*) \tag{5.11}$$

Step 2: Exchange rate at $t = 1$. In period 1 the money supplies are still $M_1 = M_0$ and $M_1^* = M_0^*$, so current fundamentals are identical to those at $t = 0$. Using the one-period forward-looking form of the monetary model:

$$S_t = \frac{1}{1+\alpha_2}\big(M_t - M_t^* - \alpha_1(Y - Y^*)\big) + \frac{\alpha_2}{1+\alpha_2}\, E_t[S_{t+1}]$$

and setting $t = 1$, we substitute the current fundamental (which equals S_0) and the expectation from (5.11):

$$\begin{aligned} S_1 &= \frac{1}{1+\alpha_2}\underbrace{\big(M_0 - M_0^* - \alpha_1(Y - Y^*)\big)}_{=S_0} + \frac{\alpha_2}{1+\alpha_2}\big(S_0 + \gamma(I - I^*)\big) \\ &= S_0 \underbrace{\left(\frac{1}{1+\alpha_2} + \frac{\alpha_2}{1+\alpha_2}\right)}_{=1} + \frac{\alpha_2}{1+\alpha_2}\,\gamma(I - I^*) \end{aligned}$$

$$\boxed{S_1 = S_0 + \frac{\alpha_2}{1+\alpha_2}\,\gamma(I - I^*)} \tag{5.12}$$

Summary of results. The three equilibrium exchange rates are collected in Table 5.2.

Period	Exchange Rate	Change from S_0	Adjustment
$t = 0$	$S_0 = M_0 - M_0^* - \alpha_1(Y - Y^*)$	0	Initial equilibrium
$t = 1$	$S_1 = S_0 + \frac{\alpha_2}{1+\alpha_2}\,\gamma(I - I^*)$	$\frac{\alpha_2}{1+\alpha_2}\,\gamma(I - I^*)$	Partial
$t = 2$	$S_2 = S_0 + \gamma(I - I^*)$	$\gamma(I - I^*)$	Full (long-run)

Table 5.2: Exchange rate equilibria under a credible signalling intervention. The domestic currency depreciates when $I > I^*$.

Interpretation. Several key results emerge from equations (5.11)–(5.12) and Table 5.2:

- **Immediate effect without changing fundamentals.** The exchange rate adjusts at $t = 1$ even though the current money supply is unchanged. The intervention operates *purely through expectations*: rational agents observe the central bank's action and immediately update their beliefs about future monetary policy.
- **Partial adjustment in period 1.** The period-1 response is discounted by $\frac{\alpha_2}{1+\alpha_2} \in (0, 1)$ relative to the full long-run change $\gamma(I - I^*)$. This discount factor is exactly the weight the model places on expected future exchange rates; the market moves towards, but does not yet reach, the new equilibrium because the policy change lies one period ahead.
- **Full adjustment from period 2 onwards.** Once the announced monetary policy is implemented, S_2 absorbs the complete change $\gamma(I - I^*)$. The exchange rate path is therefore monotonically increasing between $t = 1$ and $t = 2$: $S_0 < S_1 < S_2$ (for $I > I^*$).
- **Role of α_2.** A larger interest semi-elasticity of money demand α_2 pushes $\frac{\alpha_2}{1+\alpha_2}$ closer to unity, so the period-1 adjustment is larger. Intuitively, the more forward-looking the model, the more the present exchange rate is driven by expected future fundamentals.

- **Credibility and coordination.** The result holds only if the signal is credible ($\gamma > 0$ is believed). If credibility is partial, γ is effectively scaled down and the effect is proportionally weaker. Moreover, coordinated intervention by both central banks—$I > 0$ and $I^* < 0$ simultaneously—maximises $(I - I^*)$ and therefore produces the largest exchange rate movement.

5.5 Empirical Evidence

5.5.1 Fatum and Hutchison (2003)

Motivation and Research Question

fatum2003 ask whether sterilised foreign exchange intervention systematically affects exchange rates in the short run. The authors argue that the widespread failure to find evidence of effectiveness is a *methodological artefact*: standard time-series methods (VAR, GARCH) are ill-suited to the sporadic, clustered nature of intervention. They instead apply an **event study methodology** — standard in corporate finance for earnings announcements and M&A events — to daily Bundesbank and Fed intervention in the DEM/USD market over **1 September 1985 – 31 December 1995**, following the Plaza Agreement.

Event Definition and Window Structure

Rather than treating each intervention day as a separate observation, the authors define an *event* as a cluster of same-direction intervention days, allowing up to **15 consecutive no-intervention days** within a single episode. This yields **32 distinct events**, of which 26 are classified as *leaning-against-the-wind* (intervention opposing the pre-event trend) and 6 as leaning-with-the-wind. Each event window has three components:

1. **Pre-event window** (Δs_{i-}): the exchange rate trend prior to intervention, used to classify the central bank's policy stance.

2. **Event window**: the full intervention episode, from first to last day of central bank activity.

3. **Post-event window** (Δs_{i+}): exchange rate movement after the central bank exits the market, capturing the cumulative policy effect.

Windows of 2, 5, 10, and 15 days are considered for robustness, with the **2-day window** as the baseline, as it guarantees no overlap between adjacent events.

Success Criteria

Let E_i denote net intervention in event i (positive = USD purchase) and $\Delta s_{i\pm}$ the average daily DEM/USD percentage change in the post/pre-event window. Three criteria assess effectiveness:

Direction criterion. Intervention is successful if the post-event exchange rate moves in the intended direction:

$$\text{Success} \iff (E_i > 0 \text{ and } \Delta s_{i+} > 0) \text{ or } (E_i < 0 \text{ and } \Delta s_{i+} < 0). \tag{5.13}$$

Smoothing criterion. Intervention is successful if it satisfies the direction criterion *or* at least slows the prevailing movement:

$$\text{Success} \iff (E_i > 0 \text{ and } \Delta s_{i+} > \Delta s_{i-}) \text{ or } (E_i < 0 \text{ and } \Delta s_{i+} < \Delta s_{i-}). \tag{5.14}$$

Reversal criterion. To avoid spurious success when the central bank leans with the wind, the direction criterion is applied exclusively to leaning-against-the-wind events, where intervention explicitly opposes the pre-existing trend:

$$\text{Leaning against the wind} \iff (E_i > 0 \text{ and } \Delta s_{i-} < 0) \text{ or } (E_i < 0 \text{ and } \Delta s_{i-} > 0). \quad (5.15)$$

Statistical Tests

Two tests are employed. The **non-parametric sign test** treats the number of successes $X \sim \text{Binomial}(n, 0.5)$ under the null of no systematic effect, assessing whether observed successes exceed the random benchmark. For the smoothing criterion, the null probability is set to $p = 0.75$, reflecting the unconditional smoothing rate in non-event periods. The **matched sample t-test** tests for a significant mean shift in exchange rate changes between pre- and post-event windows. Defining $D_i = \Delta s_{i+} - \Delta s_{i-}$, the statistic

$$t = \frac{\bar{D}}{\hat{\sigma}_D/\sqrt{n}} \sim t(n-1) \quad (5.16)$$

is computed under $H_0 : \mu_D = 0$. To account for excess kurtosis in exchange rate returns, White (1980) heteroscedasticity-consistent standard errors are used as a robustness check, yielding identical inference.

Results

The results strongly support short-run effectiveness of sterilised intervention. Under the **direction criterion**, 27 of 32 events are successful, rejecting randomness at the 1% level. Disaggregating, all 14 DEM-purchase events succeed ($p < 0.01$), as do 13 of 18 DEM-sale events ($p < 0.05$). Under the **reversal criterion** — the most stringent test, restricted to leaning-against-the-wind episodes — 24 of 26 events reverse the prior trend, again significant at the 1% level. Under the **smoothing criterion**, all 26 leaning-against-the-wind events are successful, rejecting randomness at the 1% level. The matched sample test corroborates these findings: for DEM-purchase (DEM-sale) events, the mean pre-to-post shift in the daily exchange rate change is -1.26 ($+0.62$) percentage points, with t-statistics of -19.78 and 13.81 respectively, both significant at the 1% level.

These findings are robust across all window lengths of 2, 5, 10, and 15 days, and survive exclusion of events coinciding with central bank interest rate changes (21 "pure intervention" events), with significance maintained in 9 of 12 cases. Excluding small-scale interventions below USD 100 million leaves conclusions unchanged. Notably, larger-volume and longer-duration events are associated with larger subsequent exchange rate movements, and all six events lasting more than 40 days are successful. However, extending the post-event window to **30 days** eliminates all evidence of effectiveness, confirming that the impact of sterilised intervention is **strictly short-run**.

5.5.2 Fratzscher 2008

The main idea of the paper is to study whether **communications** or **Oral interventions** of the central bank are effective. The reason this is interesting is the current trend among monetary policy institutions to shift from direct intervention to communication, thereby working on market expectations. Communication is therefore important since it can influence exchange rates through:

- **Signalling Channel**: statements about exchange rates give a signal about future monetary policy.

- **Coordination**: Public statements work as a coordination device that induces markets to converge or move toward a particular direction.

The paper develops on daily communications of policy makers of **US, Japan, EA** from 1990-2003, they analyse public statements about exchange rates by gathering material from Reuters News and tracking daily exchange rate quotes at closing prices. Then they proceed with a classification which is given by:

$$IO_t = \begin{cases} +1 & \text{if "strengthening" oral statement} \\ -1 & \text{if "weakening" oral statement} \\ 0 & \text{if "ambiguous" oral statement} \end{cases}$$

They provide some descriptive statistics whic are summarized as follows:

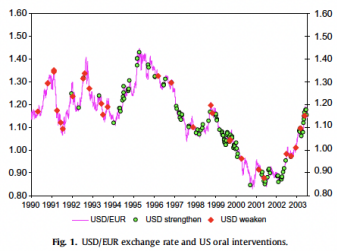


Fig. 1. USD/EUR exchange rate and US oral interventions.

(a) US: interventions often supported a strong dollar even near its peak.

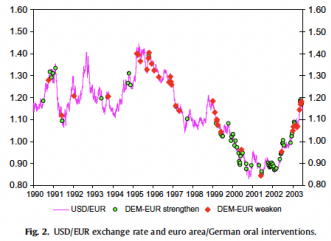


Fig. 2. USD/EUR exchange rate and euro area/German oral interventions.

(b) Euro area/Germany: general support for strong currency with occasional deviations.

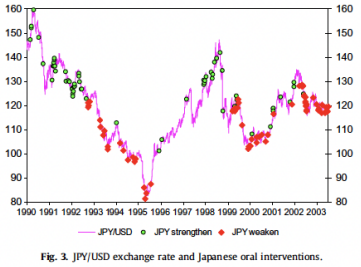


Fig. 3. JPY/USD exchange rate and Japanese oral interventions.

(c) Japan: policy shifts, alternating between strong and weak yen stance.

Figure 5.12: Exchange rates and oral interventions across countries.

This shows how actually policy makers have comunicated to the market given different policy stands:

- **USA**: almost always pursued to strengthen the dollar even during periods of a strong dollar.
- **EA**: Supporting strength of Euro even if with a lesser degree.
- **Japan**: Change in policy regimes.

The US sells foreign currency, thereby injecting foreign currency into the market and buying USD, which puts upward pressure on the dollar (appreciation).

We see here that the US and EA have stopped pursuing direct intervention after 1999, instead pursuing a communication policy. Now the model he presents is given by the following. The exchange rate is a **present value of expected future fundamentals**:

$$s_t = (1-\theta)\sum_{i=0}^{\infty} \theta^i E_t[f_{t+i} \mid \Omega_t]$$

where Ω_t is the public information set at time t and $\theta \in (0,1)$ is a discount factor. Since s_t already reflects all available information, any **change** in the spot rate must be driven by **news**, i.e., revisions to expectations about future fundamentals. Imposing UIP with a risk premium ρ, i.e., $E_t[\Delta s_{t+1} \mid \Omega_t] = i_t - i_t^* + \rho$, one can decompose:

$$\Delta s_{t+1} = \underbrace{(i_t - i_t^* + \rho)}_{\text{expected change}} + \underbrace{(1-\theta)\sum_{i=0}^{\infty} \theta^i \Big(E_{t+1}[f_{t+1+i} \mid \Omega_{t+1}] - E_t[f_{t+1+i} \mid \Omega_t] \Big)}_{\text{unexpected change (news)}}$$

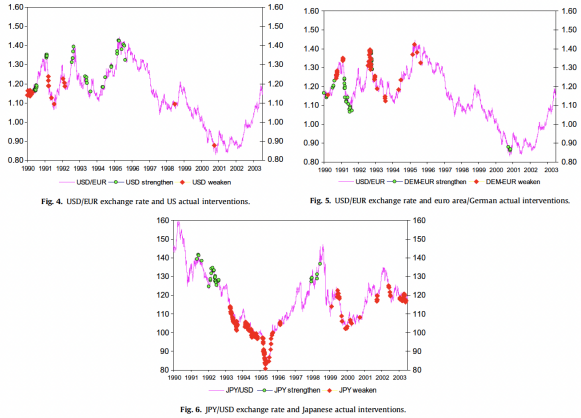


Figure 5.13: Enter Caption

The unexpected component captures all information arriving *after* t but *before* $t+1$. Central bank interventions affect s_t through three channels:

1. **Portfolio balance:** actual interventions shift the risk premium ρ
2. **Signaling:** oral and actual interventions alter expectations about future fundamentals directly
3. **Coordination:** information is not common knowledge; trading aggregates private information and makes it public

The empirical methodology that he pursues considers an EGARCH model, which solves the problem of non-normal, negative-skewed exchange rate returns.

Fratscher (2008) – Empirical Results Summary

Econometric Setup

The model is a **GARCH specification** where intervention variables enter both the conditional mean equation (do interventions move the level of s_t?) and the conditional variance equation (do they reduce exchange rate volatility?). The sample covers 1 January 1990 – 30 September 2003, split at 31 December 1998 (pre- vs. post-1999).

Results

The empirical analysis is based on a **GARCH specification** where intervention variables enter both the conditional mean and conditional variance equations, covering the period 1990–2003 and split at 1999.

Baseline effectiveness. Interventions significantly move exchange rates in the intended direction. For the USD/EUR, GE/EA actual interventions are the most powerful ($\hat{\beta} = 1.536^{***}$), while both US and GE/EA oral interventions are also significant. For the JPY/USD, US actual interventions have a large effect ($\hat{\beta} = 1.287^{***}$) and Japanese oral interventions are consistently significant. Crucially, however, actual interventions *increase* rather than reduce conditional variance, indicating that interventions are **destabilizing in volatility**. Effectiveness declines markedly post-1999 for both currency pairs, consistent with deeper and more liquid FX markets absorbing interventions more easily.

Deviations from the policy mantra. Interventions that **deviate from the prevailing exchange rate mantra** are significantly more effective than those consistent with it. US oral interventions deviating from the strong-dollar policy carry a much larger mean effect (H_0 : -1.162^{***}) compared to non-deviating ones (H_1 : -0.041), with the two coefficients statistically different. GE/EA actual interventions show the same asymmetry (H_0 : 1.725^{***} vs. H_1 : 1.175^{**}). This strongly supports the **signaling channel**: an intervention that signals a genuine shift in policy intentions conveys more information and hence moves markets more.

Signaling channel I: future monetary policy. Interventions consistent with the **direction of future monetary policy** are significantly more powerful. GE/EA oral interventions aligned with upcoming rate changes reach $\hat{\beta} = 0.270^{***}$, while misaligned ones lose significance. For the JPY/USD, Japanese actual interventions consistent with the current monetary policy trend are particularly impactful ($\hat{\beta} = 3.737^{***}$). Credibility is therefore central: an intervention reinforcing the monetary policy stance conveys credible information about future fundamentals and is priced in accordingly.

Signaling channel II: future actual interventions. Oral interventions that **precede actual interventions** within a two-week window are significantly more effective than those not followed by actual operations. GE/EA oral interventions with prior signaling reach $\hat{\beta} = 0.222^{***}$, while Japanese oral interventions with signaling are -0.234^{***}, with H_0 and H_1 statistically different. Oral interventions therefore derive much of their power from functioning as a **credible commitment device** for future actual operations.

Coordination channel: market uncertainty and positioning. Interventions have a **larger impact when FX market uncertainty is high**, proxied by both historical EGARCH volatility and implied volatility from options. US oral interventions in high-uncertainty periods are highly significant ($\hat{\beta} = -0.284^{***}$ historical; $\hat{\beta} = -0.184^{***}$ implied), while their low-uncertainty counterparts are negligible. GE/EA actual interventions exhibit the same asymmetry (H_0 : 1.637^{***} vs. H_1 : 0.067). A complementary proxy – market **positioning** via risk reversals and exchange rate trends – confirms that interventions are more powerful when markets are strongly positioned in one direction. Both results support the coordination channel: when beliefs are dispersed and uncertainty is elevated, an intervention acts as a *focal point* that coordinates expectations, generating a larger price impact.

Overall conclusion. Three main results emerge. First, interventions move exchange rates in the intended direction but increase rather than reduce volatility. Second, the **signaling channel** is the dominant mechanism: effectiveness is highest when interventions deviate from the policy mantra, align with monetary policy, or credibly signal future actual operations. Third, the **coordination channel** is empirically relevant: interventions are most powerful precisely when high uncertainty and one-sided market positioning make a credible signal most informative.

5.6 Danish Monetary Policy

Danmarks Nationalbank's monetary policy pursues a single, unambiguous objective: to maintain the Danish krone's fixed parity against the euro. This institutional arrangement reflects a deliberate application of the Mundell-Fleming trilemma — by committing to both free capital mobility and a fixed exchange rate, Denmark effectively surrenders monetary policy autonomy. As a direct consequence, the krone imports the price stability target of the European Central Bank, meaning Danish inflation dynamics are anchored externally rather than determined by domestic monetary conditions.

Within this framework, fiscal policy assumes a compensating role. The Danish government conducts its budgetary policy in a manner consistent with the exchange rate constraint, accepting that macroeconomic stabilization must operate through fiscal rather than monetary instruments. This reflects a clear institutional division of responsibilities: the central bank defends the peg, while the government manages aggregate demand.

The Nationalbank disposes of two instruments to manage pressure on the krone: foreign exchange intervention — the purchase or sale of foreign currency reserves — and adjustments to the monetary policy interest rate. To some extent, these tools act as substitutes in the short run. A mild appreciation or depreciation pressure may be absorbed through intervention alone, without altering the interest rate. However, this substitutability is limited: when pressure on the krone is sustained, intervention depletes reserves without addressing the underlying cause, and a change in the interest rate becomes necessary. This asymmetry is consistent with the predictions of the Portfolio Balance Model, in which sterilized intervention—which leaves the money supply unchanged—has limited long-run effectiveness precisely because it does not alter the monetary stance that ultimately governs capital flows and exchange rate expectations.

Summary: Spange & Sørensen (2016) — Effects of Danmarks Nationalbank's Interventions in the Foreign Exchange Market

Context and Motivation

Denmark operates a fixed exchange rate regime against the euro within the European Exchange Rate Mechanism (ERM2), with an official fluctuation band of $\pm 2.25\%$ around the central rate of DKK 746.038 per 100 EUR — though in practice deviations are far smaller. Danmarks Nationalbank's (DN) primary instrument to defend the peg is foreign exchange (FX) market intervention; interest rate adjustments constitute the second step of its monetary policy reaction function. The paper evaluates the effectiveness of DN's FX interventions over the period 2002–2016, motivated by the severe speculative pressure episode of January–February 2015.

Institutional Facts and Data

- DN intervened on approximately **12% of all trading days** between January 2002 and October 2016.
- There were **162 intervention events** in total, with an average duration of 3.1 days. The longest event lasted 28 days.
- Average intervention size: DKK 3.0bn overall (DKK 2.5bn for krone purchases, DKK 3.6bn for krone sales).
- Largest single-day intervention: DKK 35.4bn (January–February 2015).
- In January–February 2015 alone, DN sold DKK 275bn to resist krone appreciation, while cutting the deposit rate to -0.75%.

Transmission Channels

DN's interventions operate through two theoretical channels:

Portfolio Balance Channel. By altering the relative supply of krone-denominated assets, interventions shift relative asset prices, provided domestic and foreign assets are imperfect substitutes. This channel is operative regardless of whether market participants observe the intervention.

Signalling Channel. Interventions may signal DN's dissatisfaction with the exchange rate trend and induce market participants to revise expectations about future monetary policy rates. Empirically, the signal value is substantial: on a random day, the probability of a change in DN's monetary policy spread within the next five banking days is only **5%**. However, on days when DN intervenes, this probability rises to **21%**, and to **15%** if intervention occurred on any of the preceding five days.

Empirical Methodology

Event Study

Following Fatum & Hutchison (2003), the paper defines two success criteria evaluated over a pre-event window ($t-2$ to $t-1$) and a post-event window ($t+1$ to $t+2$):

- **Direction Criterion:** The exchange rate moves in the direction intended by the intervention in the post-event window.
- **Stabilisation Criterion** (weaker): The exchange rate trend weakens relative to the pre-event window, even if the direction is not fully reversed.

Regression Model with IV Estimation

A daily regression is estimated:

$$\Delta \textit{Krone rate}_t = \text{const} + \beta_1 \, \textit{Interventions}_t + \beta_2 \, \textit{Customer purchases}_t + \varepsilon_t$$

where $\Delta \textit{Krone rate}_t$ is the daily percentage change (DKK per 100 EUR), and $\textit{Interventions}_t$ denotes DN's net krone purchases in DKK billion. Since DN's interventions respond endogenously to intraday exchange rate developments, **instrumental variable (IV) estimation** is employed: the instrument is DN's interventions on the preceding five trading days, which predicts same-day intervention (due to the clustered nature of events) but has no independent effect on today's exchange rate change. Newey–West standard errors correct for autocorrelation and heteroscedasticity.

Main Results

Event Study

- **All 162 events**: stabilisation criterion met in $> 80\%$ of cases; direction criterion met in $\sim 75\%$.
- **"Leaning against the wind" events without concurrent interest rate changes** (94 events): stabilisation criterion met in **90%** of cases; direction criterion met in **68%**.
- **Statistical test** (binomial, H_0: interventions have no effect): the p-value for the stabilisation criterion is **0.0001%** and for the direction criterion is **0.03%**. The null of no effect is overwhelmingly rejected.

Regression Results

- A DN krone purchase of **DKK 1bn strengthens the krone by approximately 0.005%**, equivalent to roughly **DKK 0.40 per 100 EUR for a DKK 10bn intervention**. The effect is statistically significant.

- The estimated effect of DN's interventions is slightly *larger* than the effect of equivalent bank customer purchases. Since customer flows operate only through the portfolio balance channel, the additional impact of DN's interventions is attributed to the **signalling channel**.
- Across sub-periods, interventions were effective in all regimes, but the magnitude varied:

Sub-period	Relative Effect
Low volatility (2002–2007)	Moderate
Financial crisis (Lehman – 2009)	**Strongest**
European sovereign debt crisis (2011–2015)	Weakest
Krone pressure (Jan.–Feb. 2015)	**Strongest**

- During January–February 2015, the estimated effects of DN's interventions and bank customer purchases were indistinguishable. This is consistent with DN's declared unlimited intervention commitment suppressing the signalling channel: investors could not read interventions as a signal of imminent rate changes.
- The money market spread (CITA–EONIA) and the VIX index are **not statistically significant** in the regression, likely because their effects are transmitted indirectly through customer order flows already included in the model.

Key Takeaways

1. FX intervention is an **effective instrument** for stabilising the krone, even in periods of intense speculative pressure.
2. Interventions operate primarily through the **portfolio balance channel**, with the signalling channel providing an incremental effect relative to private order flows.
3. The signalling channel is **state-dependent**: it is inoperative when DN commits to unlimited intervention, as occurred in 2015, because the interest rate signal is absent.
4. Interest rate adjustments remain the **decisive instrument** for counteracting sustained exchange rate pressure, consistent with the Mundell–Fleming framework: under a credible peg and open capital account, monetary policy must ultimately serve the exchange rate objective.

.1 UIP – Utility Maximization

In this section we derive the Uncovered Interest Parity (UIP) condition starting from a utility maximization problem. We consider an infinitely lived representative agent operating in a small open economy, who maximizes expected lifetime utility given by

$$U = E_0 \left[\sum_{t=0}^{\infty} \beta^t \, u\left(c_t, \frac{M_t}{P_t} \right) \right],$$

where c_t denotes consumption, M_t/P_t real money balances, and $\beta \in (0,1)$ is the subjective discount factor.

The agent faces the following period-t budget constraint:

$$S_t B_{t+1} + B^*_{t+1} = S_t B_t (1+R_{t-1}) + B^*_t (1+R^*_{t-1}) + S_t (M_t - M_{t+1}) + x_{t-1}(F_{t-1} - S_t) + S_t P_t (Y_t - c_t). \tag{17}$$

This constraint states that the value of next-period domestic and foreign bond holdings (left-hand side) must equal total available resources (right-hand side). These resources consist of: returns from previously held domestic and foreign bonds, net changes in money holdings, gains or losses from forward market positions, and labor income net of consumption expenditures.

To make the optimization problem explicit in terms of the control variable c_t, we rearrange the budget constraint and solve for consumption:

$$c_t = Y_t - \frac{1}{S_t P_t} \Big[S_t B_{t+1} + B^*_{t+1} - S_t B_t (1+R_{t-1}) - B^*_t (1+R^*_{t-1}) - S_t (M_t - M_{t+1}) - x_{t-1}(F_{t-1} - S_t) \Big]. \tag{18}$$

This expression shows that consumption is equal to output minus net savings in financial assets, adjusted for exchange rates, money holdings, and forward market payoffs. This formulation allows us to proceed with the agent's intertemporal optimization and derive the Euler equations underlying the UIP condition.

Now we can set the Bellman equation as follows:

$$V(B_t, B^*_t, M_t) = \max_{B_{t+1}, B^*_{t+1}, M_{t+1}} \left[u(c_t, M_t/P_t) + \beta E\big[V(B_{t+1}, B^*_{t+1}, M_{t+1}) \big] \right]. \tag{19}$$

The value function depends on the current holdings of domestic bonds, foreign bonds, and money balances, while the control variables are next-period asset positions.

We start from the **FOCs**. Therefore, we obtain:

$$\text{FOC (1)}: \quad V^{t'}_{B_{t+1}} := \frac{u'_c}{P_t} + \beta E\Big[V^{t+1'}_{B_{t+1}} \Big],$$

and

$$\text{FOC (2)}: \quad V^{t'}_{B^*_{t+1}} := \frac{u'_c}{P_t S_t} + \beta E\Big[V^{t+1'}_{B^*_{t+1}} \Big].$$

These conditions equate the marginal utility cost of acquiring an additional unit of domestic or foreign bonds to the discounted expected marginal value of holding those assets in the next period.

Now, by means of the envelope theorem, we obtain:

$$V'_{B_t} = \frac{u'_c}{P_t}(1 + R_{t-1}),$$

and

$$V'_{B^*_t} = \frac{u'_c}{P_t S_t}(1 + R^*_{t-1}).$$

These expressions reflect how an additional unit of wealth in domestic or foreign bonds affects the value function through its gross return.

Shifting the envelope conditions one period ahead and using the FOCs, we obtain:

$$-\frac{u_c^{t'}}{P_t} = \beta E\left[\frac{u_c^{t+1'}}{P_{t+1}}(1+R_{t-1})\right],$$

and

$$-\frac{u_c^{t'}}{S_t P_t} = \beta E\left[\frac{u_c^{t+1'}}{P_{t+1}S_{t+1}}(1+R^*_{t-1})\right].$$

These Euler equations characterize optimal intertemporal allocation between current consumption and domestic or foreign bond holdings.

Dividing the second condition by the first, we obtain:

$$(1+R_t) = E\left[\frac{S_{t+1}}{S_t}\right](1+R^*_t), \tag{20}$$

which is the Uncovered Interest Parity condition.